EX.NO:01 DATE: 9/7/26	Bio-data

AIM:

To write a program Bio-data.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Bio-Data</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
    <h1>BIO-DATA</h1>

    <table>
        <tr>
            <td class="label">Name</td>
            <td>Vijay Pradhap</td>
        </tr>
        <tr>
            <td class="label">Father's Name</td>
            <td>srinivasan</td>
        </tr>
        <tr>
            <td class="label">Date of Birth</td>
            <td>27-11-2004</td>
        </tr>
        <tr>
            <td class="label">Gender</td>
            <td>Male</td>
        </tr>
        <tr>
            <td class="label">Address</td>
            <td>Dindigul, Tamil Nadu</td>
        </tr>
        <tr>
            <td class="label">Mobile</td>
            <td>9080441431</td>
        </tr>
        <tr>
            <td class="label">Email</td>
            <td>vijay@example.com</td>
        </tr>
        <tr>
            <td class="label">Qualification</td>
            <td>MCA,Bsc.CS</td>
        </tr>
        <tr>
            <td class="label">Skills</td>
            <td>Python, Java, web</td>
        </tr>
        <tr>
            <td class="label">Languages Known</td>
            <td>Tamil, English,kannada</td>
        </tr>
    </table>
</div>

</body>
</html>

/* File: style.css */
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}

.container {
    width: 700px;
    margin: 30px auto;
    background-color: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 0 10px gray;
}

h1 {
```

```
    text-align: center;
    color: darkblue;
}

table {
    width: 100%;
    border-collapse: collapse;
}

table, td {
    border: 1px solid black;
}

td {
    padding: 10px;
}

.label {
    font-weight: bold;
    width: 35%;
    background-color: #d9edf7;
}
```

OUTPUT :



John Doe

Contact: john.doe@email.com | (555) 123-4567

Summary

Highly motivated and results-oriented professional with 5 years of experience in Software Development. Proven ability to deliver high-quality projects on time.

Experience

- **Senior Developer** - Tech Solutions Inc. (2020 - Present)
  - Led a team of 3 developers on the new e-commerce platform.
  - Reduced server response time by 20% through optimization.

Education

- **M.S. Computer Science** - State University (2018)
- **B.E. Information Technology** - City College (2016)

Skills

Python, JavaScript, HTML, CSS, SQL, AWS, Git

RESULT :

Thus the above program was executed successfully.

EX.NO:02 DATE: 13/7/26	Application form

AIM:

To write a program Application form.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Job Application Form</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">

<h2>Job Application Form</h2>

<form>

<label>Full Name</label>
<input type="text" placeholder="Enter your name" required>

<label>Email</label>
<input type="email" placeholder="Enter your email" required>

<label>Mobile Number</label>
<input type="tel" placeholder="Enter mobile number">

<label>Age</label>
<input type="number" min="18" max="60">

<label>Date of Birth</label>
<input type="date">

<label>Gender</label>
<div class="option-group">
    <input type="radio" name="gender"> Male
    <input type="radio" name="gender"> Female
    <input type="radio" name="gender"> Other
</div>

<label>Job Position</label>
<select>
    <option>Select Position</option>
    <option>Software Developer</option>
    <option>Web Designer</option>
    <option>Data Analyst</option>
    <option>System Administrator</option>
    <option>Tester</option>
</select>

<label>Skills</label>
<div class="option-group">
    <input type="checkbox"> HTML
    <input type="checkbox"> CSS
    <input type="checkbox"> JavaScript
    <input type="checkbox"> Java
    <input type="checkbox"> Python
</div>

<label>Experience</label>
<select>
    <option>Fresher</option>
    <option>1-2 Years</option>
    <option>3-5 Years</option>
    <option>5+ Years</option>
</select>

<label>Upload Resume</label>
<input type="file">

<label>Address</label>
<textarea rows="4" placeholder="Enter your address"></textarea>
<br>

<input type="submit" value="Submit Application">
<input type="reset" value="Reset">
</form>
```

```
</div>

<script>
const checkboxes = document.querySelectorAll('input[name="skills"]');

checkboxes.forEach(checkbox => {
    checkbox.addEventListener('change', () => {
        const checked = document.querySelectorAll('input[name="skills"]:checked');

        if (checked.length > 3) {
            checkbox.checked = false;
            alert("You can select a maximum of 3 skills.");
        }
    });
});
</script>
</body>
</html>
body {
    font-family: Arial;
    background: #f2f2f2;
    margin: 0;
    padding: 20px;}
.container {
    width: 600px;
    margin: auto;
    background: white;
    padding: 20px;
    border-radius: 8px;}

h2 {
    text-align: center;
    color: blue;}
label {
    display: block;
    margin-top: 10px;
    font-weight: bold;}
input,
select,textarea {
    width: 100%;
    padding: 8px;
    margin-top: 5px;}
textarea {
    height: 80px;}
.option-group label {
    display: inline;
    font-weight: normal;}
input[type="submit"],
input[type="reset"] {
    padding: 10px 20px;
    margin-top: 15px;
    color: white;
    border: none;
    cursor: pointer;}
 input[type="submit"] {
    background: green;}
input[type="reset"] {
    background: red;}
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:03 DATE: 13/7/26	Display two items

AIM:

To write a program Display two items.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Featured Products</title>
<style>
    body{
        font-family: Arial, sans-serif;
        margin: 20px;}
    h1{
        text-align: center;}
    .products{
        display: flex;
        justify-content: center;
        gap: 30px;
        flex-wrap: wrap;}
    .product{
        width: 300px;
        border: 1px solid #ccc;
        padding: 15px;
        border-radius: 10px;
        text-align: center;
        box-shadow: 0 2px 5px rgba(0,0,0,0.2);}
    .product img{
        width: 100%;
        height: 200px;
        object-fit: cover;
        border-radius: 8px;
    }</style></head><body>
<h1>Featured Products</h1>
 <div class="products">
    <div class="product">
        <img src="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcTuAbxJg56uoWYQxMxXNM_CyOUQvcJ6YYok60flxyb_Sg&s=10" alt="Wireless Headphones">
        <h2>Product A: Wireless Headphones</h2>
        <p><strong>Price:</strong> $79.99</p>
        <p><strong>Description:</strong> Noise-cancelling over-ear headphones with 20 hours of battery life. Available
in Black and White.</p>
    </div>
    <div class="product">
        <img src="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQh-FumD5_tosdem258YueAG8x5pa8QCvTvw5iB-
U4amw&s=10" alt="Smart Watch">
        <h2>Product B: Smart Watch</h2>
        <p><strong>Price:</strong> $199.00</p>
        <p><strong>Description:</strong> Tracks fitness, sleep, and heart rate. Features a durable aluminum case and
waterproof design.</p>
    </div>
 </div> </body></html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:04 DATE: 13/7/26	All text & image format

AIM:

To write a program All text & image format.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>Formatting Tags Demo</title>
</head>
<body>

    <h1>Text and Image Formatting Tags</h1>

    <p>This is a paragraph</p>
    <i>italic</i>
    <mark>mark</mark>
    <small>small</small>
    <ins>inserted</ins>
    <del>discontinue</del>
    <code>code</code>



    <hr>

    <h2>Image Tag Example</h2>

    <img src=""
         alt="Placeholder Image"
         style="border:2px solid blue; padding:5px; width:50%;">



</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:05 DATE: 13/7/26	iFrame

AIM:

To write a program iFrame.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>iFrame Example</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            text-align: center;
            margin: 20px;
        }

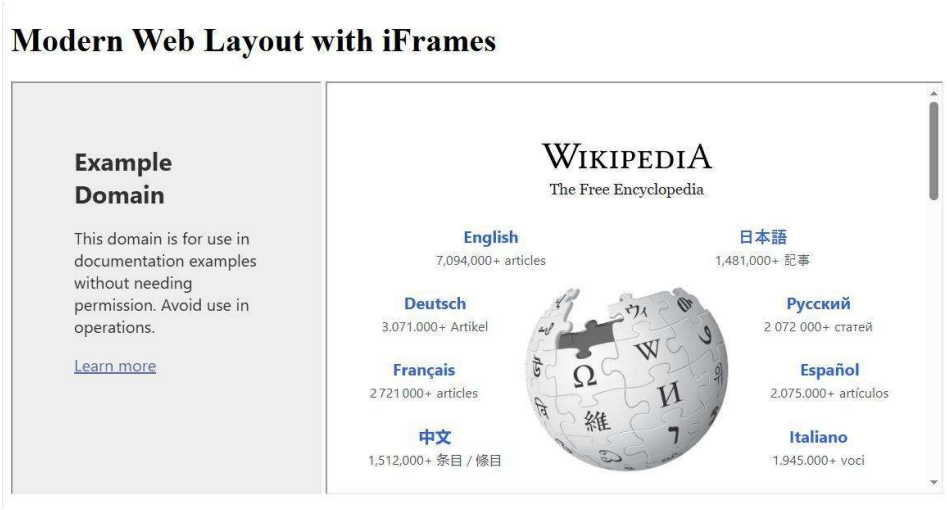
        h1 {
            color: #333;
        }

        iframe {
            border: 2px solid #007BFF;
            border-radius: 5px;
            margin: 10px;
        }
    </style>
</head>
<body>
    <h1>Modern Web Layout with iFrames</h1>

    <iframe src="https://www.example.com"
            width="300"
            height="400"
            title="Example Website 1">
    </iframe>

    <iframe src="https://www.wikipedia.org"
            width="600"
            height="400"
            title="Example Website 2">
    </iframe>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:06 DATE: 13/7/26	HTML Table & Lists

AIM:

To write a program HTML Table & Lists.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>Tables and Lists Demo</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
            background-color: #f4f4f4;
        }

        h1 {
            text-align: center;
            color: #333;
        }

        h2 {
            color: #0066cc;
        }

        table {
            width: 60%;
            border-collapse: collapse;
            background-color: white;
        }

        th, td {
            border: 1px solid black;
            padding: 10px;
            text-align: center;
        }

        th {
            background-color: #0066cc;
            color: white;
        }

        ul {
            background-color: white;
            width: 200px;
            padding: 15px;
            border: 1px solid #ccc;
        }

        li {
            margin: 5px 0;
        }
    </style>
</head>

<body>

    <h1>Table and List Examples</h1>

    <h2>Simple Data Table</h2>

    <table>
        <tr>
            <th>Name</th>
            <th>Age</th>
            <th>City</th>
        </tr>
        <tr>
            <td>Alice</td>
            <td>30</td>
            <td>New York</td>
        </tr>
        <tr>
            <td>Bob</td>
            <td>25</td>
            <td>London</td>
        </tr>
    </table>
```

```
        <hr>

        <h2>Shopping List</h2>

        <ul>
            <li>Apples</li>
            <li>Bananas</li>
            <li>Milk</li>
            <li>Bread</li>
        </ul>

    </body>
    </html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:07 DATE: 13/7/26	CSS (inline, internal, external)

AIM:

To write a program CSS (inline, internal, external).

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>CSS Implementation Demo</title>

    <!-- External CSS Files -->
    <link rel="stylesheet" href="external-styles.css">
    <link rel="stylesheet" href="multiple-styles.css">

    <!-- Internal CSS -->
    <style>
        h1 {
            color: purple;
            border-bottom: 2px solid purple;
        }

        .internal-style {
            background-color: lightyellow;
            padding: 10px;
        }
    </style>
</head>

<body>

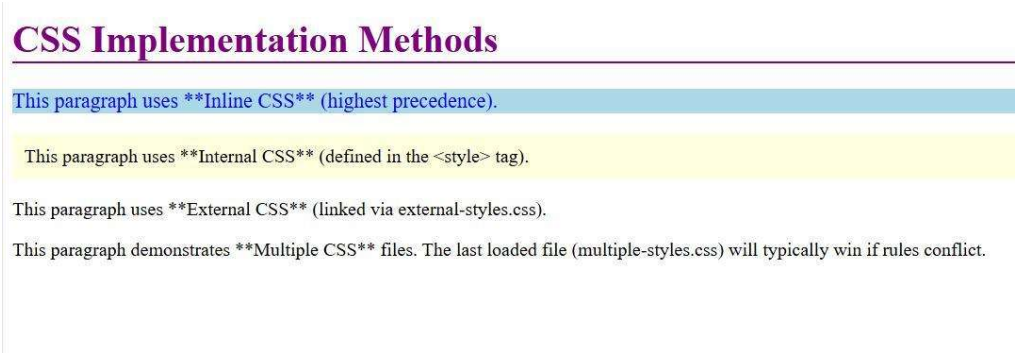
    <h1>CSS Implementation Methods</h1>

    <!-- Inline CSS -->
    <p style="color: blue; font-size: 18px; background-color: lightblue; padding:10px;">
        This paragraph uses <b>Inline CSS</b> (highest precedence).
    </p>
    <p class="internal-style">
        This paragraph uses <b>Internal CSS</b> (defined in the &lt;style&gt; tag).
    </p>

    <p id="external-style">
        This paragraph uses <b>External CSS</b> (linked via external-styles.css).
    </p>

    <p class="multiple-style">
        This paragraph demonstrates <b>Multiple CSS</b> files. The last loaded file
        (multiple-styles.css) will typically override earlier rules if they conflict.
    </p>
</body>
</html>
.multiple-style {
    color: orange;
    background-color: #fff3cd;
    padding: 10px;
}
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:08 DATE: 16/7/26	Registration form

AIM:

To write a program Registration form.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Registration Form</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background: #f2f2f2;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }

        .card {
            background: white;
            padding: 20px;
            width: 300px;
            border-radius: 8px;
            box-shadow: 0 0 10px gray;
        }

        h2 {
            text-align: center;
        }

        label {
            display: block;
            margin-top: 10px;
        }

        input {
            width: 100%;
            padding: 8px;
            margin-top: 5px;
            box-sizing: border-box;
        }

        button {
            width: 100%;
            padding: 10px;
            margin-top: 15px;
            background: blue;
            color: white;
            border: none;
            cursor: pointer;
        }

        button:hover {
            background: darkblue;
        }
    </style>
</head>

<body>

<form class="card" onsubmit="register(event)">
    <h2>Registration Form</h2>

    <label>Full Name</label>
    <input type="text" id="name" required>

    <label>Email</label>
    <input type="email" id="email" required>

    <label>Password</label>
    <input type="password" id="password" required>

    <button type="submit">Register</button>
</form>
```

```
<script>
function register(event) {
    event.preventDefault();

    const name = document.getElementById("name").value.trim();
    const email = document.getElementById("email").value.trim();
    const password = document.getElementById("password").value;

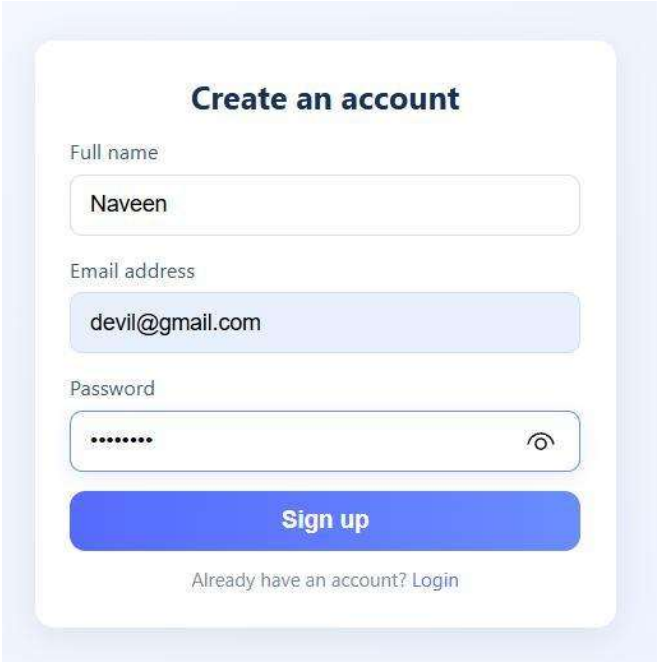
    // Email validation (basic)
    const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailRegex.test(email)) {
        alert("Please enter a valid email address.");
        return;
    }

    // Password validation (at least 8 chars, at least 1 letter and 1 number)
    const passRegex = /^(?=.*[A-Za-z])(?=.*\d).{8,}$/;
    if (!passRegex.test(password)) {
        alert("Password must be at least 8 characters and include at least one letter and one number.");
        return;
    }

    alert("Registration Successful!\n\nName: " + name + "\nEmail: " + email);
}
</script>

</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:09 DATE: 16/7/26	Emoji

AIM:

To write a program Emoji.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>HTML Emojis</title>
</head>
<body>

    <h2>Using Unicode Emojis</h2>

    <div>
        😀 Grinning Face
    </div>

    <div>
        👍 Thumbs Up
    </div>

    <div>
        💖 Red Heart
    </div>

    <hr>

    <h2>Using HTML Entity (Hex)</h2>

    <div>
        &#x1F601; Grinning Face with Smiling Eyes
    </div>

    <div>
        &#x1F4A1; Light Bulb
    </div>

    <div>
        &#x1F680; Rocket
    </div>

</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:10 DATE: 16/7/26	Menus

AIM:

To write a program Menus.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple Demo</title>
    <style>
        body{
            font-family: Arial;
            margin:20px;}
        a{
            text-decoration:none;
            margin:10px;
            color:blue;}
        .box{
            width:60px;
            height:60px;
            border:3px solid red;
            display:inline-block;
            text-align:center;
            line-height:60px;
            margin:10px;}
        .shadow{
            padding:15px;
            width:120px;
            box-shadow:3px 3px 5px gray;
            margin:10px 0;}
        .circle{
            width:60px;
            height:60px;
            background:orange;
            border-radius:50%;}
        .triangle{
            width:0;
            height:0;
            border-left:30px solid transparent;
            border-right:30px solid transparent;
            border-bottom:55px solid green;}</style>
</head>
<body>
<h2>Navigation Bar</h2>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Contact</a>
<a href="https://www.google.com/">ANOTHER WEBSITE</a>
<h2>Border</h2>
<div class="box">Box</div>
<h2>Shadow</h2>
<div class="shadow">Shadow Box</div>
<h2>Shape</h2>
<div class="circle"></div> <hr>
<div class="triangle"></div>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:11 DATE: 20/7/26	Validation of register page

AIM:

To write a program Validation of register page.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>User Registration</title>
<style>
body{font-family:Arial;text-align:center;margin-top:50px;}
form{display:inline-block;padding:20px;border:1px solid #ccc;}
input{display:block;width:220px;margin:10px;padding:8px;}
button{padding:8px 20px;}
</style>
</head>
<body>

<h2>User Registration Form</h2>

<form onsubmit="return validate()">
<input type="text" id="name" placeholder="Enter Name">
<input type="email" id="email" placeholder="Enter Email">
<input type="password" id="password" placeholder="Enter Password">
<button type="submit">Register</button>
</form>

<script>
function validate(){
let name=document.getElementById("name").value;
let email=document.getElementById("email").value;
let password=document.getElementById("password").value;

if(name==""){ alert("Enter Name"); return false; }

if(email==""){ alert("Enter Email"); return false; }

let pattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;
if(!pattern.test(email)){ alert("Invalid Email"); return false; }

if(password.length<6){
alert("Password must be at least 6 characters");
return false;
}

alert("Registration Successful!");
return true;
}
</script>

</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:12 DATE: 20/7/26	User data stored into database

AIM:

To write a program User data stored into database.

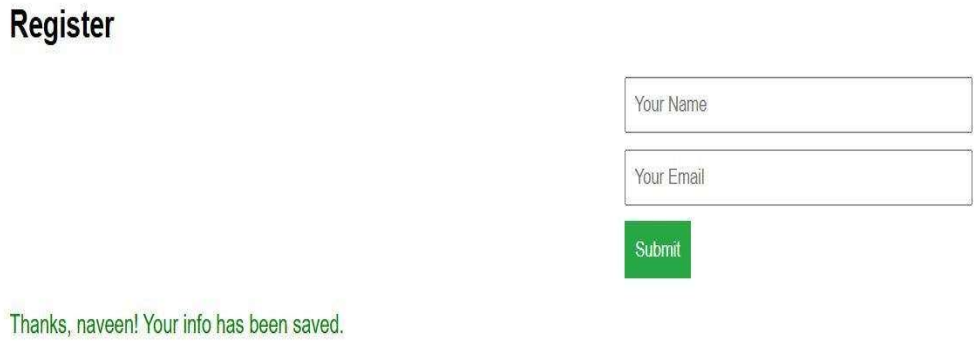
PROGRAM:

```
/* File: db.sql */
CREATE DATABASE awpdb;
USE awpdb;
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50)
);
```

```
<!DOCTYPE html>
<html>
<head>
<title>User Form</title>
<style>
body {
background: #f3f3f3;
font-family: Arial;
}
.container {
width: 350px;
background: white;
padding: 20px;
margin: 50px auto;
border-radius: 10px;
box-shadow: 0 0 10px gray;
}
input {
width: 95%;
padding: 10px;
margin-top: 10px;
border-radius: 5px;
border: 1px solid gray;
}
button {
padding: 10px;
width: 100%;
background: blue;
color: white;
border: none;
margin-top: 15px;
border-radius: 5px;
cursor: pointer;
}
button:hover {
background: darkblue;
}
</style>
<script>
function validateForm() {
let name = document.getElementById("name").value;
let email = document.getElementById("email").value;
if (name == "" || email == "") {
alert("Both fields are required!");
return false;
}
if (!email.includes("@")) {
alert("Enter a valid email ID!");
return false;
}
return true;
}
</script>
</head>
<body>
<div class="container">
<h2>Save User</h2>
<form action="save.php" method="post" onsubmit="return validateForm()">
<input type="text" id="name" name="name" placeholder="Enter Name">
<input type="text" id="email" name="email" placeholder="Enter Email">
<button type="submit">Save</button>
</form>
</div>
</body>
</html>
```

```
/* File: save.php */
<?php
$con = mysqli_connect("localhost", "root", "", "awpdb");
$name = $_POST["name"];
$email = $_POST["email"];
$sql = "INSERT INTO users(name, email) VALUES('$name', '$email')";
mysqli_query($con, $sql);
echo "<h2>User Saved Successfully!</h2>";
echo "<a href='index.html'>Go Back</a>";
?>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:13 DATE: 20/7/26	XML File

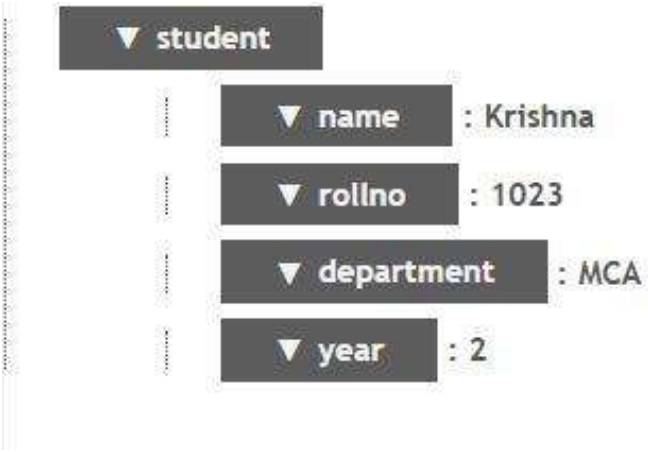
AIM:

To write a program XML File.

PROGRAM:

```
<?xml version="1.0" encoding="UTF-8"?>
<student>
    <name>Krishna</name>
    <rollno>1023</rollno>
    <department>MCA</department>
    <year>2</year>
</student>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:14 DATE: 20/7/26	XML with internal DTD

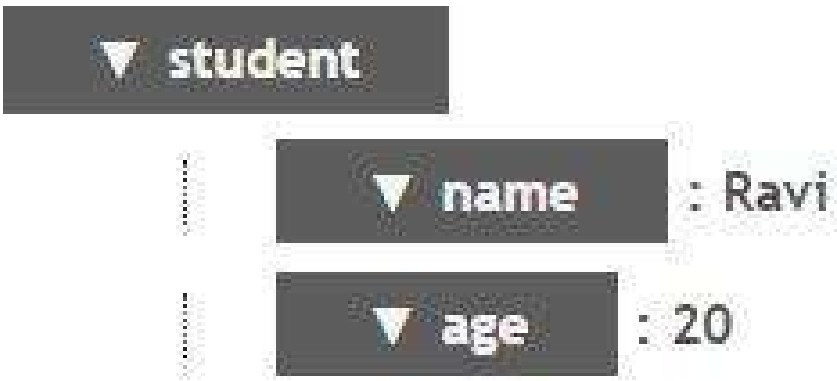
AIM:

To write a program XML with internal DTD.

PROGRAM:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE student [
<!ELEMENT student (name, age)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT age (#PCDATA)>
]>
<student>
    <name>Ravi</name>
    <age>20</age>
</student>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:15 DATE: 20/7/26	XML with External DTD

AIM:

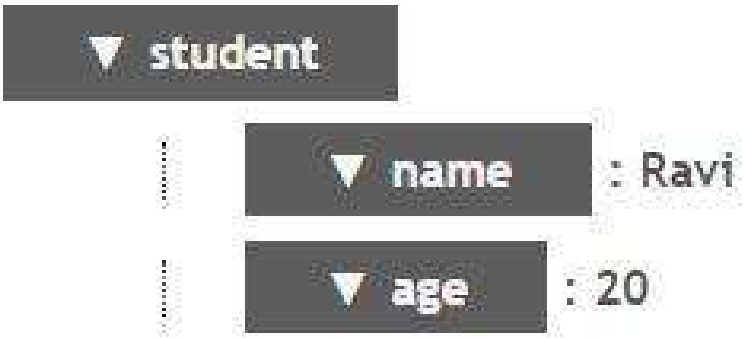
To write a program XML with External DTD.

PROGRAM:

```
/* File: student.dtd */
<!ELEMENT student (name, age)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT age (#PCDATA)>

/* File: student.xml */
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE student SYSTEM "student.dtd">
<student>
  <name>Ravi</name>
  <age>20</age>
</student>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:16 DATE: 20/7/26	Implement all web API

AIM:

To write a program Implement all web API.

PROGRAM:

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8"><meta name="viewport" content="width=device-width,initial- scale=1">
<title>Mini Web APIs Demo</title>
<style>
body{font-family:Arial;margin:18px;background:#f7fbff;color:#123}
.box{width:80px;height:80px;background:#6bbcff;border-radius:8px;display:grid;place-
items:center;color:#fff;cursor:grab}
.drop{width:200px;height:100px;border:2px dashed #9ab;border-radius:8px;display:grid;place-items:center;margin-
top:8px}
.row{display:flex;gap:12px;flex-wrap:wrap}
.card{background:#fff;padding:10px;border-radius:8px;box-shadow:0 6px 18px rgba(0,30,80,.06);min-width:240px}
small{color:#556;font-size:12px}
</style>
</head>
<body>
<h3>Compact Web APIs demo</h3>
<div class="row">
<!-- Drag & Drop -->
<div class="card">
<strong>Drag & Drop</strong>
<div id="drag" class="box" draggable="true">DRAG</div>
<div id="drop" class="drop">Drop here</div>
<small id="dmsg"></small>
</div>
<!-- Geolocation -->
<div class="card">
<strong>Geolocation</strong>
<button id="geoBtn">Get Location</button>
<div id="loc">—</div>
<small>Uses navigator.geolocation</small>
</div>
<!-- Third-party (Lodash) -->
<div class="card">
<strong>Third-party (Lodash)</strong>
<input id="txt" placeholder="type name"><button id="cap">Cap</button>
<div id="capout">—</div>
<small>CDN script + _.capitalize</small>
</div>
<!-- Validation + Storage -->
<div class="card">
<strong>Validation & Storage</strong>
<form id="f">
<input id="name" placeholder="name (letters only)" required pattern="^[A-Za-z ]+$"><br><br>
<input id="age" placeholder="age (18-99)" required pattern="^(1[89]|[2-9][0-9])$"><br><br>
<button>Submit</button>
</form>
<div><button id="save" type="button">Save name to localStorage</button> <button id="load"
type="button">Load</button></div>
<div id="stored">—</div>
</div>
<!-- History API -->
<div class="card">
<strong>History API</strong>
<div id="page">Page A</div>
<button id="goB" type="button">Go to B</button>
<small>Use browser Back to return</small>
</div>
<!-- Web Worker -->
<div class="card">
<strong>Web Worker</strong>
<button id="startw" type="button">Start Worker (calc)</button>
<div id="wout">—</div>
<small>Worker sums 1..N in background</small>
</div>
</div>
<!-- Lodash CDN (third-party) -->
<script src="https://cdn.jsdelivr.net/npm/lodash@4.17.21/lodash.min.js"></script>
<script>
// --- Drag & Drop ---
const drag = document.getElementById('drag'), drop = document.getElementById('drop'), dmsg =
document.getElementById('dmsg');
drag.addEventListener('dragstart', e => e.dataTransfer.setData('text/plain','dragged'));
drop.addEventListener('dragover', e => e.preventDefault());
```

```
drop.addEventListener('drop', e => { e.preventDefault(); dmsg.textContent = 'Dropped!'; });

// --- Geolocation ---
document.getElementById('geoBtn').onclick = () => {
  const out = document.getElementById('loc');
  if (!navigator.geolocation) return out.textContent = 'Not supported';
  navigator.geolocation.getCurrentPosition(
    p => out.textContent = `Lat:${p.coords.latitude.toFixed(4)}, Lon:${p.coords.longitude.toFixed(4)}`,
    () => out.textContent = 'Denied / error'
  );
};

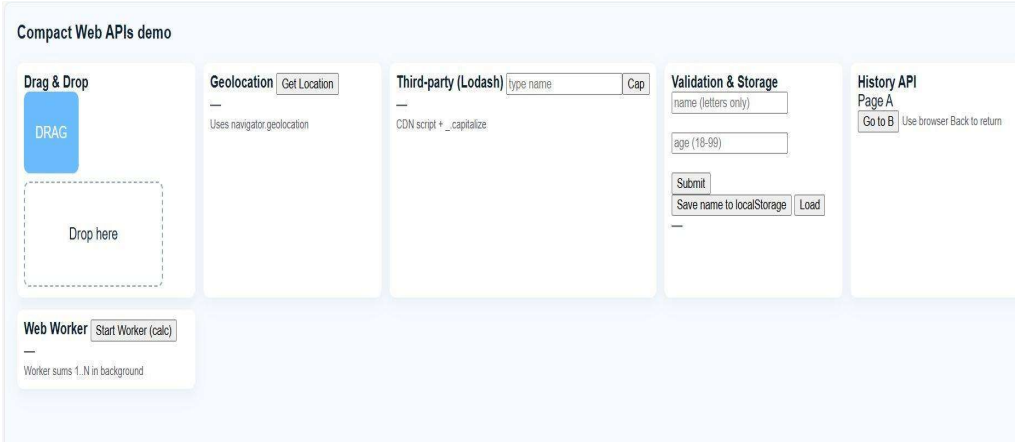
// --- Lodash usage (third-party) ---
document.getElementById('cap').onclick = () => {
  const v = document.getElementById('txt').value || '';
  document.getElementById('capout').textContent = _.capitalize(v);
};

// --- Validation + Storage ---
const form = document.getElementById('f');
form.addEventListener('submit', e => {
  e.preventDefault();
  if (!form.checkValidity()) { alert('Fix fields'); return; }
  alert('Form OK — name: ' + document.getElementById('name').value);
});
document.getElementById('save').onclick = () => {
  const n = document.getElementById('name').value;
  if(n) localStorage.setItem('demo_name', n);
  document.getElementById('stored').textContent = 'Saved';
};
document.getElementById('load').onclick = () => {
  document.getElementById('stored').textContent = localStorage.getItem('demo_name') || 'Noname';
};

// --- History API ---
const page = document.getElementById('page');
document.getElementById('goB').onclick = () => {
  history.pushState({page:'B'}, 'B', '?page=B');
  page.textContent = 'Page B';
};
window.addEventListener('popstate', e => {
  page.textContent = (e.state && e.state.page==='B') ? 'Page B' : 'Page A';
});

// --- Web Worker via Blob ---
const wout = document.getElementById('wout');
document.getElementById('startw').onclick = () => {
  // small worker that sums numbers up to N (N=1000000)
  const code = `self.onmessage=e=>{const n=e.data;let s=0;for(let i=1;i<=n;i++)s+=i;self.postMessage(s)}`;
  const worker = new Worker(URL.createObjectURL(new Blob([code],{type:'text/javascript'})));
  wout.textContent = 'Working...';
  worker.onmessage = m => { wout.textContent = 'Sum=' + m.data; worker.terminate(); };
  worker.postMessage(1000000);
};
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:17 DATE: 20/7/26	XML file in HTML Document

AIM:

To write a program XML file in HTML Document.

PROGRAM:

```
/* File: data.xml */
<?xml version="1.0" encoding="UTF-8"?>
<items>
  <item>
    <name>Pen</name>
    <price>10</price>
  </item>
  <item>
    <name>Book</name>
    <price>40</price>
  </item>
</items>

/* File: display.html */
<!DOCTYPE html>
<html>
<body>
<h2>Items List</h2>
<table border="1" id="tbl"></table>
<script>
var x = new XMLHttpRequest();
x.open("GET", "data.xml", false);
x.send();
var xml = x.responseXML;
var items = xml.getElementsByTagName("item");
let out = "<tr><th>Name</th><th>Price</th></tr>";
for (let i = 0; i < items.length; i++) {
  let name = items[i].getElementsByTagName("name")[0].textContent;
  let price = items[i].getElementsByTagName("price")[0].textContent;
  out += "<tr><td>" + name + "</td><td>" + price + "</td></tr>";
}
document.getElementById("tbl").innerHTML = out;
</script>
</body>
</html>
```

OUTPUT :

Items List

Name	Price
Pen	10
Book	40

RESULT :

Thus the above program was executed successfully.

EX.NO:18 DATE: 23/7/26	HTML Form to XML File

AIM:

To write a program HTML Form to XML File.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
  <title>Send Form Data to XML</title>
</head>
<body>
  <h2>Send HTML Form Data to XML File</h2>
  <form onsubmit="sendData(); return false;">
    Name: <input type="text" id="name"><br><br>
    Email: <input type="text" id="email"><br><br>
    <button type="submit">Save to XML</button>
  </form>

  <h3>XML Created:</h3>
  <pre id="output"></pre>

  <script>
    function sendData() {
      let name = document.getElementById("name").value;
      let email = document.getElementById("email").value;

      // Create XML string
      let xml =
        "<user>\n" +
        " <name>" + name + "</name>\n" +
        " <email>" + email + "</email>\n" +
        "</user>";

      document.getElementById("output").textContent = xml;

      // Send XML to PHP for saving
      let x = new XMLHttpRequest();
      x.open("POST", "savexml.php", true);
      x.setRequestHeader("Content-Type", "application/x-www-form-urlencoded");
      x.send("xmlData=" + encodeURIComponent(xml));
    }
  </script>
</body>
</html>
/* File: savexml.php */
<?php
$xml = $_POST["xmlData"];
// Save the XML string into a file
file_put_contents("user.xml", $xml);
echo "XML File Saved Successfully!";
?>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:19 DATE: 23/7/26	XML to HTML Form

AIM:

To write a program XML to HTML Form.

PROGRAM:

```
/* File: form.html */
<!DOCTYPE html>
<html>
<body>
  Name: <input type="text" id="name">
  <script>
    var x = new XMLHttpRequest();
    x.open("GET", "user.xml", false);
    x.send();

    // Try responseXML first (works when served from a server)
    var xml = x.responseXML;
    if (xml && xml.getElementsByTagName("name")[0]) {
      document.getElementById("name").value = xml.getElementsByTagName("name")[0].textContent;
    } else if (x.responseText) {
      // Fallback: parse responseText with DOMParser (works with local files)
      var parser = new DOMParser();
      var doc = parser.parseFromString(x.responseText, "text/xml");
      var nameNode = doc.getElementsByTagName("name")[0];
      if (nameNode) {
        document.getElementById("name").value = nameNode.textContent;
      }
    }
  </script>
</body>
</html>

/* File: user.xml */
<user>
  <name>Raja</name>
</user>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:20 DATE: 30/7/26	Simple data type (XML Schema / XSD)

AIM:

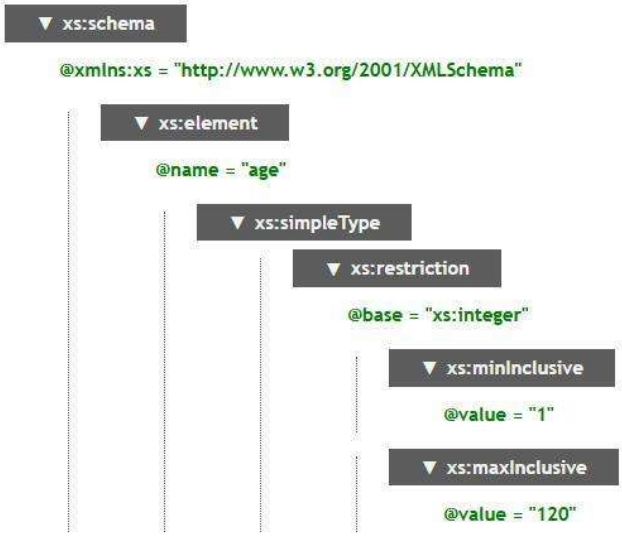
To write a program Simple data type (XML Schema / XSD).

PROGRAM:

```
/* File: student.xml */
<?xml version="1.0" encoding="UTF-8"?>
<student xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:noNamespaceSchemaLocation="student.xsd">
  <name>Ravi Kumar</name>
  <department>Computer Science</department>
  <year>2</year>
</student>

/* File: student.xsd */
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="student">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="name" type="xs:string"/>
        <xs:element name="department" type="xs:string"/>
        <xs:element name="year" type="xs:integer"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:21 DATE: 30/7/26	Complex data type (XML Schema / XSD)

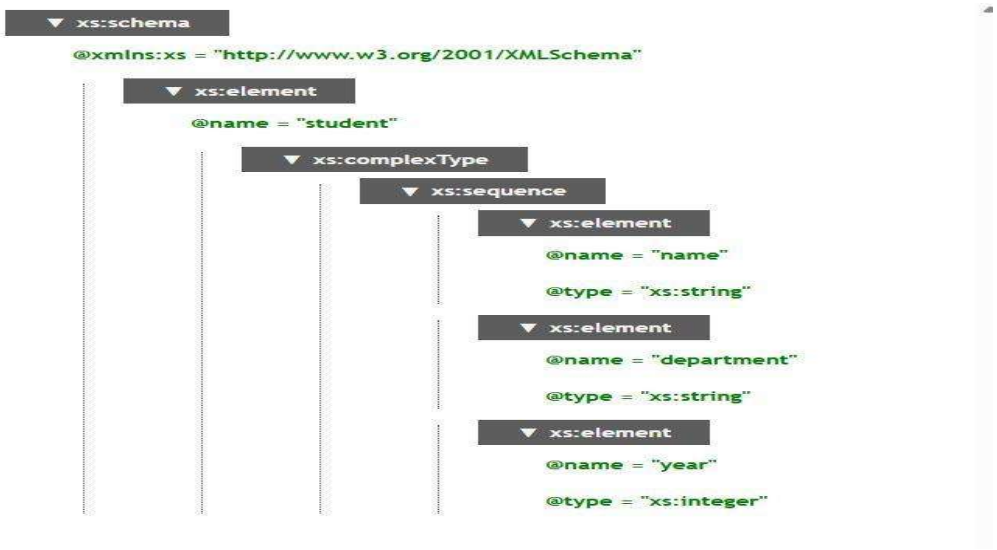
AIM:

To write a program Complex data type (XML Schema / XSD).

PROGRAM:

```
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="student">
<xs:complexType>
<xs:sequence>
<xs:element name="name" type="xs:string"/>
<xs:element name="department" type="xs:string"/>
<xs:element name="year" type="xs:integer"/>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:22 DATE: 31/8/26	Read XML data using PHP display

AIM:

To write a program Read XML data using PHP display.

PROGRAM:

```
<?php
$xmlString = <<<XML
<bookstore>
  <book category="fiction">
    <title lang="en">END</title>
    <author>vijay malik</author>
    <year>1925</year>
    <price>12.99</price>
  </book>
</bookstore>
XML;

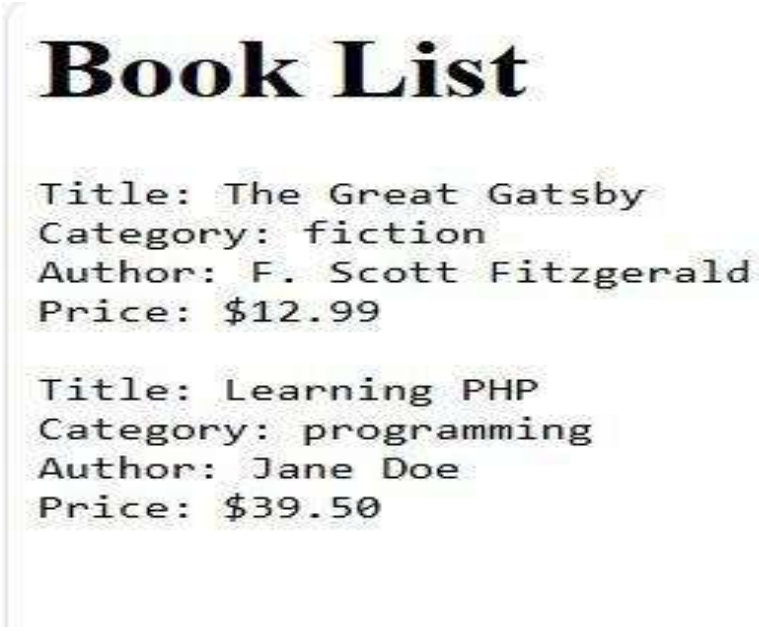
$xml = simplexml_load_string($xmlString);

if ($xml !== false) {
    echo "<h1>Book List</h1><pre>";

    foreach ($xml->book as $book) {
        echo "Title: " . (string)$book->title . "\n";
        echo "Category: " . (string)$book['category'] . "\n";
        echo "Author: " . (string)$book->author . "\n";
        echo "Year: " . (string)$book->year . "\n";
        echo "Price: $" . (string)$book->price . "\n\n";
    }

    echo "</pre>";
} else {
    echo "Failed to load XML.";
}
?>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:23 DATE: 31/8/26	Read XML data to display in HTML

AIM:

To write a program Read XML data to display in HTML.

PROGRAM:

```
<?php
$xmlData = <<<XML
<?xml version="1.0" encoding="UTF-8"?>
<user>
  <name>Devil Kumar</name>
  <email>devil.kumar@example.com</email>
  <city>Madurai</city>
</user>
XML;

$xml = simplexml_load_string($xmlData);
echo "Name: " . $xml->name . "<br>";
echo "Email: " . $xml->email . "<br>";
echo "City: " . $xml->city . "<br>";
?>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:24 DATE: 8/8/26	Display XML data using PHP

AIM:

To write a program Display XML data using PHP.

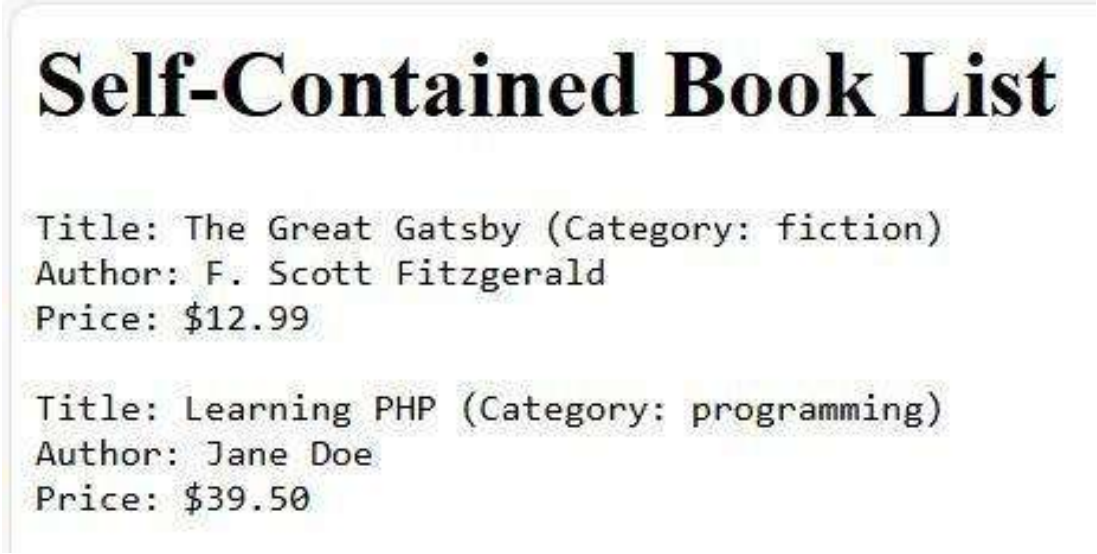
PROGRAM:

```
<?php
$xmlString = <<<XML
<bookstore>
  <book category="fiction">
    <title lang="en">The Great Gatsby</title>
    <author>F. Scott Fitzgerald</author>
    <price>12.99</price>
  </book>
  <book category="programming">
    <title lang="en">Learning PHP</title>
    <author>Jane Doe</author>
    <price>39.50</price>
  </book>
</bookstore>
XML;

$xml = simplexml_load_string($xmlString);

if ($xml !== false) {
    echo "<h1>Self-Contained Book List</h1><pre>";
    foreach ($xml->book as $book) {
        echo "Title: " . (string)$book->title . " (Category: " . (string)$book['category'] . ")\n";
        echo "Author: " . (string)$book->author . "\n";
        echo "Price: $" . (string)$book->price . "\n\n";
    }
    echo "</pre>";
} else {
    echo "Failed to load XML from string.";
}
?>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:25 DATE: 8/8/26	XML File using PHP display statement

AIM:

To write a program XML File using PHP display statement.

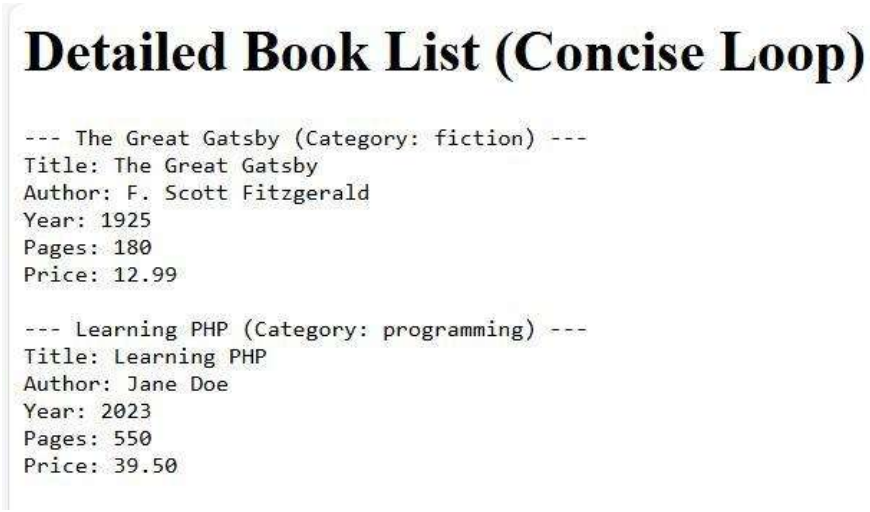
PROGRAM:

```
<?php
$xmlString = <<<XML
<bookstore>
  <book category="fiction">
    <title lang="en">The Great Gatsby</title>
    <author>F. Scott Fitzgerald</author>
    <year>1925</year>
    <pages>180</pages>
    <price>12.99</price>
  </book>
  <book category="programming">
    <title lang="en">Learning PHP</title>
    <author>Jane Doe</author>
    <year>2023</year>
    <pages>550</pages>
    <price>39.50</price>
  </book>
</bookstore>
XML;

$xml = simplexml_load_string($xmlString);

if ($xml !== false) {
    echo "<h1>Detailed Book List (Concise Loop)</h1><pre>";
    foreach ($xml->book as $book) {
        echo "--- " . (string)$book->title . " (Category: " . (string)$book['category'] . ") ---\n";
        foreach ($book as $key => $value) {
            echo ucfirst($key) . ": " . (string)$value . "\n";
        }
        echo "\n";
    }
    echo "</pre>";
} else {
    echo "Failed to load XML from string.";
}
?>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:26 DATE: 8/8/26	JavaScript object using JSON, Illustrate

AIM:

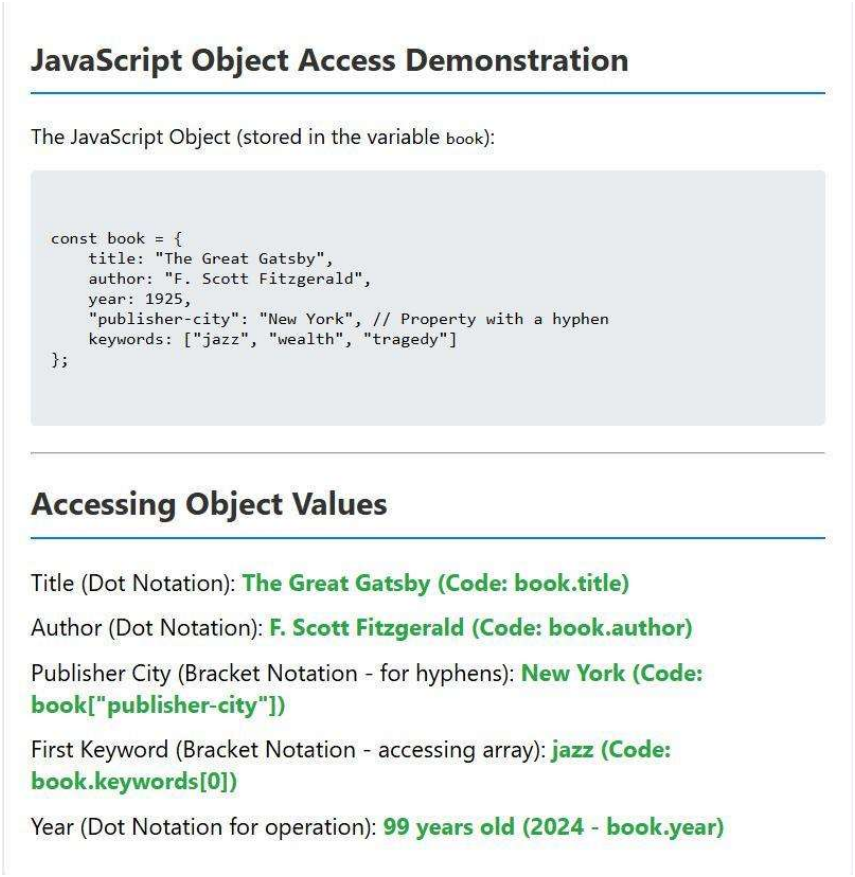
To write a program JavaScript object using JSON, Illustrate.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>JavaScript Object Literal Demo</title>
<style>
body {
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
margin: 20px;
background-color: #f4f4f9;
}
.container {
max-width: 600px;
margin: 0 auto;
background-color: #fff;
padding: 20px;
border-radius: 8px;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
h2 {
color: #333;
border-bottom: 2px solid #007bff;
padding-bottom: 10px;
}
.code-box {
background-color: #e9ecef;
padding: 15px;
border-radius: 4px;
margin-bottom: 20px;
overflow-x: auto;
}
.output-section p {
font-size: 1.1em;
margin: 10px 0;
}
.highlight {
font-weight: bold;
color: #28a745;
}
</style>
</head>
<body>
<div class="container">
<h2>JavaScript Object Access Demonstration</h2>
<p>The JavaScript Object (stored in the variable <code>book</code>):</p>
<div class="code-box">
<pre><code>
const book = {
title: "The Great Gatsby", author: "F. Scott Fitzgerald", year: 1925, "publisher-city": "New York", // Property with a
hyphen
keywords: ["jazz", "wealth", "tragedy"]
};
</code></pre>
</div>
<hr>
<div class="output-section">
<h2>Accessing Object Values</h2>
<p>Title (Dot Notation): <span id="output-title" class="highlight"></span></p>
<p>Author (Dot Notation): <span id="output-author" class="highlight"></span></p><p>Publisher City (Bracket Notation -
for hyphens): <span id="output-publisher"class="highlight"></span></p>
<p>First Keyword (Bracket Notation - accessing array): <span id="output-keyword"class="highlight"></span></p>
<p>Year (Dot Notation for operation): <span id="output-operation"class="highlight"></span></p>
</div>
</div>
<script>
const book = {
title: "The Great Gatsby", author: "F. Scott Fitzgerald", year: 1925, "publisher-city": "New York", keywords: ["jazz",
"wealth", "tragedy"]
};
const bookTitle = book.title;
const bookAuthor = book.author
const publisherCity = book["publisher-city"];
const firstKeyword = book.keywords[0];
const age = 2024 - book.year;
document.getElementById('output-title').textContent = bookTitle + " (Code: book.title)";
document.getElementById('output-author').textContent = bookAuthor + " (Code: book.author)";
```

```
document.getElementById('output-publisher').textContent = publisherCity + " (Code:
book[\"publisher-city\"])";
document.getElementById('output-keyword').textContent = firstKeyword + " (Code:
book.keywords[0])";
document.getElementById('output-operation').textContent = age + " years old (2024- book.year)";
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:27 DATE: 8/8/26	JavaScript array by passing a JSON string value

AIM:

To write a program JavaScript array by passing a JSON string value.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>JSON to Array Demo</title>
</head>
<body>
<h1>Parsing JSON String to JavaScript Array</h1>
<p>JSON String (input): <code id="json-string"></code></p>
<p>Parsed Array Length: <span id="array-length"></span></p>
<p>First Element (Index 0): <span id="first-element"></span></p>
<p>Last Element (Index 2): <span id="last-element"></span></p>
<script>
const jsonString = '[ "Apple", "Banana", "Cherry", "Date" ]';
document.getElementById('json-string').textContent = jsonString;
const fruitArray = JSON.parse(jsonString);
document.getElementById('array-length').textContent = fruitArray.length;
document.getElementById('first-element').textContent = fruitArray[0];
document.getElementById('last-element').textContent = fruitArray[fruitArray.length- 1];
console.log("Parsed Array:", fruitArray);
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:28 DATE: 17/8/26	JSON File using your own File

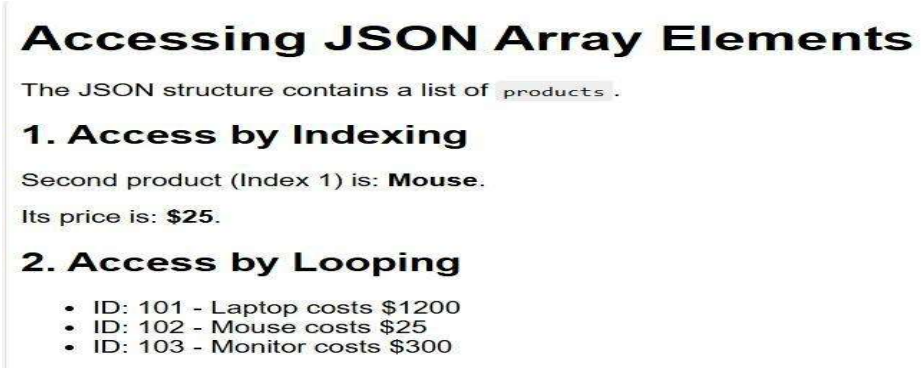
AIM:

To write a program JSON File using your own File.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>JSON Array Access Demo</title>
<style>
body { font-family: Arial, sans-serif; }
code { background-color: #eee; padding: 2px 4px; border-radius: 3px; }
.output-box { border: 1px solid #ccc; padding: 10px; margin-top: 10px; }
</style>
</head>
<body>
<h1>Accessing JSON Array Elements</h1>
<p>The JSON structure contains a list of <code>books</code>.</p>
<div id="indexed-output">
<h2>1. Access by Indexing</h2>
</div>
<div id="looped-output">
<h2>2. Access by Looping</h2>
<ul id="product-list">
</ul>
</div>
<script>
const jsonString = `{
"storeName": "ABC Book Store",
"location": "Chennai",
"books": [
{ "id": 201, "name": "Java Programming", "price": 450 },
{ "id": 202, "name": "Python Programming", "price": 500 },
{ "id": 203, "name": "Web Development", "price": 600 }
]
}`;
const data = JSON.parse(jsonString);
const booksArray = data.books;
const secondBookName = booksArray[1].name;
const secondBookPrice = booksArray[1].price;
document.getElementById('indexed-output').innerHTML += `
<p>Second book (Index 1) is:
<strong>${secondBookName}</strong>.</p>
<p>Its price is:
<strong>₹${secondBookPrice}</strong>.</p>
`;
const listContainer = document.getElementById('product-list');
for (let i = 1; i < booksArray.length; i++) {
const book = booksArray[i];
const listItem = document.createElement('li');
listItem.textContent =
`ID: ${book.id} - ${book.name} costs ₹${book.price}`;
listContainer.appendChild(listItem);
 }</script></body></html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:29 DATE: 17/8/26	Usage of Date & Function in JSON

AIM:

To write a program Usage of Date & Function in JSON.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>JSON Date and Function Example</title>
</head>

<body>
    <h1>JSON Date and Function Demo</h1>
    <p>Check the browser console (F12) for the results.</p>

    <script>
        const student = {
            studentId: 101,
            studentName: "Arun",
            admissionDate: new Date(),
            calculateMark: function(x) {
                return x + 10;
            }
        };

        const jsonString = JSON.stringify(student, null, 2);

        console.log("--- Original JavaScript Object ---");
        console.log(student);

        console.log("\n--- JSON.stringify() Result ---");
        console.log(jsonString);

        const parsedStudent = JSON.parse(jsonString);

        const dateFromJSON = new Date(parsedStudent.admissionDate);

        console.log("\n--- After JSON.parse() ---");
        console.log(
            "Admission Date type in parsed JSON:",
            typeof parsedStudent.admissionDate
        );
        console.log(
            "Re-created JavaScript Date:",
            dateFromJSON.getFullYear()        );
        console.log(
            "Function property (calculateMark):",
            parsedStudent.calculateMark
        );
    </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:30 DATE: 24/8/26	JSON.stringify() on array, date & function

AIM:

To write a program JSON.stringify() on array, date & function.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>JSON.stringify() Usage Demo</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            padding: 20px;
        }

        pre {
            background-color: #f4f4f4;
            padding: 15px;
            border: 1px solid #ddd;
            overflow-x: auto;
        }

        .result-box {
            margin-bottom: 20px;
            padding: 10px;
            border-left: 3px solid #007bff;
        }

        .label {
            font-weight: bold;
            color: #333;
        }
    </style>
</head>

<body>

    <h1>Using JSON.stringify()</h1>

    <div id="output"></div>

    <script>
        // 1. Array
        const myArray = [10, "hello", true, null];
        const stringifiedArray = JSON.stringify(myArray);

        // 2. Date Object
        const myDate = new Date();
        const stringifiedDate = JSON.stringify(myDate);

        // 3. Object containing a function
        const myObjectWithFunction = {
            id: 1,
            name: "Test",
            calculate: function(a, b) {
                return a + b;
            }
        };

        const stringifiedObject =
            JSON.stringify(myObjectWithFunction);


        function displayResult(title, original, stringified) {

            const outputDiv = document.getElementById("output");

            let originalDisplay;

            if (original instanceof Date) {
                originalDisplay = original.toString();
            } else if (typeof original === "object" && original !== null) {
                originalDisplay = JSON.stringify(original, null, 2);
            } else {
                originalDisplay = String(original);
            }

            const conclusion = title.includes("Function")
                ? "The function was <strong>removed</strong> from the final JSON string."
```

```
                : "The data was successfully converted into a JSON <strong>string representation</strong>.";

            outputDiv.innerHTML += `
                <div class="result-box">

                    <h3>${title}</h3>

                    <p class="label">Original Data Type:</p>
                    <pre><code>${originalDisplay}</code></pre>

                    <p class="label">Stringified JSON Output:</p>
                    <pre><code>${stringified}</code></pre>

                </div>
            `;
        }


        // Display results
        displayResult(
            "1. Array ([ ])",
            myArray,
            stringifiedArray
        );

        displayResult(
            "2. Date Object (new Date())",
            myDate,
            stringifiedDate
        );

        displayResult(
            "3. Object with Function",
            myObjectWithFunction,
            stringifiedObject
        );
    </script>

</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:31 DATE: 24/8/26	Send HTML form data to JSON File

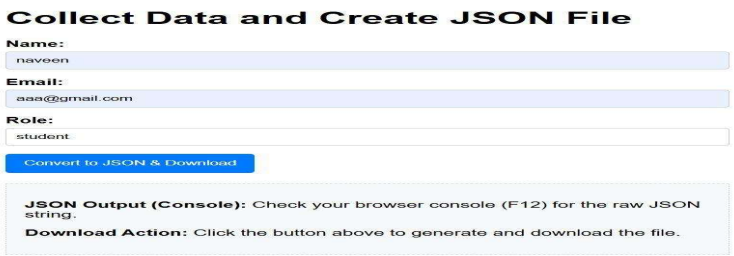
AIM:

To write a program Send HTML form data to JSON File.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>Form Data to JSON Download</title>
<style>
body { font-family: Arial, sans-serif; padding: 20px; max-width: 600px; margin: auto; }label { display: block; margin-
bottom: 5px; font-weight: bold; }
input[type="text"], input[type="email"] { width: 100%; padding: 8px; margin-bottom:
15px; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
button { padding: 10px 15px; background-color: #007bff; color: white; border: none;
border-radius: 4px; cursor: pointer; }
button:hover { background-color: #0056b3; }
.output-box { margin-top: 20px; padding: 15px; background-color: #f8f9fa; border: 1pxdashed #ced4da; }
</style>
</head>
<body>
<h1>Collect Data and Create JSON File</h1>
<form id="dataForm">
<label for="userName">Name:</label>
<input type="text" id="userName" name="name" required>
<label for="userEmail">Email:</label>
<input type="email" id="userEmail" name="email" required>
<label for="userRole">Role:</label>
<input type="text" id="userRole" name="role" required>
<button type="submit">Convert to JSON & Download</button>
</form>
<div class="output-box">
<p><strong>JSON Output (Console):</strong> Check your browser console (F12) for
the raw JSON string.</p>
</div>
<script>
document.getElementById('dataForm').addEventListener('submit', function(event) {
event.preventDefault();
const formData = new FormData(event.target);
const dataObject = {};
formData.forEach((value, key) => {
dataObject[key] = value;
});
dataObject.submissionDate = new Date().toISOString();
const jsonString = JSON.stringify(dataObject, null, 2);
console.log("JSON Data to be Saved:", jsonString);
const blob = new Blob([jsonString], { type: 'application/json' });
const url = URL.createObjectURL(blob);
const a = document.createElement('a');
a.href = url;
a.download = 'form_data_' + Date.now() + '.json';
document.body.appendChild(a);
a.click();
document.body.removeChild(a);
URL.revokeObjectURL(url);
});
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:32 DATE: 24/8/26	Data from JSON File to HTML Forms

AIM:

To write a program Data from JSON File to HTML Forms.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
    <title>Multiple Record JSON</title>
</head>
<body>

<form id="recordForm">
    Name: <input type="text" id="name" required><br><br>
    Age: <input type="number" id="age" required><br><br>

    <input type="submit" value="Save Record">
</form>

<script>
document.getElementById("recordForm").addEventListener("submit", function(event) {
    event.preventDefault();

    const name = document.getElementById("name").value;
    const age = document.getElementById("age").value;

    const data = {
        name: name,
        age: age
    };

    const jsonData = JSON.stringify(data, null, 2);

    const blob = new Blob([jsonData], {
        type: "application/json"
    });

    const url = URL.createObjectURL(blob);

    const link = document.createElement("a");
    link.href = url;
    link.download = "record_" + Date.now() + ".json";

    link.click();

    URL.revokeObjectURL(url);
});
</script>

</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:33 DATE: 24/8/26	Store HTML form data as Separate JSON File

AIM:

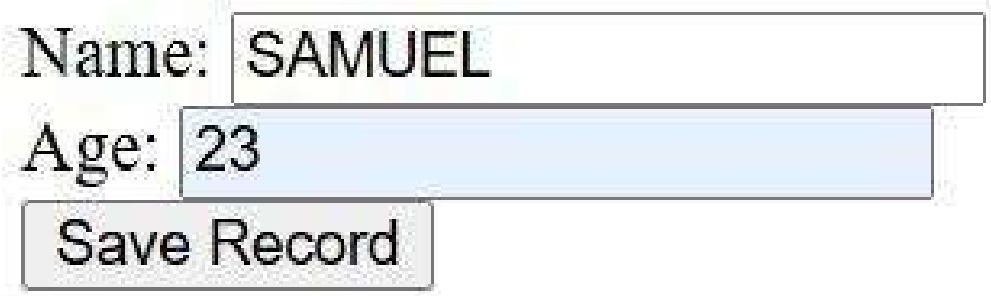
To write a program Store HTML form data as Separate JSON File.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head><title>Multiple Record JSON</title></head>
<body>
<form method="post" action="save_multi.php">
Name: <input type="text" name="name"><br>
Age: <input type="text" name="age"><br>
<input type="submit" value="Save Record">
</form>
</body>
</html>

/* File: save_multi.php */
<?php
$name = $_POST['name'];
$age = $_POST['age'];
$data = array("name"=>$name, "age"=>$age);
$fileName = "record_" . time() . ".json";
file_put_contents($fileName, json_encode($data));
echo "Record saved as $fileName";
?>
```

OUTPUT :




```
Record saved as record_1764424134.json
```

RESULT :

Thus the above program was executed successfully.

EX.NO:34 DATE: 10/9/26	Types of Expression

AIM:

To write a program Types of Expression.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Expression</title>
</head>
<body>
    <h2>Expression</h2>
    <p>Numeric:{{10+3}}</p>
    <p>string:{{"hello"+"world"}}</p>
    <p>Boolean:{{10 < 2}}</p>
    <p>Numeric:{{{'name':'vijay','age':22}name}}</p>
    <p>array:{{[10,20,50][1]}}</p>
    <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
</body>
</html>
```

OUTPUT :

Expression

Numeric:{{10+3}}

string:{{"hello"+"world"}}

Boolean:{{10 < 2}}

Numeric:{{{'name':'vijay','age':22}name}}

array:{{[10,20,50][1]}}

RESULT :

Thus the above program was executed successfully.

EX.NO:35 DATE: 10/9/26	ng-model Directive

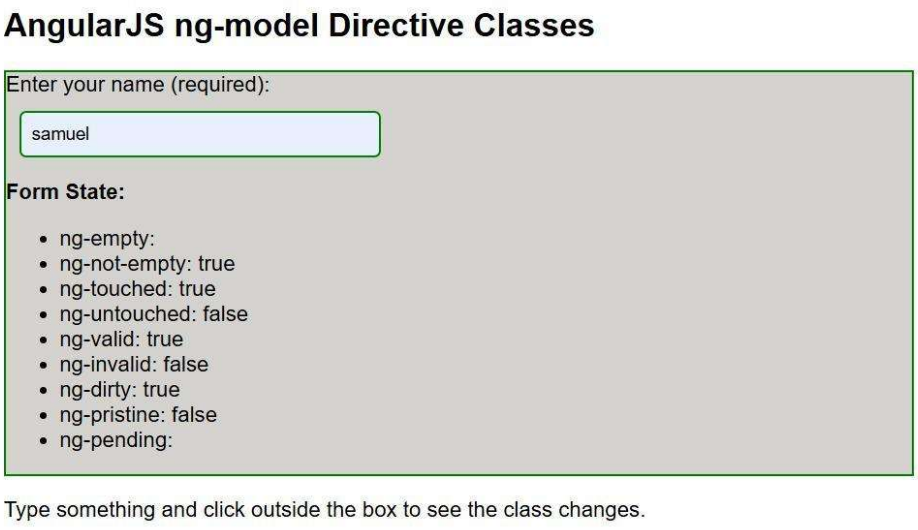
AIM:

To write a program ng-model Directive.

PROGRAM:

```
<!DOCTYPE html><html lang="en" ng-app><head><meta charset="UTF-8">
<title>ng-model Directive Classes</title>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script><style>
body { font-family: Arial; margin: 30px; }
input {display: block;margin: 10px 10px;padding: 8px;
width: 250px;border: 2px solid gray;border-radius: 5px;}
.ng-empty { background-color: #f8d7da; }
.ng-not-empty { background-color: #d4edda; }
.ng-touched { border-color: blue; }
.ng-untouched { border-color: orange; }
.ng-valid { border: 2px solid green; }
.ng-invalid { border: 2px solid red; }
.ng-dirty { background-color: #d4d3d0; }
.ng-pristine { background-color: #e2e3e5; }
</style>
</head>
<body>
<h2>AngularJS ng-model Directive Classes</h2>
<form name="myForm">
<label>Enter your name (required):</label>
<input type="text" name="username" ng-model="user.name" required>
<p><b>Form State:</b></p>
<ul>
<li>ng-empty: {{ myForm.username.$error.required }}</li>
<li>ng-not-empty: {{ !myForm.username.$error.required }}</li>
<li>ng-touched: {{ myForm.username.$touched }}</li>
<li>ng-untouched: {{ myForm.username.$untouched }}</li>
<li>ng-valid: {{ myForm.username.$valid }}</li>
<li>ng-invalid: {{ myForm.username.$invalid }}</li>
<li>ng-dirty: {{ myForm.username.$dirty }}</li>
<li>ng-pristine: {{ myForm.username.$pristine }}</li>
<li>ng-pending: {{ myForm.username.$pending }}</li>
</ul>
</form>
<p>Type something and click outside the box to see the class changes. </p>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:36 DATE: 10/9/26	Types of Data Binding

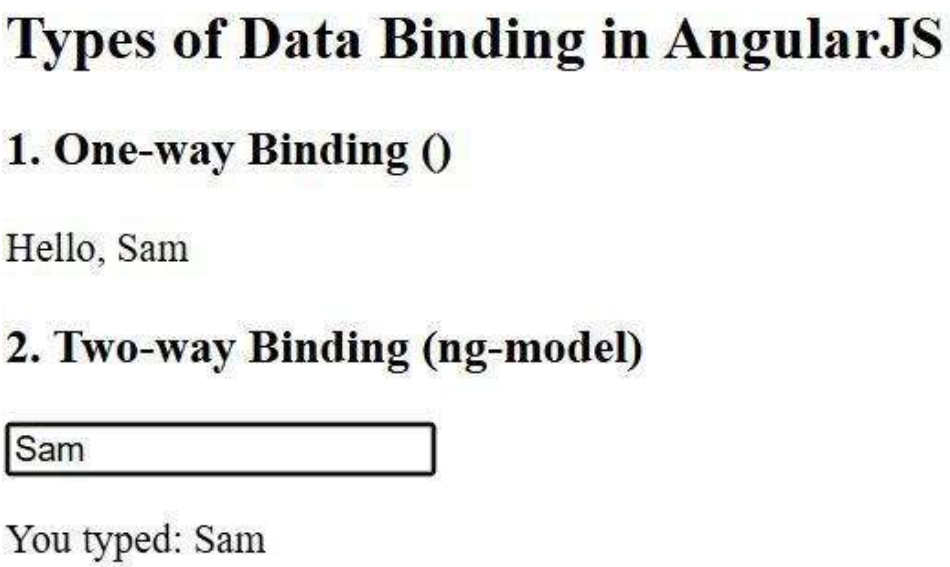
AIM:

To write a program Types of Data Binding.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="bindApp">
<head>
<title>Types of Data Binding</title>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script></head>
<body ng-controller="bindCtrl">
<h2>Types of Data Binding in AngularJS</h2>
<!-- One-way binding -->
<h3>1. One-way Binding ({{ }})</h3>
<p>Hello, {{ name }}</p>
<!-- Two-way binding -->
<h3>2. Two-way Binding (ng-model)</h3>
<input type="text" ng-model="message">
<p>You typed: {{ message }}</p>
<script>
var app = angular.module("bindApp", []);
app.controller("bindCtrl", function($scope) {
$scope.name = "Sam";
$scope.message = "Type something...";
})
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:37 DATE: 10/9/26	Modules & Controller directives

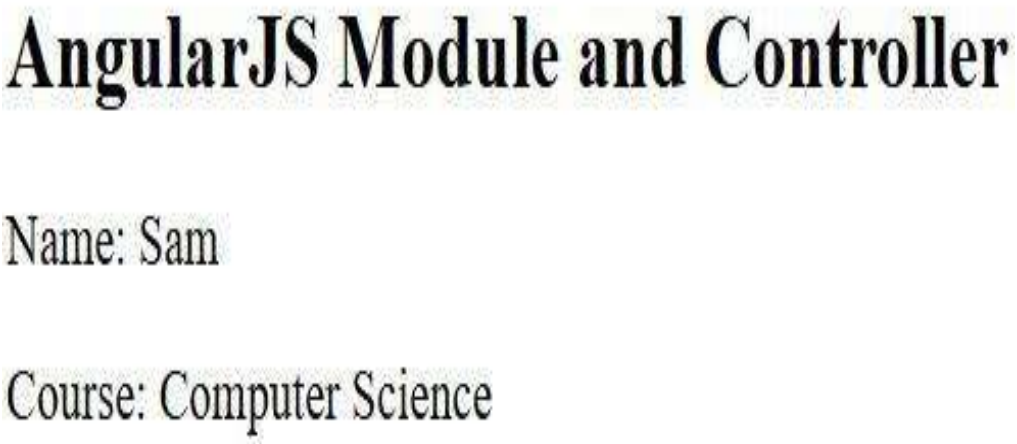
AIM:

To write a program Modules & Controller directives.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="studentApp">
<head><title>Modules and Controllers</title></head>
<body ng-controller="studentCtrl">
<h2>AngularJS Module and Controller</h2>
<p>Name: {{student.name}}</p>
<p>Course: {{student.course}}</p>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script><script>
angular.module("studentApp", [])
.controller("studentCtrl", function($scope) {
$scope.student = {name:"Sam", course:"Computer Science"};
});
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:38 DATE: 17/9/26	Array for AngularJS

AIM:

To write a program Array for AngularJS.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="filterApp" ng-controller="filterCtrl">
<head><title>AngularJS Filters</title></head>
<body>
<h2>AngularJS Filters Example</h2>
<p>1. Uppercase: {{name | uppercase}}</p>
<p>2. Lowercase: {{name | lowercase}}</p>
<p>3. Currency: {{price | currency:"₹"}}</p>
<p>4. Number: {{num | number:2}}</p>
<p>5. LimitTo: {{names | limitTo:3}}</p>
<p>6. OrderBy: {{students | orderBy:'marks'}}</p>
<p>7. Filter: {{students | filter:'A'}}</p>
<p>8. JSON: {{students | json}}</p>
<p>9. Date: {{today | date:'medium'}}</p>
<p>10. Custom Filter: {{text | reverse}}</p>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script><script>
angular.module("filterApp", [])
.controller("filterCtrl", function($scope) {
$scope.name = "Sam";
$scope.price = 99.99;
$scope.num = 1234.567;
$scope.names = ["Asha", "Ravi", "Sam", "Kiran"];
$scope.students = [{name:"A", marks:70},{name:"B",marks:90}];
$scope.today = new Date();
$scope.text = "Hello";
})
.filter('reverse', function() {
return function(input) {
return input.split("").reverse().join("");
};
});
</script>
</body>
</html>
```

OUTPUT :

AngularJS Filters Example

1. Uppercase: SAM
2. Lowercase: sam
3. Currency: ₹99.99
4. Number: 1,234.57
5. LimitTo: ["Asha","Ravi","Sam"]
6. OrderBy: [{"name":"A","marks":70},{"name":"B","marks":90}]
7. Filter: [{"name":"A","marks":70}]
8. JSON: [{ "name": "A", "marks": 70 }, { "name": "B", "marks": 90 }]
9. Date: Nov 29, 2025 6:32:40 PM
10. Custom Filter: olleH

RESULT :

Thus the above program was executed successfully.

EX.NO:39 DATE: 17/9/26	Services of $http

AIM:

To write a program Services of $http.

PROGRAM:

```
<!DOCTYPE html><html lang="en" ng-app="httpApp"><head><meta charset="UTF-8">
<title>$http Services Example</title>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script><style>
body { font-family: Arial; margin: 30px; }
button { margin: 5px; padding: 8px 12px; cursor: pointer; }
pre { background: #f4f4f4; padding: 10px; border-radius: 5px; }</style></head>
<body ng-controller="httpCtrl">
<h2>$http Services in AngularJS</h2>
<p>Click the buttons below to simulate different $http methods.</p>
<button ng-click="getData()">$http.get()</button>
<button ng-click="headData()">$http.head()</button>
<button ng-click="postData()">$http.post()</button>
<button ng-click="putData()">$http.put()</button>
<button ng-click="deleteData()">$http.delete()</button>
<button ng-click="jsonpData()">$http.jsonp()</button>
<button ng-click="patchData()">$http.patch()</button>
<h3>Response:</h3>
<pre>{{response | json}}</pre>
<script>angular.module("httpApp", [])
.controller("httpCtrl", function($scope, $http) {
const url = "https://jsonplaceholder.typicode.com/posts/1"; // free test API
$scope.getData = function() {
$http.get(url).then(function(res) {
$scope.response = res.data;});};
$scope.headData = function() {
$http.head(url).then(function(res) {
$scope.response = res.headers();});};
$scope.postData = function() {
$http.post("https://jsonplaceholder.typicode.com/posts", {
title: "New Post", body: "Sample content", userId: 1
}).then(function(res) {
$scope.response = res.data;});};
$scope.putData = function() {
$http.put(url, { title: "Updated Post", body: "Edited content", userId: 1 })
.then(function(res) {
$scope.response = res.data;});};
$scope.deleteData = function() {
$http.delete(url).then(function(res) {
$scope.response = { message: "Data deleted successfully!" };});};
$scope.jsonpData = function() {
$http.jsonp("https://api.github.com/users/angular?callback=JSON_CALLBACK")
.then(function(res) {
$scope.response = res.data;});};
$scope.patchData = function() {
$http.patch(url, { title: "Patched Title" }).then(function(res) {
$scope.response = res.data;
});};});
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:40 DATE: 17/9/26	ng-repeat Directive (with filter & without filter)

AIM:

To write a program ng-repeat Directive (with filter & without filter).

PROGRAM:

```
/* 1. Static Table */
<!DOCTYPE html>
<html ng-app="tableApp">
<head>
<meta charset="utf-8" />
<title>ng-repeat – Static Table</title>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script><style>
body { font-family: Arial, sans-serif; margin: 30px; }
table { border-collapse: collapse; width: 80%; max-width: 800px; }
th, td { padding: 10px 12px; border: 1px solid #ccc; text-align: left; }
thead { background: #2c7be5; color: white; }
tbody tr:nth-child(even) { background: #f6f8fb; }
caption { font-size: 1.2em; margin-bottom: 8px; font-weight: bold; }
</style>
</head>
<body ng-controller="TableCtrl">
<table>
<caption>Student List (Static)</caption>
<thead>
<tr><th>#</th><th>Name</th><th>Age</th><th>Course</th></tr>
</thead>
<tbody>
<tr ng-repeat="s in students track by $index">
<td>{{ $index + 1 }}</td>
<td>{{ s.name }}</td>
<td>{{ s.age }}</td>
<td>{{ s.course }}</td>
</tr>
</tbody>
</table>
<script>
angular.module('tableApp', [])
.controller('TableCtrl', function($scope) {
$scope.students = [
{ name: 'Sam', age: 20, course: 'CSE' }, { name: 'Asha', age: 21, course: 'ECE' }, { name: 'Ravi', age: 22, course:
'ME' }, { name: 'Kiran', age: 19, course: 'IT' }
];
});
</script>
</body>
</html>

/* 2. Server Table */
<!DOCTYPE html>
<html ng-app="tableAppServer">
<head>
<meta charset="utf-8" />
<title>ng-repeat – Table from PHP (JSON)</title>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
<style>
body { font-family: Arial, sans-serif; margin: 30px; }
table { border-collapse: collapse; width: 90%; max-width: 900px; }
th, td { padding: 10px 12px; border: 1px solid #ddd; text-align: left; }
thead { background: #28a745; color: #fff; }
tbody tr:hover { background: #eef9f0; }
caption { font-weight: bold; margin-bottom: 8px; }
.loading { color: #666; margin-top: 12px; }
.error { color: red; margin-top: 12px; }
</style>
</head>
<body ng-controller="ServerTableCtrl">
<table ng-if="students.length">
<caption>Student List (From PHP)</caption>
<thead>
<tr><th>#</th><th>Name</th><th>Age</th><th>Course</th></tr>
</thead>
<tbody>
<tr ng-repeat="s in students track by $index">
<td>{{ $index + 1 }}</td>
<td>{{ s.name }}</td>
<td>{{ s.age }}</td>
<td>{{ s.course }}</td>
</tr>
</tbody>
</table>
```

```
<div class="loading" ng-if="loading">Loading data from server...</div>
<div class="error" ng-if="error">{{ error }}</div>
<script>
angular.module('tableAppServer', [])
.controller('ServerTableCtrl', function($scope, $http) {
$scope.students = [];
$scope.loading = true;
$scope.error = null;
// Adjust path if PHP file is in a different folder
$http.get('students.php')
.then(function(response){
$scope.students = response.data;
$scope.loading = false;
})
.catch(function(err){
$scope.error = 'Failed to load data. Check students.php and server.';
$scope.loading = false;
console.error(err);
});
});
</script>
</body>
</html>
```

OUTPUT :

Student List (Static)

#	Name	Age	Course
1	Sam	20	CSE
2	Asha	21	ECE
3	Ravi	22	ME
4	Kiran	19	IT

RESULT :

Thus the above program was executed successfully.

EX.NO:41 DATE: 17/9/26	Binding application to Data attribute of HTML DOM

AIM:

To write a program Binding application to Data attribute of HTML DOM.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="bindApp" ng-controller="bindCtrl">
<head><title>Attribute Binding</title></head>
<body>
<h2>Binding Data to Attributes</h2>
<img ng-src="{{image}}" width="150">
<br>
<a ng-href="{{link}}">Go to Google</a>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script><script>
angular.module("bindApp", [])
.controller("bindCtrl", function($scope) {
$scope.image =
"https://th.bing.com/th/id/R.fa21749d39d8ff8622c2c237ceb5f748?rik=f3h%2b%2fFY2Mm6WVg&riu=http%3a%2f%2fww
w.themarysue.com%2fwp- content%2fuploads%2f2015%2f04%2fspider-
man.jpg&ehk=rp3sX0qoTecfB0bn3ODwXhKXeCl6JKb%2brMxg59H6n4E%3d&risl=&pid=ImgRaw&r=0";
$scope.link = "https://www.google.com";
});
</script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:42 DATE: 21/9/26	Dynamic Student Profile Application

AIM:

To write a program Dynamic Student Profile Application.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="profileApp">
<head>
  <meta charset="UTF-8">
  <title>Program 42 - Dynamic Student Profile</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="ProfileCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#42</span>
        <div class="header-title">
          <h1>Dynamic Student Profile</h1>
          <p>AngularJS Expressions, Modules & Controllers</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <div class="grid-2">
      <!-- Input Card -->
      <div class="card">
        <div class="card-title"><span>Student Information</span><span class="badge badge-info">ng-model</span></div>
        <div class="form-group">
          <label class="form-label">Full Name</label>
          <input type="text" class="form-control" ng-model="student.name" placeholder="Enter name">
        </div>
        <div class="form-group">
          <label class="form-label">Roll Number</label>
          <input type="text" class="form-control" ng-model="student.rollNo" placeholder="Enter roll number">
        </div>
        <div class="form-group">
          <label class="form-label">Department</label>
          <select class="form-control" ng-model="student.dept">
            <option>Computer Science</option>
            <option>Information Technology</option>
            <option>Electronics</option>
          </select>
        </div>
        <div class="grid-3">
          <div class="form-group">
            <label class="form-label">HTML5</label>
            <input type="number" class="form-control" ng-model="student.m1" min="0" max="100">
          </div>
          <div class="form-group">
            <label class="form-label">CSS3</label>
            <input type="number" class="form-control" ng-model="student.m2" min="0" max="100">
          </div>
          <div class="form-group">
            <label class="form-label">JavaScript</label>
            <input type="number" class="form-control" ng-model="student.m3" min="0" max="100">
          </div>
        </div>
      </div>

      <!-- Preview Card -->
      <div class="card">
        <div class="card-title"><span>Live Profile Preview</span><span class="badge badge-success">Evaluated via {{
}}</span></div>
        <div style="margin-bottom: 20px;">
          <h2 style="color: #fff;">{{ (student.name || 'Student Name') | uppercase }}</h2>
          <p class="text-muted">{{ student.rollNo || 'Roll Number' }} • {{ student.dept }}</p>
        </div>

        <div class="grid-3 mb-3">
          <div class="stat-box">
            <div class="stat-val text-info">{{ getTotal() }}</div>
            <div class="stat-label">Total (/300)</div>
          </div>
          <div class="stat-box">
            <div class="stat-val text-success">{{ getAvg() | number:1 }}%</div>
```

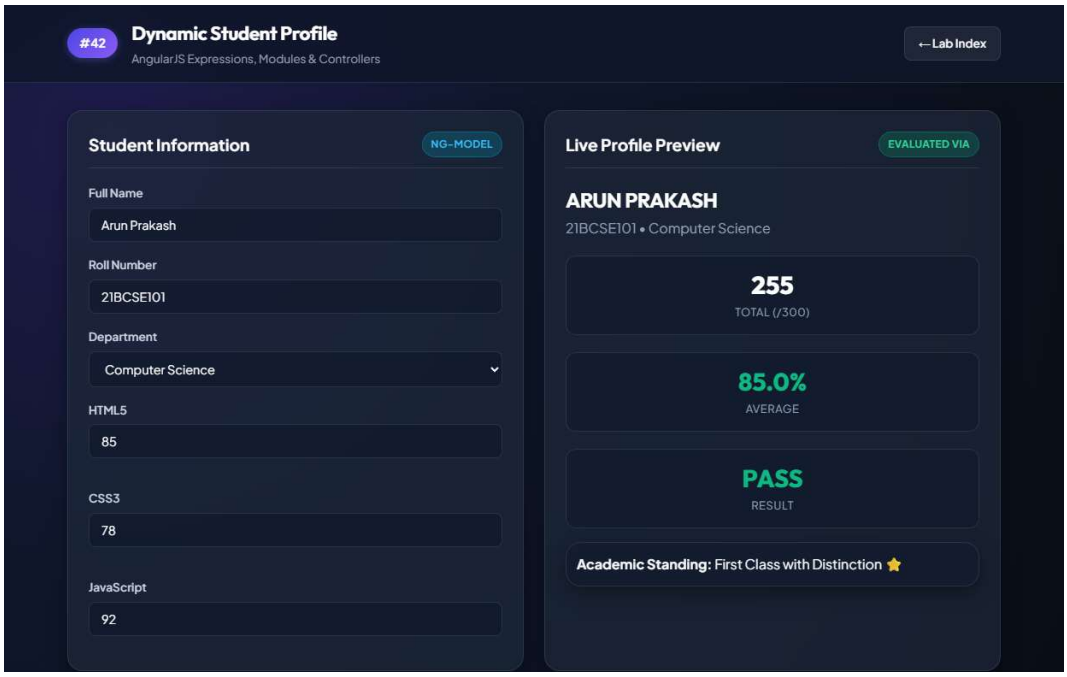
```
            <div class="stat-label">Average</div>
          </div>
          <div class="stat-box">
            <div class="stat-val" ng-class="getAvg() >= 50 ? 'text-success' : 'text-danger'">
              {{ getAvg() >= 50 ? 'PASS' : 'FAIL' }}
            </div>
            <div class="stat-label">Result</div>
          </div>
        </div>

        <div class="card" style="background: rgba(15,23,42,0.6); padding: 12px; margin: 0;">
          <strong>Academic Standing:</strong>
          <span>{{ getStanding() }}</span>
        </div>
      </div>
    </div>
  </main>

  <script>
    angular.module('profileApp', [])
      .controller('ProfileCtrl', function($scope) {
        $scope.student = { name: 'Arun Prakash', rollNo: '21BCSE101', dept: 'Computer Science', m1: 85, m2: 78, m3: 92
};

        $scope.getTotal = function() {
          return (Number($scope.student.m1) || 0) + (Number($scope.student.m2) || 0) + (Number($scope.student.m3) ||
0);
        };
        $scope.getAvg = function() { return $scope.getTotal() / 3; };
        $scope.getStanding = function() {
          var a = $scope.getAvg();
          return a >= 75 ? 'First Class with Distinction ⭐' : a >= 50 ? 'First Class ✓' : 'Needs Improvement ⚠';
        };
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:43 DATE: 21/9/26	College Management Modules

AIM:

To write a program College Management Modules.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="collegeApp">
<head>
  <meta charset="UTF-8">
  <title>Program 43 - College Management</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="CollegeCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#43</span>
        <div class="header-title">
          <h1>College Management System</h1>
          <p>Modular Architecture with Sub-Modules</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <div class="card mb-4 text-center">
      <h2 style="color: #fff;">{{ collegeName }}</h2>
      <p class="text-muted">Master Module: <code>angular.module('collegeApp', ['studentModule',
'facultyModule'])</code></p>
    </div>

    <div class="grid-2">
      <!-- Student Sub-Module -->
      <div class="card" ng-controller="StudentCtrl">
        <div class="card-title"><span>Student Sub-Module</span><span class="badge badge-
info">studentModule</span></div>
        <form ng-submit="addStudent()" style="display: flex; gap: 8px; margin-bottom: 16px;">
          <input type="text" class="form-control" ng-model="newName" placeholder="Student name" required>
          <input type="text" class="form-control" ng-model="newRoll" placeholder="Roll No" style="width: 120px;"
required>
          <button class="btn btn-primary btn-sm">Add</button>
        </form>
        <div class="table-container">
          <table class="custom-table">
            <thead><tr><th>#</th><th>Roll No</th><th>Student Name</th><th>Action</th></tr></thead>
            <tbody>
              <tr ng-repeat="s in students">
                <td>{{ $index + 1 }}</td>
                <td>{{ s.roll }}</td>
                <td>{{ s.name }}</td>
                <td><button class="btn btn-danger btn-sm" ng-click="students.splice($index, 1)">×</button></td>
              </tr>
            </tbody>
          </table>
        </div>
      </div>

      <!-- Faculty Sub-Module -->
      <div class="card" ng-controller="FacultyCtrl">
        <div class="card-title"><span>Faculty Sub-Module</span><span class="badge badge-
success">facultyModule</span></div>
        <form ng-submit="addFaculty()" style="display: flex; gap: 8px; margin-bottom: 16px;">
          <input type="text" class="form-control" ng-model="newName" placeholder="Faculty name" required>
          <input type="text" class="form-control" ng-model="newDept" placeholder="Department" style="width: 120px;"
required>
          <button class="btn btn-primary btn-sm">Add</button>
        </form>
        <div class="table-container">
          <table class="custom-table">
            <thead><tr><th>#</th><th>Faculty Name</th><th>Department</th><th>Action</th></tr></thead>
            <tbody>
              <tr ng-repeat="f in facultyList">
                <td>{{ $index + 1 }}</td>
                <td>{{ f.name }}</td>
                <td>{{ f.dept }}</td>
                <td><button class="btn btn-danger btn-sm" ng-click="facultyList.splice($index, 1)">×</button></td>
              </tr>
```

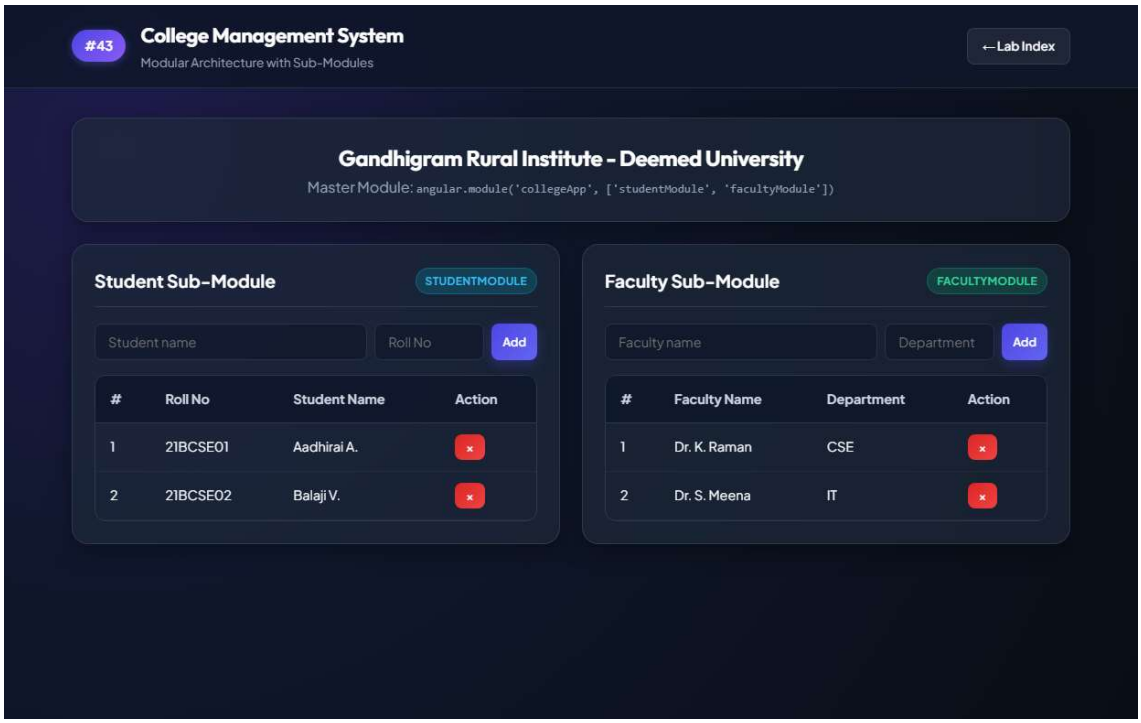
```
            </tbody>
          </table>
        </div>
      </div>
    </div>
  </main>

  <script>
    // Sub-Module 1: Students
    angular.module('studentModule', [])
      .controller('StudentCtrl', function($scope) {
        $scope.students = [{ roll: '21BCSE01', name: 'Aadhirai A.' }, { roll: '21BCSE02', name: 'Balaji V.' }];
        $scope.addStudent = function() {
          $scope.students.push({ name: $scope.newName, roll: $scope.newRoll });
          $scope.newName = $scope.newRoll = '';
        };
      });

    // Sub-Module 2: Faculty
    angular.module('facultyModule', [])
      .controller('FacultyCtrl', function($scope) {
        $scope.facultyList = [{ name: 'Dr. K. Raman', dept: 'CSE' }, { name: 'Dr. S. Meena', dept: 'IT' }];
        $scope.addFaculty = function() {
          $scope.facultyList.push({ name: $scope.newName, dept: $scope.newDept });
          $scope.newName = $scope.newDept = '';
        };
      });

    // Master App Module injecting sub-modules
    angular.module('collegeApp', ['studentModule', 'facultyModule'])
      .controller('CollegeCtrl', function($scope) {
        $scope.collegeName = 'Gandhigram Rural Institute - Deemed University';
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:44 DATE: 21/9/26	Student Attendance Application

AIM:

To write a program Student Attendance Application.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="attendanceApp">
<head>
  <meta charset="UTF-8">
  <title>Program 44 - Student Attendance</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="AttendanceCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#44</span>
        <div class="header-title">
          <h1>Student Attendance Application</h1>
          <p>Conditional & Repetition Directives (ng-repeat, ng-switch, ng-if, ng-show)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <!-- ng-if Alert Directive -->
    <div ng-if="count('Absent') > 0" class="card mb-4" style="background: rgba(239, 68, 68, 0.15); border-color:
#ef4444;">
      <strong style="color: #fca5a5;">⚠ ng-if Alert:</strong> {{ count('Absent') }} student(s) marked Absent!
    </div>

    <!-- Controls & ng-show Summary -->
    <div class="card mb-4 flex-between">
      <button class="btn btn-secondary btn-sm" ng-click="showSummary = !showSummary">
        {{ showSummary ? 'Hide Summary (ng-show)' : 'Show Summary (ng-show)' }}
      </button>
      <div style="display: flex; gap: 8px;">
        <button class="btn btn-success btn-sm" ng-click="setAll('Present')">All Present</button>
        <button class="btn btn-danger btn-sm" ng-click="setAll('Absent')">All Absent</button>
      </div>
    </div>

    <!-- ng-show Summary Box -->
    <div ng-show="showSummary" class="grid-3 mb-4">
      <div class="stat-box"><div class="stat-val text-info">{{ students.length }}</div><div class="stat-label">Total
Students</div></div>
      <div class="stat-box"><div class="stat-val text-success">{{ count('Present') }}</div><div class="stat-
label">Present</div></div>
      <div class="stat-box"><div class="stat-val text-danger">{{ count('Absent') }}</div><div class="stat-
label">Absent</div></div>
    </div>

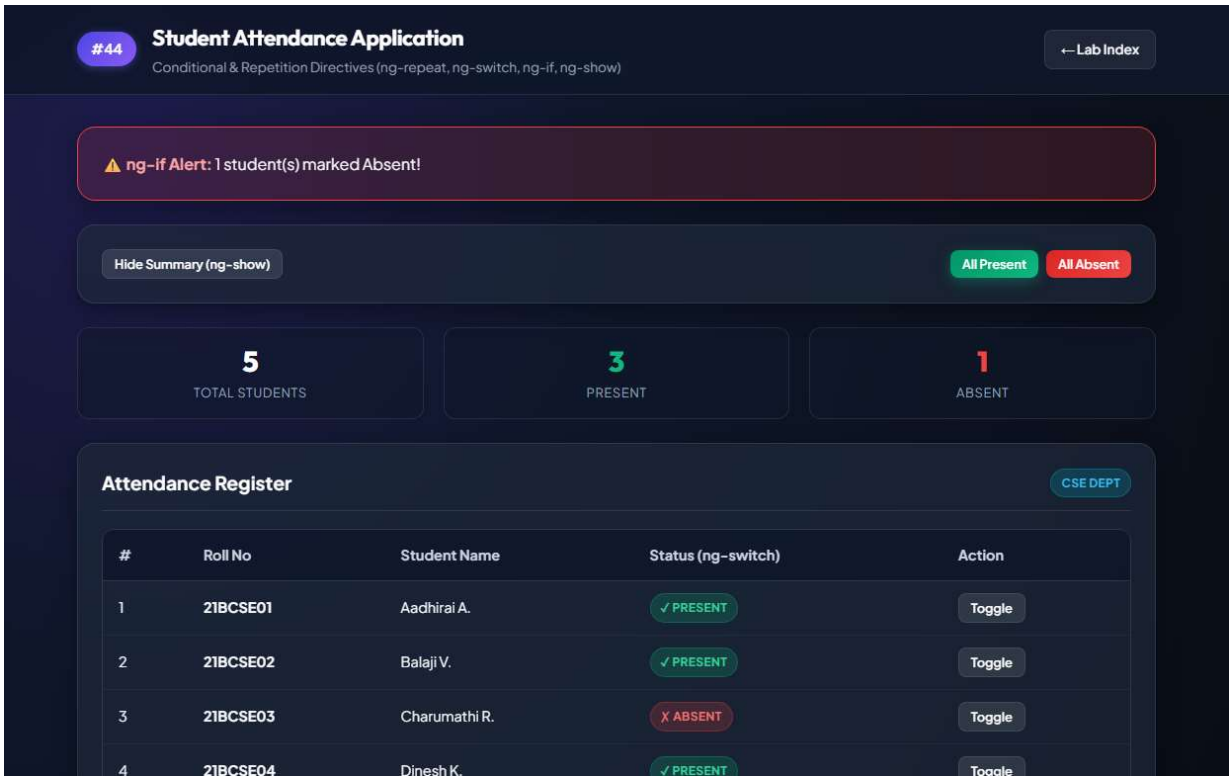
    <!-- Table with ng-repeat & ng-switch -->
    <div class="card">
      <div class="card-title"><span>Attendance Register</span><span class="badge badge-info">CSE Dept</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead><tr><th>#</th><th>Roll No</th><th>Student Name</th><th>Status (ng-
switch)</th><th>Action</th></tr></thead>
          <tbody>
            <tr ng-repeat="s in students" ng-class="{'row-even': $even, 'row-odd': $odd}">
              <td>{{ $index + 1 }}</td>
              <td><strong>{{ s.rollNo }}</strong></td>
              <td>{{ s.name }}</td>
              <td>
                <div ng-switch="s.status">
                  <span ng-switch-when="Present" class="badge badge-success">✓ Present</span>
                  <span ng-switch-when="Absent" class="badge badge-danger">✗ Absent</span>
                  <span ng-switch-when="On Duty" class="badge badge-info">★ On Duty</span>
                </div>
              </td>
              <td><button class="btn btn-secondary btn-sm" ng-click="cycle(s)">Toggle</button></td>
            </tr>
          </tbody>
        </table>
      </div>
    </div>
```

```
  </main>

  <script>
    angular.module('attendanceApp', [])
      .controller('AttendanceCtrl', function($scope) {
        $scope.showSummary = true;
        var states = ['Present', 'Absent', 'On Duty'];
        $scope.students = [
          { rollNo: '21BCSE01', name: 'Aadhirai A.', status: 'Present' },
          { rollNo: '21BCSE02', name: 'Balaji V.', status: 'Present' },
          { rollNo: '21BCSE03', name: 'Charumathi R.', status: 'Absent' },
          { rollNo: '21BCSE04', name: 'Dinesh K.', status: 'Present' },
          { rollNo: '21BCSE05', name: 'Elango S.', status: 'On Duty' }
        ];

        $scope.cycle = function(s) { s.status = states[(states.indexOf(s.status) + 1) % states.length]; };
        $scope.setAll = function(val) { $scope.students.forEach(function(s) { s.status = val; }); };
        $scope.count = function(val) { return $scope.students.filter(function(s) { return s.status === val; }).length;
};
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:45 DATE: 21/9/26	Product Billing Application

AIM:

To write a program Product Billing Application.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="billingApp">
<head>
  <meta charset="UTF-8">
  <title>Program 45 - Product Billing</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="BillingCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#45</span>
        <div class="header-title">
          <h1>Product Billing Application</h1>
          <p>Real-Time Two-Way Data Binding (ng-model)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <!-- Add Item Form -->
    <div class="card mb-4">
      <div class="card-title"><span>Add New Product</span><span class="badge badge-info">Two-Way Bound</span></div>
      <form ng-submit="addItem()" style="display: flex; gap: 12px; flex-wrap: wrap;">
        <input type="text" class="form-control" ng-model="newItem.name" placeholder="Item Name" required style="flex:
2; min-width: 160px;">
        <input type="number" class="form-control" ng-model="newItem.price" placeholder="Price (₹)" min="1" required
style="flex: 1; min-width: 100px;">
        <input type="number" class="form-control" ng-model="newItem.qty" placeholder="Qty" min="1" required
style="flex: 1; min-width: 80px;">
        <button type="submit" class="btn btn-primary">+ Add Item</button>
      </form>
    </div>

    <!-- Items Table -->
    <div class="card mb-4">
      <div class="card-title"><span>Billed Items</span><span class="badge badge-success">{{ items.length }}
Item(s)</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead><tr><th>#</th><th>Item Description</th><th>Price (₹)</th><th>Quantity</th><th>Total
(₹)</th><th>Action</th></tr></thead>
          <tbody>
            <tr ng-repeat="item in items">
              <td>{{ $index + 1 }}</td>
              <td><strong>{{ item.name }}</strong></td>
              <td><input type="number" class="form-control" ng-model="item.price" min="1" style="width: 100px;
padding: 4px 8px;"></td>
              <td><input type="number" class="form-control" ng-model="item.qty" min="1" style="width: 80px; padding:
4px 8px;"></td>
              <td class="text-info font-heading">₹{{ item.price * item.qty }}</td>
              <td><button class="btn btn-danger btn-sm" ng-click="items.splice($index, 1)">×</button></td>
            </tr>
          </tbody>
        </table>
      </div>
    </div>

    <!-- Summary Box -->
    <div class="card">
      <div class="grid-3">
        <div class="stat-box"><div class="stat-val text-info">₹{{ subtotal() }}</div><div class="stat-
label">Subtotal</div></div>
        <div class="stat-box"><div class="stat-val text-warning">₹{{ (subtotal() * 0.05) | number:2 }}</div><div
class="stat-label">GST (5%)</div></div>
        <div class="stat-box"><div class="stat-val text-success">₹{{ (subtotal() * 1.05) | number:2 }}</div><div
class="stat-label">Grand Total</div></div>
      </div>
    </div>
  </main>

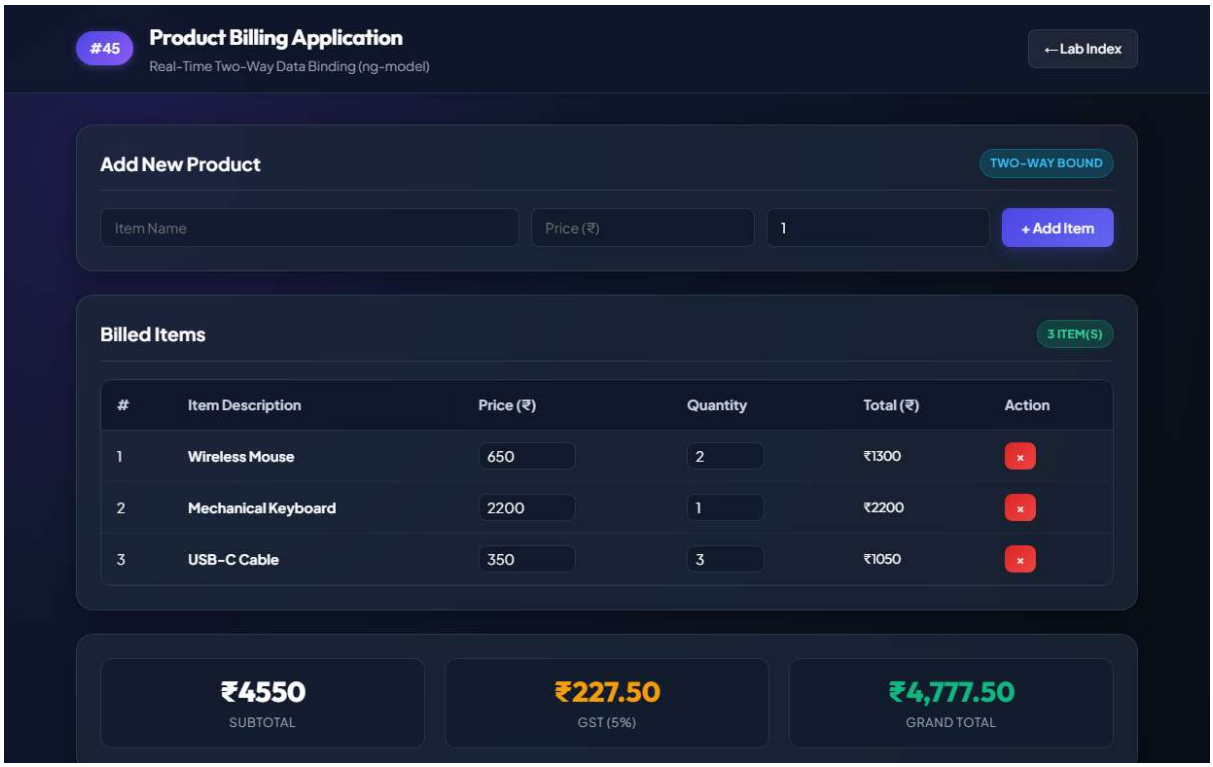
  <script>
```

```
    angular.module('billingApp', [])
      .controller('BillingCtrl', function($scope) {
        $scope.items = [
          { name: 'Wireless Mouse', price: 650, qty: 2 },
          { name: 'Mechanical Keyboard', price: 2200, qty: 1 },
          { name: 'USB-C Cable', price: 350, qty: 3 }
        ];
        $scope.newItem = { name: '', price: '', qty: 1 };

        $scope.addItem = function() {
          $scope.items.push({ name: $scope.newItem.name, price: Number($scope.newItem.price), qty:
Number($scope.newItem.qty) });
          $scope.newItem = { name: '', price: '', qty: 1 };
        };

        $scope.subtotal = function() {
          return $scope.items.reduce(function(sum, item) { return sum + (item.price * item.qty); }, 0);
        };
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:46 DATE: 21/9/26	Library Management Application

AIM:

To write a program Library Management Application.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="libraryApp">
<head>
  <meta charset="UTF-8">
  <title>Program 46 - Library Management</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body>
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#46</span>
        <div class="header-title">
          <h1>Library Management Application</h1>
          <p>Multi-Controller Architecture with Shared Service</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

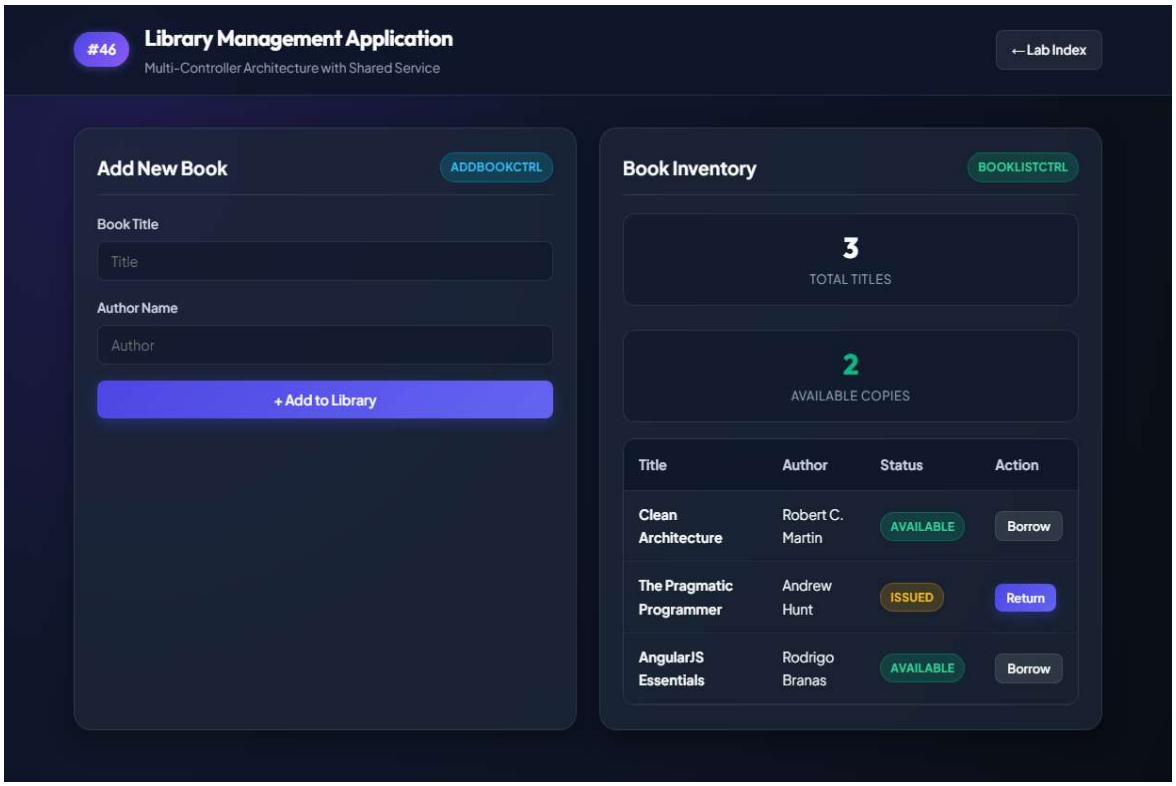
  <main class="container">
    <div class="grid-2">
      <!-- Controller 1: AddBookCtrl -->
      <div class="card" ng-controller="AddBookCtrl">
        <div class="card-title"><span>Add New Book</span><span class="badge badge-info">AddBookCtrl</span></div>
        <form ng-submit="add()">
          <div class="form-group">
            <label class="form-label">Book Title</label>
            <input type="text" class="form-control" ng-model="title" placeholder="Title" required>
          </div>
          <div class="form-group">
            <label class="form-label">Author Name</label>
            <input type="text" class="form-control" ng-model="author" placeholder="Author" required>
          </div>
          <button type="submit" class="btn btn-primary" style="width: 100%;">+ Add to Library</button>
        </form>
      </div>

      <!-- Controller 2: BookListCtrl -->
      <div class="card" ng-controller="BookListCtrl">
        <div class="card-title"><span>Book Inventory</span><span class="badge badge-success">BookListCtrl</span></div>
        <div class="grid-2 mb-3">
          <div class="stat-box"><div class="stat-val text-info">{{ books.length }}</div><div class="stat-label">Total
Titles</div></div>
          <div class="stat-box"><div class="stat-val text-success">{{ getAvailable() }}</div><div class="stat-
label">Available Copies</div></div>
        </div>
        <div class="table-container">
          <table class="custom-table">
            <thead><tr><th>Title</th><th>Author</th><th>Status</th><th>Action</th></tr></thead>
            <tbody>
              <tr ng-repeat="b in books">
                <td><strong>{{ b.title }}</strong></td>
                <td>{{ b.author }}</td>
                <td><span class="badge" ng-class="b.available ? 'badge-success' : 'badge-warning'">{{ b.available ?
'Available' : 'Issued' }}</span></td>
                <td><button class="btn btn-sm" ng-class="b.available ? 'btn-secondary' : 'btn-primary'" ng-
click="b.available = !b.available">{{ b.available ? 'Borrow' : 'Return' }}</button></td>
              </tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>
  </main>

  <script>
    angular.module('libraryApp', [])
      .factory('LibraryService', function() {
        var books = [
          { title: 'Clean Architecture', author: 'Robert C. Martin', available: true },
          { title: 'The Pragmatic Programmer', author: 'Andrew Hunt', available: false },
          { title: 'AngularJS Essentials', author: 'Rodrigo Branas', available: true }
```

```
        ];
        return {
          books: books,
          add: function(title, author) { books.push({ title: title, author: author, available: true }); }
        };
      })
      .controller('AddBookCtrl', function($scope, LibraryService) {
        $scope.add = function() {
          LibraryService.add($scope.title, $scope.author);
          $scope.title = $scope.author = '';
        };
      })
      .controller('BookListCtrl', function($scope, LibraryService) {
        $scope.books = LibraryService.books;
        $scope.getAvailable = function() { return $scope.books.filter(function(b) { return b.available; }).length; };
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:47 DATE: 21/9/26	Student Mark Analysis System

AIM:

To write a program Student Mark Analysis System.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="markAnalysisApp">
<head>
  <meta charset="UTF-8">
  <title>Program 47 - Student Mark Analysis</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="MarkAnalysisCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#47</span>
        <div class="header-title">
          <h1>Student Mark Analysis System</h1>
          <p>Two-Way Binding & Dynamic Styling (ng-style)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <!-- Student Header -->
    <div class="card mb-4">
      <div class="grid-2">
        <div class="form-group" style="margin: 0;">
          <label class="form-label">Student Name</label>
          <input type="text" class="form-control" ng-model="student.name">
        </div>
        <div class="form-group" style="margin: 0;">
          <label class="form-label">Register Number</label>
          <input type="text" class="form-control" ng-model="student.regNo">
        </div>
      </div>
    </div>

    <!-- Marks Table -->
    <div class="card mb-4">
      <div class="card-title"><span>Subject Marks Evaluation</span><span class="badge badge-info">Internal (40) +
External (60)</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead>
            <tr><th>Subject</th><th>Internal (40)</th><th>External (60)</th><th>Total (100)</th><th>Progress (ng-
style)</th><th>Status</th></tr>
          </thead>
          <tbody>
            <tr ng-repeat="s in student.subjects">
              <td><strong>{{ s.name }}</strong></td>
              <td><input type="number" class="form-control" ng-model="s.internal" min="0" max="40" style="width:
80px;"></td>
              <td><input type="number" class="form-control" ng-model="s.external" min="0" max="60" style="width:
80px;"></td>
              <td class="font-heading text-info">{{ s.internal + s.external }}</td>
              <td style="width: 160px;">
                <div style="background: rgba(255,255,255,0.1); border-radius: 4px; height: 16px; overflow: hidden;">
                  <div ng-style="{'width': (s.internal + s.external) + '%', 'background': (s.internal + s.external) >=
50 ? '#10b981' : '#ef4444'}" style="height: 100%;"></div>
                </div>
              </td>
              <td>
                <span class="badge" ng-class="(s.internal + s.external) >= 50 ? 'badge-success' : 'badge-danger'">
                  {{ (s.internal + s.external) >= 50 ? 'PASS' : 'FAIL' }}
                </span>
              </td>
            </tr>
          </tbody>
        </table>
      </div>
    </div>

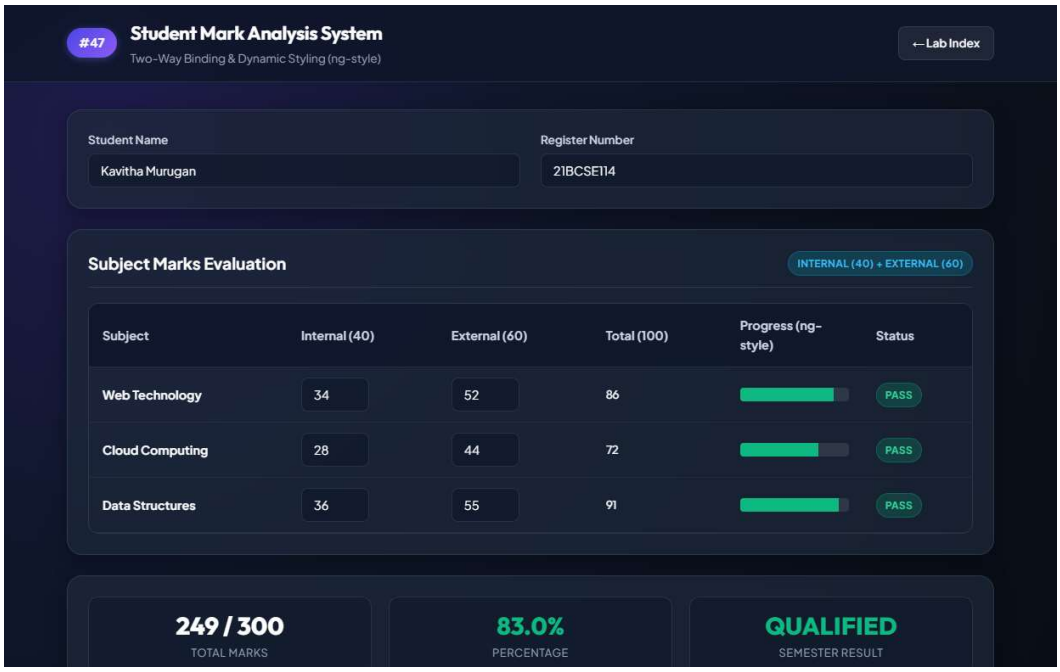
    <!-- Summary -->
    <div class="card">
      <div class="grid-3">
```

```
        <div class="stat-box"><div class="stat-val text-info">{{ getTotal() }} / 300</div><div class="stat-
label">Total Marks</div></div>
        <div class="stat-box"><div class="stat-val text-success">{{ (getTotal() / 3) | number:1 }}%</div><div
class="stat-label">Percentage</div></div>
        <div class="stat-box">
          <div class="stat-val" ng-class="isPassed() ? 'text-success' : 'text-danger'">{{ isPassed() ? 'QUALIFIED' :
'ARREAR' }}</div>
          <div class="stat-label">Semester Result</div>
        </div>
      </div>
    </div>
  </main>

  <script>
    angular.module('markAnalysisApp', [])
      .controller('MarkAnalysisCtrl', function($scope) {
        $scope.student = {
          name: 'Kavitha Murugan', regNo: '21BCSE114',
          subjects: [
            { name: 'Web Technology', internal: 34, external: 52 },
            { name: 'Cloud Computing', internal: 28, external: 44 },
            { name: 'Data Structures', internal: 36, external: 55 }
          ]
        };

        $scope.getTotal = function() {
          return $scope.student.subjects.reduce(function(sum, s) { return sum + (Number(s.internal) || 0) +
(Number(s.external) || 0); }, 0);
        };
        $scope.isPassed = function() {
          return $scope.student.subjects.every(function(s) { return (s.internal + s.external) >= 50; });
        };
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:48 DATE: 21/9/26	Product Catalog

AIM:

To write a program Product Catalog.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="catalogApp">
<head>
  <meta charset="UTF-8">
  <title>Program 48 - Product Catalog</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="CatalogCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#48</span>
        <div class="header-title">
          <h1>Dynamic Product Catalog</h1>
          <p>Built-in & Custom Directives (&lt;stock-badge&gt;)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

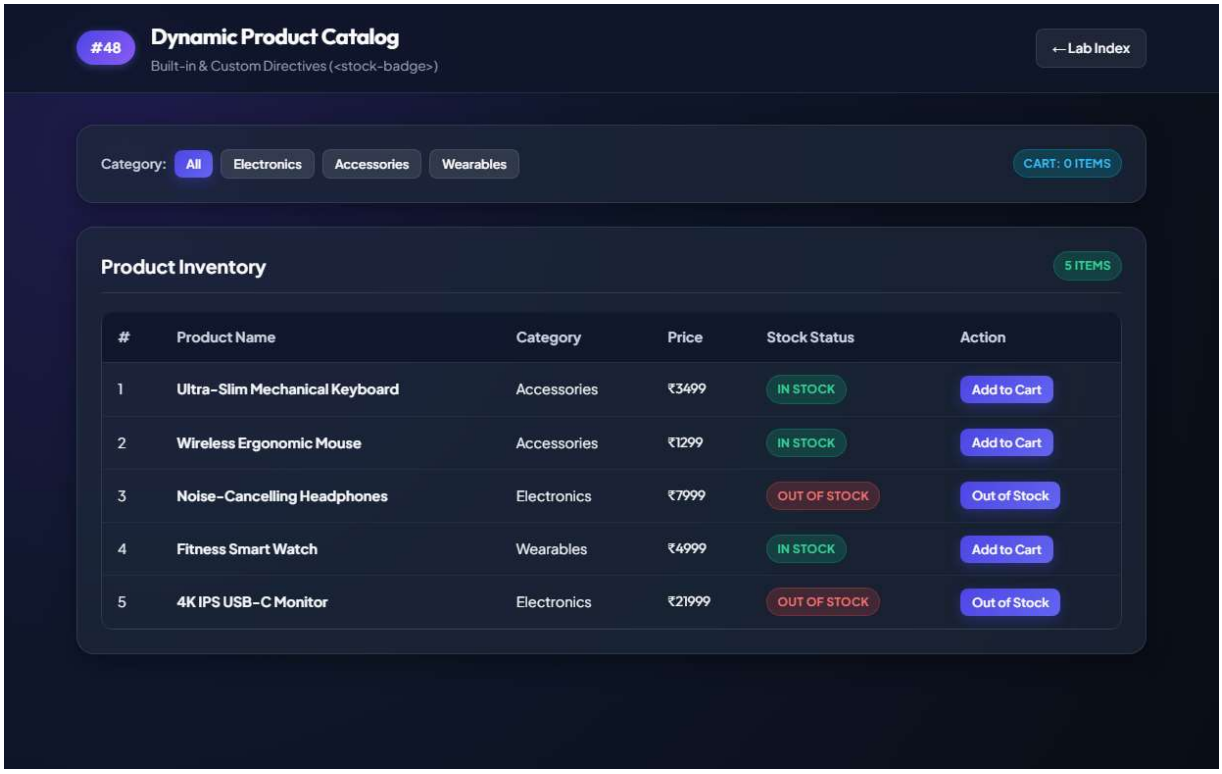
  <main class="container">
    <!-- Filter Controls -->
    <div class="card mb-4 flex-between">
      <div style="display: flex; gap: 8px; align-items: center; flex-wrap: wrap;">
        <label class="form-label" style="margin: 0;">Category:</label>
        <button class="btn btn-sm" ng-class="activeCat === '' ? 'btn-primary' : 'btn-secondary'" ng-click="activeCat =
''">All</button>
        <button class="btn btn-sm" ng-repeat="c in categories" ng-class="activeCat === c ? 'btn-primary' : 'btn-
secondary'" ng-click="$parent.activeCat = c">{{ c }}</button>
      </div>
      <span class="badge badge-info">Cart: {{ cartCount }} items</span>
    </div>

    <!-- Table -->
    <div class="card">
      <div class="card-title"><span>Product Inventory</span><span class="badge badge-success">{{ (products |
filter:(activeCat ? {category: activeCat} : '')).length }} Items</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead><tr><th>#</th><th>Product Name</th><th>Category</th><th>Price</th><th>Stock
Status</th><th>Action</th></tr></thead>
          <tbody>
            <tr ng-repeat="p in products | filter:(activeCat ? {category: activeCat} : '')">
              <td>{{ $index + 1 }}</td>
              <td><strong>{{ p.name }}</strong></td>
              <td>{{ p.category }}</td>
              <td class="text-info font-heading">₹{{ p.price }}</td>
              <td><stock-badge in-stock="p.available"></stock-badge></td>
              <td>
                <button class="btn btn-sm btn-primary" ng-disabled="!p.available" ng-click="cartCount = cartCount +
1">
                  {{ p.available ? 'Add to Cart' : 'Out of Stock' }}
                </button>
              </td>
            </tr>
          </tbody>
        </table>
      </div>
    </div>
  </main>

  <script>
    angular.module('catalogApp', [])
      .directive('stockBadge', function() {
        return {
          restrict: 'E',
          scope: { inStock: '=' },
          template: '<span class="badge" ng-class="inStock ? \'badge-success\' : \'badge-danger\'">{{ inStock ? "In
Stock" : "Out of Stock" }}</span>'
        };
      })
      .controller('CatalogCtrl', function($scope) {
        $scope.cartCount = 0;
```

```
        $scope.activeCat = '';
        $scope.categories = ['Electronics', 'Accessories', 'Wearables'];
        $scope.products = [
          { name: 'Ultra-Slim Mechanical Keyboard', category: 'Accessories', price: 3499, available: true },
          { name: 'Wireless Ergonomic Mouse', category: 'Accessories', price: 1299, available: true },
          { name: 'Noise-Cancelling Headphones', category: 'Electronics', price: 7999, available: false },
          { name: 'Fitness Smart Watch', category: 'Wearables', price: 4999, available: true },
          { name: '4K IPS USB-C Monitor', category: 'Electronics', price: 21999, available: false }
        ];
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:49 DATE: 21/9/26	Student Performance System

AIM:

To write a program Student Performance System.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="performanceApp">
<head>
  <meta charset="UTF-8">
  <title>Program 49 - Student Performance</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="PerformanceCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#49</span>
        <div class="header-title">
          <h1>Student Performance System</h1>
          <p>AngularJS Scope Watchers ($scope.$watch)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <div class="grid-2">
      <!-- Input Card -->
      <div class="card">
        <div class="card-title"><span>Performance Monitor</span><span class="badge badge-info">$watch
Active</span></div>
        <div class="form-group">
          <label class="form-label">Student Name</label>
          <input type="text" class="form-control" ng-model="student.name">
        </div>
        <div class="form-group">
          <label class="form-label">Internal Marks (/100)</label>
          <input type="number" class="form-control" ng-model="student.marks" min="0" max="100">
        </div>
        <div class="form-group">
          <label class="form-label">Attendance (%)</label>
          <input type="number" class="form-control" ng-model="student.attendance" min="0" max="100">
        </div>
        <div ng-if="warning" class="card" style="background: rgba(239,68,68,0.2); border-color: #ef4444; padding:
10px; margin-top: 10px;">
          <strong style="color: #fca5a5;">⚠ Warning:</strong>
          <span style="color: #fecaca; font-size: 0.85rem;">{{ warning }}</span>
        </div>
      </div>

      <!-- Scope Evaluation Card -->
      <div class="card">
        <div class="card-title"><span>Real-Time Evaluation</span><span class="badge badge-success">Watch Count: {{
watchCount }}</span></div>
        <div class="grid-2 mb-3">
          <div class="stat-box"><div class="stat-val text-info">{{ student.marks }}</div><div class="stat-
label">Marks</div></div>
          <div class="stat-box"><div class="stat-val" ng-class="student.attendance >= 75 ? 'text-success' : 'text-
danger'">{{ student.attendance }}%</div><div class="stat-label">Attendance</div></div>
        </div>

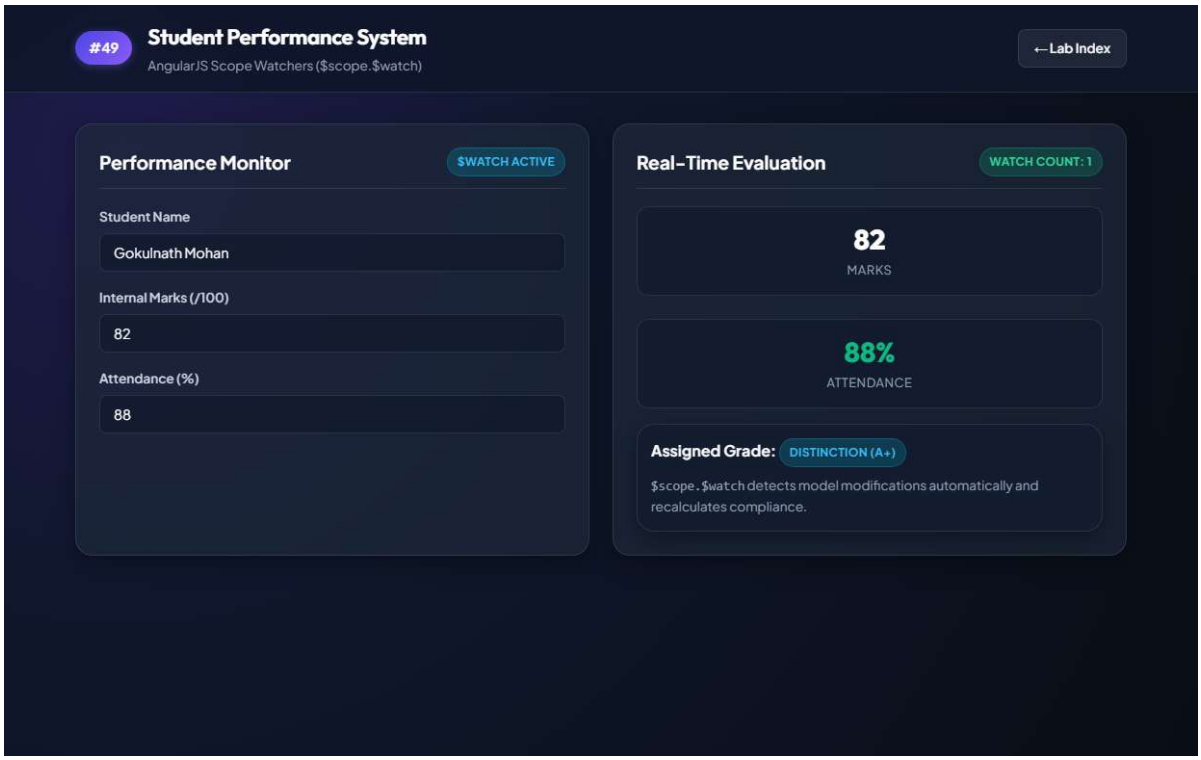
        <div class="card" style="background: rgba(15,23,42,0.6); padding: 14px; margin: 0;">
          <p><strong>Assigned Grade:</strong> <span class="badge badge-info">{{ grade }}</span></p>
          <p class="text-muted" style="font-size: 0.85rem; margin-top: 8px;">
            <code>$scope.$watch</code> detects model modifications automatically and recalculates compliance.
          </p>
        </div>
      </div>
    </div>
  </main>

  <script>
    angular.module('performanceApp', [])
      .controller('PerformanceCtrl', function($scope) {
        $scope.watchCount = 0;
        $scope.student = { name: 'Gokulnath Mohan', marks: 82, attendance: 88 };

        $scope.$watch('student', function(val) {
```

```
            $scope.watchCount++;
            $scope.grade = val.marks >= 80 ? 'Distinction (A+)' : val.marks >= 60 ? 'First Class (A)' : val.marks >= 50
? 'Second Class (B)' : 'Fail (F)';
            $scope.warning = (val.attendance < 75 && val.marks < 50) ? 'Low attendance (<75%) AND low score (<50%)!'
                           : val.attendance < 75 ? 'Attendance below mandatory 75% threshold.'
                           : val.marks < 50 ? 'Marks are below passing mark (50).' : '';
          }, true);
        });
    </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:50 DATE: 21/9/26	Department Nested Scopes

AIM:

To write a program Department Nested Scopes.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="nestedScopeApp">
<head>
  <meta charset="UTF-8">
  <title>Program 50 - Department Scopes</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body>
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#50</span>
        <div class="header-title">
          <h1>Department Nested Scopes</h1>
          <p>Scope Hierarchy, Prototypal Inheritance & $parent Access</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <!-- Parent Scope (DeanCtrl) -->
    <div class="card mb-4" ng-controller="DeanCtrl" style="border: 2px solid var(--primary);">
      <div class="card-title"><span>Level 1: Parent Scope (DeanCtrl)</span><span class="badge badge-
info">Parent</span></div>
      <div class="grid-3 mb-3">
        <div class="form-group">
          <label class="form-label">College Name (Parent)</label>
          <input type="text" class="form-control" ng-model="collegeName">
        </div>
        <div class="form-group">
          <label class="form-label">Dean Name</label>
          <input type="text" class="form-control" ng-model="deanName">
        </div>
        <div class="form-group">
          <label class="form-label">Institutional Budget (₹)</label>
          <input type="number" class="form-control" ng-model="budget">
        </div>
      </div>

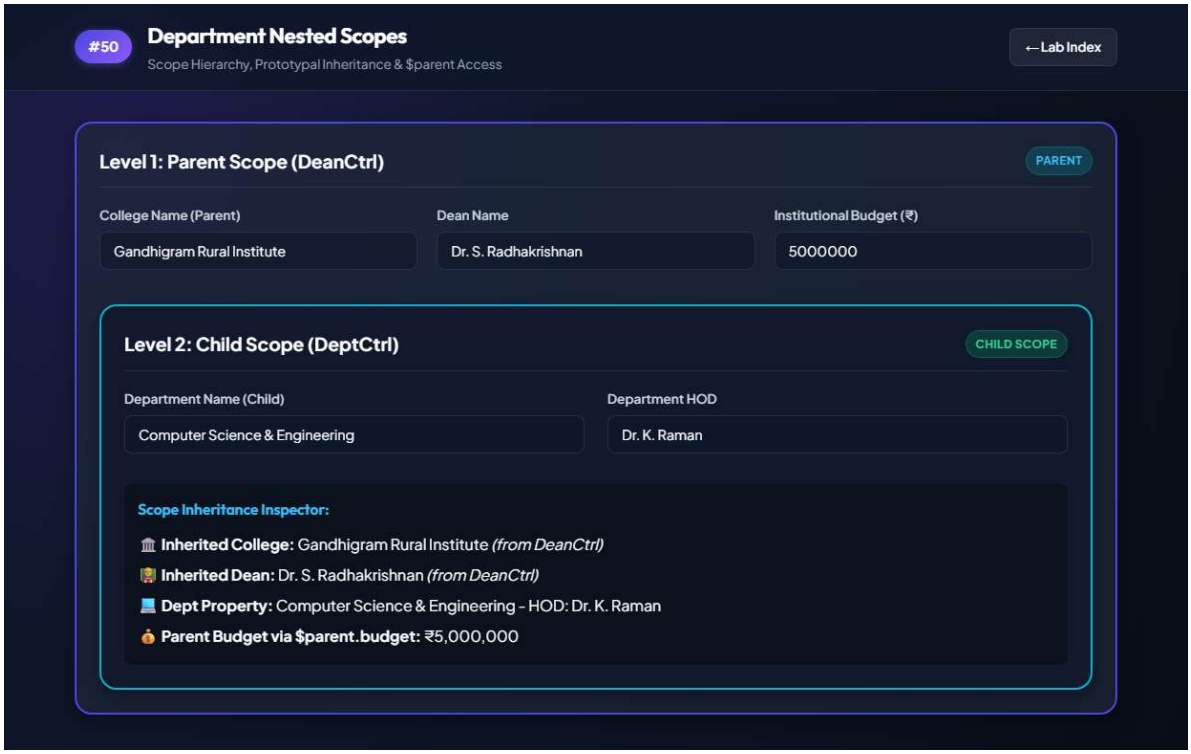
      <!-- Child Scope (DeptCtrl nested inside DeanCtrl) -->
      <div class="card" ng-controller="DeptCtrl" style="background: rgba(15,23,42,0.85); border: 2px solid var(--
secondary); margin-top: 20px;">
        <div class="card-title"><span>Level 2: Child Scope (DeptCtrl)</span><span class="badge badge-success">Child
Scope</span></div>
        <div class="grid-2 mb-3">
          <div class="form-group">
            <label class="form-label">Department Name (Child)</label>
            <input type="text" class="form-control" ng-model="deptName">
          </div>
          <div class="form-group">
            <label class="form-label">Department HOD</label>
            <input type="text" class="form-control" ng-model="hodName">
          </div>
        </div>

        <div style="background: rgba(0,0,0,0.3); padding: 14px; border-radius: var(--radius-sm);">
          <h4 style="color: #38bdf8; margin-bottom: 8px;">Scope Inheritance Inspector:</h4>
          <ul style="list-style: none; line-height: 2;">
            <li>🏛 <strong>Inherited College:</strong> {{ collegeName }} <em>(from DeanCtrl)</em></li>
            <li>👨‍🎓 <strong>Inherited Dean:</strong> {{ deanName }} <em>(from DeanCtrl)</em></li>
            <li>💻 <strong>Dept Property:</strong> {{ deptName }} - HOD: {{ hodName }}</li>
            <li>💰 <strong>Parent Budget via $parent.budget:</strong> ₹{{ $parent.budget | number }}</li>
          </ul>
        </div>
      </div>
    </div>
  </main>

  <script>
    angular.module('nestedScopeApp', [])
      .controller('DeanCtrl', function($scope) {
        $scope.collegeName = 'Gandhigram Rural Institute';
```

```
        $scope.deanName = 'Dr. S. Radhakrishnan';
        $scope.budget = 5000000;
      })
      .controller('DeptCtrl', function($scope) {
        $scope.deptName = 'Computer Science & Engineering';
        $scope.hodName = 'Dr. K. Raman';
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:51 DATE: 21/9/26	Employee Directory

AIM:

To write a program Employee Directory.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="employeeApp">
<head>
  <meta charset="UTF-8">
  <title>Program 51 - Employee Directory</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="EmployeeCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#51</span>
        <div class="header-title">
          <h1>Employee Directory</h1>
          <p>Built-in & Custom Filters (filter, orderBy, currency, date, uppercase)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

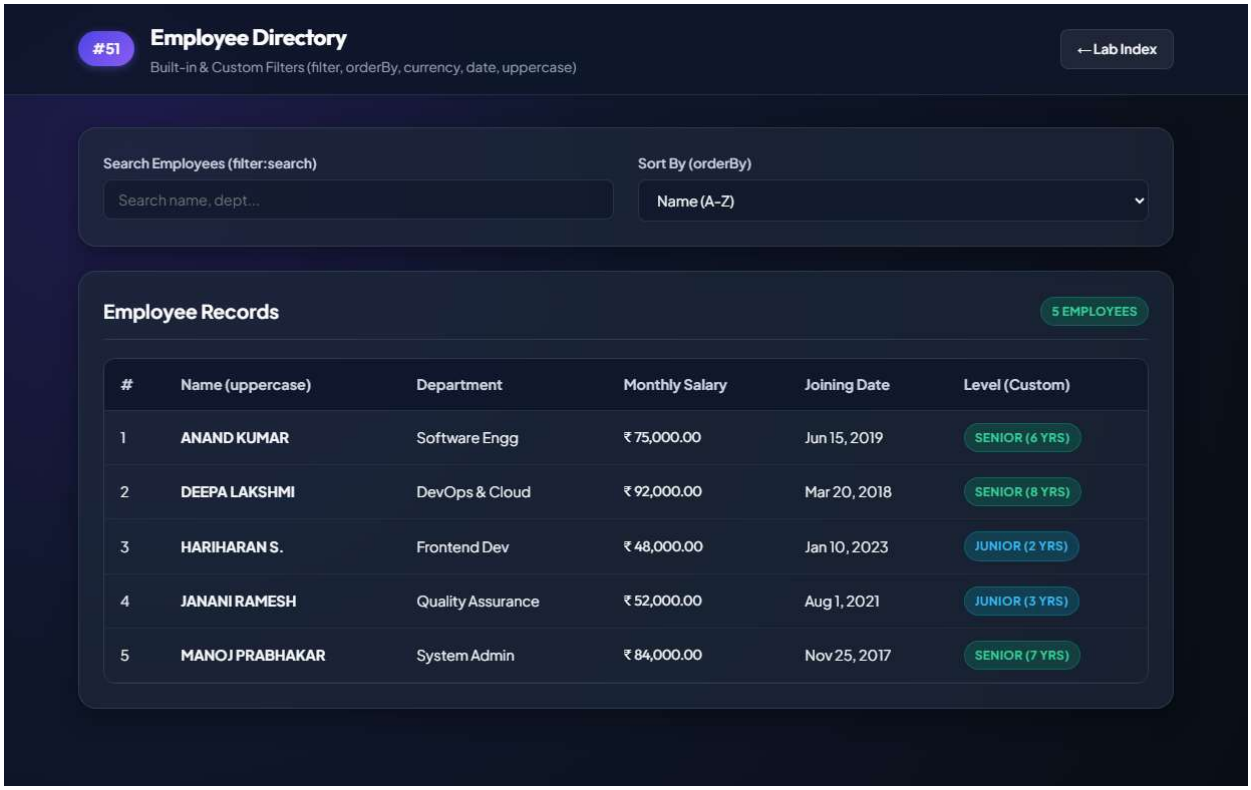
  <main class="container">
    <!-- Controls -->
    <div class="card mb-4">
      <div class="grid-2">
        <div class="form-group" style="margin: 0;">
          <label class="form-label">Search Employees (filter:search)</label>
          <input type="text" class="form-control" ng-model="searchText" placeholder="Search name, dept...">
        </div>
        <div class="form-group" style="margin: 0;">
          <label class="form-label">Sort By (orderBy)</label>
          <select class="form-control" ng-model="sortCol">
            <option value="name">Name (A-Z)</option>
            <option value="-salary">Salary (High to Low)</option>
            <option value="salary">Salary (Low to High)</option>
            <option value="-exp">Experience (Senior first)</option>
          </select>
        </div>
      </div>
    </div>

    <!-- Table -->
    <div class="card">
      <div class="card-title"><span>Employee Records</span><span class="badge badge-success">{{ (employees |
filter:searchText).length }} Employees</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead><tr><th>#</th><th>Name (uppercase)</th><th>Department</th><th>Monthly Salary</th><th>Joining
Date</th><th>Level (Custom)</th></tr></thead>
          <tbody>
            <tr ng-repeat="e in employees | filter:searchText | orderBy:sortCol">
              <td>{{ $index + 1 }}</td>
              <td><strong>{{ e.name | uppercase }}</strong></td>
              <td>{{ e.dept }}</td>
              <td class="text-info font-heading">{{ e.salary | currency:'₹ ' }}</td>
              <td>{{ e.joinDate | date:'mediumDate' }}</td>
              <td>
                <span class="badge" ng-class="e.exp >= 5 ? 'badge-success' : 'badge-info'">{{ e.exp | experienceLevel
}}</span>
              </td>
            </tr>
          </tbody>
        </table>
      </div>
    </div>
  </main>

  <script>
    angular.module('employeeApp', [])
      .filter('experienceLevel', function() {
        return function(exp) { return (exp >= 5 ? 'Senior (' : 'Junior (') + exp + ' yrs)'; };
      })
      .controller('EmployeeCtrl', function($scope) {
        $scope.sortCol = 'name';
```

```
        $scope.employees = [
          { name: 'Anand Kumar', dept: 'Software Engg', salary: 75000, joinDate: new Date('2019-06-15'), exp: 6 },
          { name: 'Deepa Lakshmi', dept: 'DevOps & Cloud', salary: 92000, joinDate: new Date('2018-03-20'), exp: 8 },
          { name: 'Hariharan S.', dept: 'Frontend Dev', salary: 48000, joinDate: new Date('2023-01-10'), exp: 2 },
          { name: 'Janani Ramesh', dept: 'Quality Assurance', salary: 52000, joinDate: new Date('2021-08-01'), exp: 3
},
          { name: 'Manoj Prabhakar', dept: 'System Admin', salary: 84000, joinDate: new Date('2017-11-25'), exp: 7 }
        ];
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:52 DATE: 21/9/26	Selectors Application

AIM:

To write a program Selectors Application.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="cascadingApp">
<head>
  <meta charset="UTF-8">
  <title>Program 53 - Cascading Selection</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="CascadingCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#53</span>
        <div class="header-title">
          <h1>Cascading Selection Application</h1>
          <p>Dependent Dropdowns with ng-options & ng-change</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <div class="card mb-4">
      <div class="card-title"><span>Academic Program Hierarchy</span><span class="badge badge-info">3-Tier Dependent
Selection</span></div>
      <div class="grid-3">
        <!-- Tier 1: Department -->
        <div class="form-group">
          <label class="form-label">1. Department:</label>
          <select class="form-control" ng-model="selDept" ng-options="d.name for d in departments" ng-change="selProg
= selCourse = null">
            <option value="">-- Choose Department --</option>
          </select>
        </div>

        <!-- Tier 2: Program -->
        <div class="form-group">
          <label class="form-label">2. Program / Degree:</label>
          <select class="form-control" ng-model="selProg" ng-disabled="!selDept" ng-options="p.title for p in
selDept.programs" ng-change="selCourse = null">
            <option value="">-- Choose Program --</option>
          </select>
        </div>

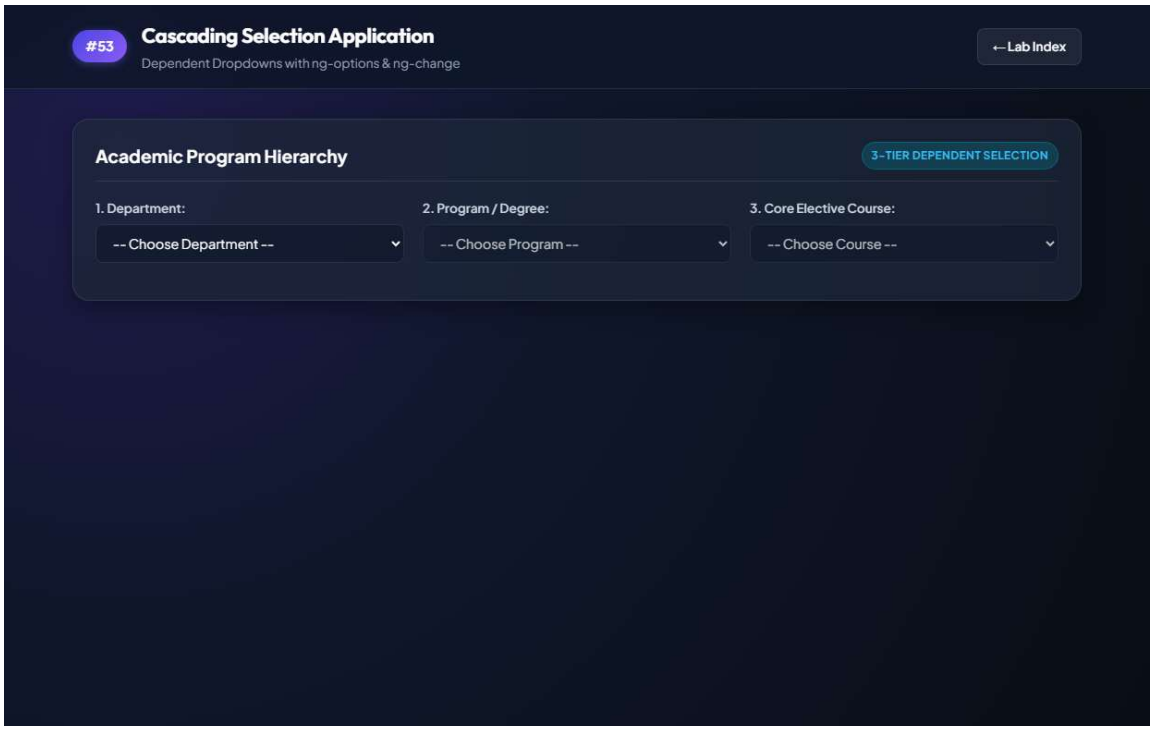
        <!-- Tier 3: Course -->
        <div class="form-group">
          <label class="form-label">3. Core Elective Course:</label>
          <select class="form-control" ng-model="selCourse" ng-disabled="!selProg" ng-options="c.code + ' - ' + c.name
for c in selProg.courses">
            <option value="">-- Choose Course --</option>
          </select>
        </div>
      </div>
    </div>

    <!-- Summary -->
    <div class="card" ng-if="selCourse">
      <div class="card-title"><span>Enrollment Summary Sheet</span><span class="badge badge-
success">Selected</span></div>
      <div style="background: rgba(15,23,42,0.6); padding: 16px; border-radius: var(--radius-sm);">
        <h3 style="color: #fff; margin-bottom: 6px;">{{ selCourse.name }}</h3>
        <p class="text-muted">Code: <strong style="color: #38bdf8;">{{ selCourse.code }}</strong> | Credits: {{
selCourse.credits }}</p>
        <p class="text-muted">Dept: <strong>{{ selDept.name }}</strong> | Degree: <strong>{{ selProg.title
}}</strong></p>
      </div>
    </div>
  </main>

  <script>
    angular.module('cascadingApp', [])
      .controller('CascadingCtrl', function($scope) {
        $scope.departments = [
          {
```

```
            name: 'Computer Science',
            programs: [
              { title: 'B.Sc Computer Science', courses: [{ code: 'CS101', name: 'Web Programming & AngularJS',
credits: 4 }, { code: 'CS102', name: 'Database Management Systems', credits: 4 }] },
              { title: 'M.Sc Information Tech', courses: [{ code: 'IT201', name: 'Cloud Computing & Microservices',
credits: 4 }, { code: 'IT202', name: 'Machine Learning', credits: 4 }] }
            ]
          },
          {
            name: 'Electronics & Comm',
            programs: [
              { title: 'B.Tech ECE', courses: [{ code: 'EC101', name: 'Digital Signal Processing', credits: 4 }, {
code: 'EC102', name: 'VLSI Circuit Design', credits: 4 }] }
            ]
          }
        ];
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:53 DATE: 21/9/26	Student Result Analysis

AIM:

To write a program Student Result Analysis.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="resultApp">
<head>
  <meta charset="UTF-8">
  <title>Program 52 - Student Result Analysis</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="ResultCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#52</span>
        <div class="header-title">
          <h1>Student Result Analysis</h1>
          <p>Multi-Criteria Filtering & Performance Analytics</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container">
    <!-- Controls -->
    <div class="card mb-4">
      <div class="grid-2">
        <div class="form-group" style="margin: 0;">
          <label class="form-label">Department:</label>
          <select class="form-control" ng-model="selDept">
            <option value="">All Departments</option>
            <option value="CSE">Computer Science (CSE)</option>
            <option value="IT">Information Technology (IT)</option>
            <option value="ECE">Electronics (ECE)</option>
          </select>
        </div>
        <div class="form-group" style="margin: 0;">
          <label class="form-label">Result Status:</label>
          <select class="form-control" ng-model="selStatus">
            <option value="">All Statuses</option>
            <option value="Pass">Pass Only</option>
            <option value="Fail">Fail Only</option>
          </select>
        </div>
      </div>
    </div>

    <!-- Analytics Cards -->
    <div class="grid-4 mb-4">
      <div class="stat-box"><div class="stat-val text-info">{{ filtered.length }}</div><div class="stat-
label">Displayed</div></div>
      <div class="stat-box"><div class="stat-val text-success">{{ getCount('Pass') }}</div><div class="stat-
label">Passed</div></div>
      <div class="stat-box"><div class="stat-val text-danger">{{ getCount('Fail') }}</div><div class="stat-
label">Failed</div></div>
      <div class="stat-box"><div class="stat-val" style="color: #a855f7;">{{ getAvg() | number:1 }}%</div><div
class="stat-label">Class Avg</div></div>
    </div>

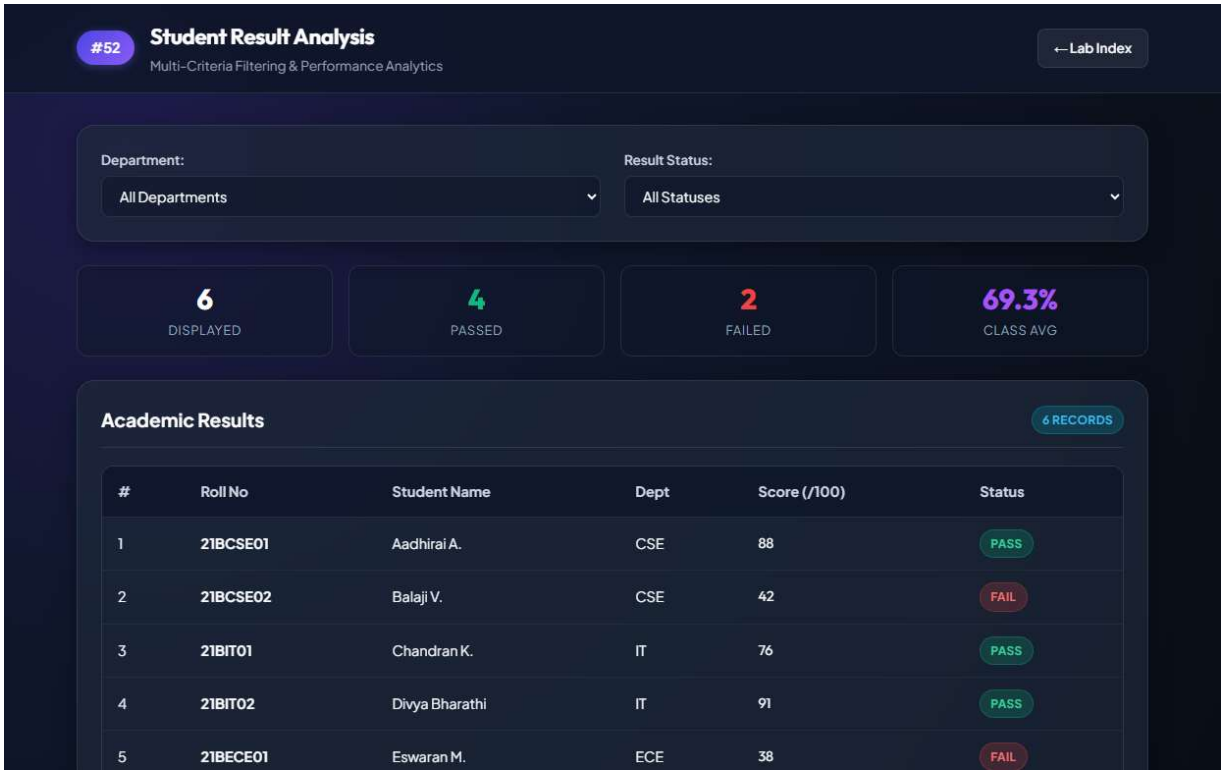
    <!-- Table -->
    <div class="card">
      <div class="card-title"><span>Academic Results</span><span class="badge badge-info">{{ filtered.length }}
Records</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead><tr><th>#</th><th>Roll No</th><th>Student Name</th><th>Dept</th><th>Score
(/100)</th><th>Status</th></tr></thead>
          <tbody>
            <tr ng-repeat="s in (filtered = (students | filter:(selDept ? {dept: selDept} : '') | filter:(selStatus ?
{status: selStatus} : '')))">
              <td>{{ $index + 1 }}</td>
              <td><strong>{{ s.rollNo }}</strong></td>
              <td>{{ s.name }}</td>
              <td>{{ s.dept }}</td>
              <td class="font-heading text-info">{{ s.score }}</td>
```

```
            <td><span class="badge" ng-class="s.status === 'Pass' ? 'badge-success' : 'badge-danger'">{{ s.status
}}</span></td>
          </tr>
        </tbody>
      </table>
    </div>
  </div>
</main>

<script>
  angular.module('resultApp', [])
    .controller('ResultCtrl', function($scope) {
      $scope.students = [
        { rollNo: '21BCSE01', name: 'Aadhirai A.', dept: 'CSE', score: 88, status: 'Pass' },
        { rollNo: '21BCSE02', name: 'Balaji V.', dept: 'CSE', score: 42, status: 'Fail' },
        { rollNo: '21BIT01', name: 'Chandran K.', dept: 'IT', score: 76, status: 'Pass' },
        { rollNo: '21BIT02', name: 'Divya Bharathi', dept: 'IT', score: 91, status: 'Pass' },
        { rollNo: '21BECE01', name: 'Eswaran M.', dept: 'ECE', score: 38, status: 'Fail' },
        { rollNo: '21BECE02', name: 'Farhan Ali', dept: 'ECE', score: 81, status: 'Pass' }
      ];

      $scope.getCount = function(st) { return ($scope.filtered || []).filter(function(s) { return s.status === st;
}).length; };
      $scope.getAvg = function() {
        var list = $scope.filtered || [];
        return list.length ? (list.reduce(function(sum, s) { return sum + s.score; }, 0) / list.length) : 0;
      };
    });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:54 DATE: 24/9/26	To-do List

AIM:

To write a program To-do List.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="todoApp">
<head>
  <meta charset="UTF-8">
  <title>Program 54 - To-Do Management</title>
  <link rel="stylesheet" href="css/style.css">
  <style>.task-done { text-decoration: line-through; color: var(--text-muted); }</style>
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="TodoCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#54</span>
        <div class="header-title">
          <h1>Interactive To-Do Application</h1>
          <p>AngularJS DOM Event Directives (ng-click, ng-keydown)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container" style="max-width: 750px;">
    <!-- Add Task -->
    <div class="card mb-4">
      <div class="card-title"><span>Add Task</span><span class="badge badge-info">Press Enter or Add</span></div>
      <div style="display: flex; gap: 10px;">
        <input type="text" class="form-control" ng-model="newText" ng-keydown="$event.keyCode === 13 && add()"
placeholder="What needs to be done?">
        <button class="btn btn-primary" ng-click="add()">Add Task</button>
      </div>
    </div>

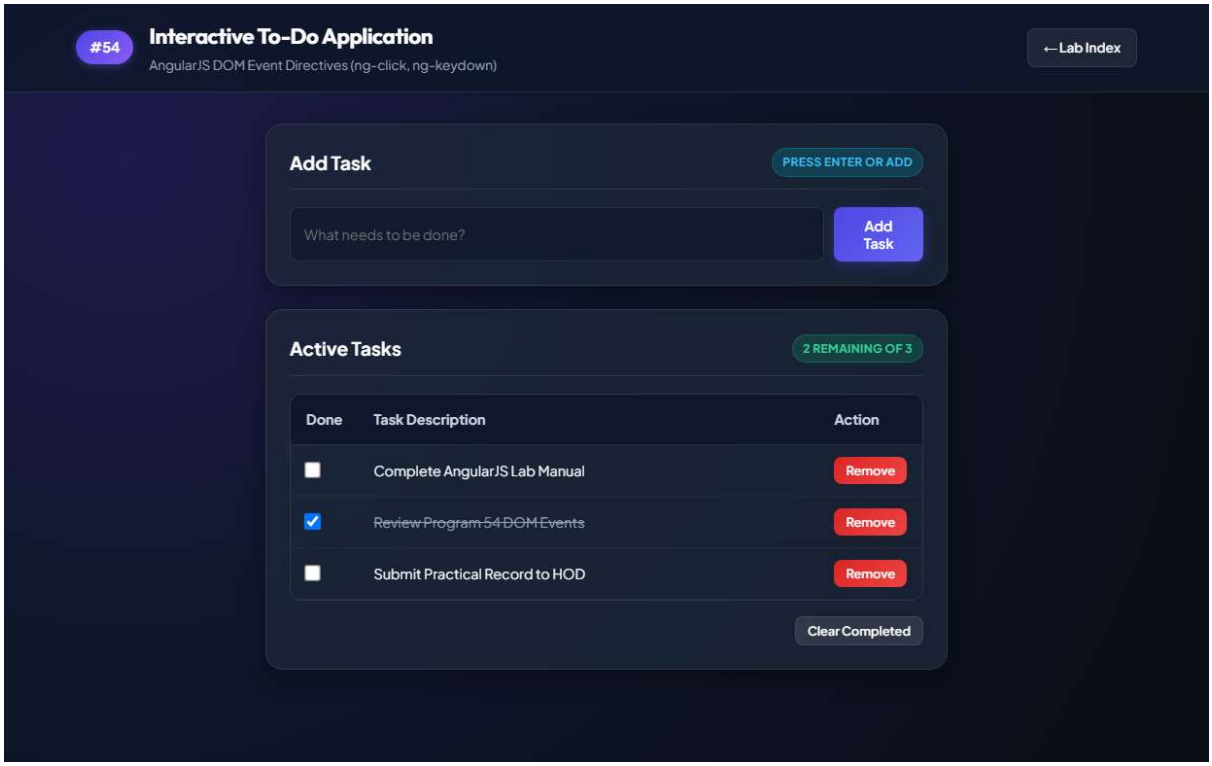
    <!-- Task List -->
    <div class="card">
      <div class="card-title"><span>Active Tasks</span><span class="badge badge-success">{{ remaining() }} Remaining
of {{ tasks.length }}</span></div>
      <div class="table-container">
        <table class="custom-table">
          <thead><tr><th style="width: 50px;">Done</th><th>Task Description</th><th style="width:
90px;">Action</th></tr></thead>
          <tbody>
            <tr ng-repeat="t in tasks">
              <td><input type="checkbox" ng-model="t.done" style="transform: scale(1.3); cursor: pointer;"></td>
              <td><span ng-class="{'task-done': t.done}">{{ t.text }}</span></td>
              <td><button class="btn btn-danger btn-sm" ng-click="tasks.splice($index, 1)">Remove</button></td>
            </tr>
          </tbody>
        </table>
      </div>
      <div style="margin-top: 16px; text-align: right;">
        <button class="btn btn-secondary btn-sm" ng-click="clear()">Clear Completed</button>
      </div>
    </div>
  </main>

  <script>
    angular.module('todoApp', [])
      .controller('TodoCtrl', function($scope) {
        $scope.tasks = [
          { text: 'Complete AngularJS Lab Manual', done: false },
          { text: 'Review Program 54 DOM Events', done: true },
          { text: 'Submit Practical Record to HOD', done: false }
        ];

        $scope.add = function() {
          if ($scope.newText && $scope.newText.trim()) {
            $scope.tasks.push({ text: $scope.newText.trim(), done: false });
            $scope.newText = '';
          }
        };
        $scope.clear = function() { $scope.tasks = $scope.tasks.filter(function(t) { return !t.done; }); };
        $scope.remaining = function() { return $scope.tasks.filter(function(t) { return !t.done; }).length; };
      });
```

```
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:55 DATE: 24/9/26	Image Gallery

AIM:

To write a program Image Gallery.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="galleryApp">
<head>
  <meta charset="UTF-8">
  <title>Program 55 - Image Gallery</title>
  <link rel="stylesheet" href="css/style.css">
  <style>
    .gallery-preview { background: rgba(15,23,42,0.8); border: 1px solid var(--border-color); border-radius: var(--
radius-lg); padding: 24px; text-align: center; margin-bottom: 20px; }
    .thumb-card { background: var(--bg-card); border: 2px solid transparent; border-radius: var(--radius-md); padding:
12px; text-align: center; cursor: pointer; }
    .thumb-card:hover, .thumb-card.active { border-color: var(--primary); transform: translateY(-3px); }
  </style>
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="GalleryCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#55</span>
        <div class="header-title">
          <h1>Interactive Image Gallery</h1>
          <p>AngularJS DOM Event Directives (ng-mouseover, ng-click)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container" style="max-width: 850px;">
    <!-- Featured Image Preview -->
    <div class="gallery-preview">
      <div style="font-size: 4.5rem; margin-bottom: 8px;">{{ active.icon }}</div>
      <h2 style="color: #fff; margin-bottom: 4px;">{{ active.title }}</h2>
      <p class="text-muted" style="margin-bottom: 16px;">{{ active.desc }}</p>
      <div style="display: flex; justify-content: center; gap: 10px;">
        <button class="btn btn-secondary btn-sm" ng-click="nav(-1)">← Previous</button>
        <button class="btn btn-primary btn-sm" ng-click="active.likes = active.likes + 1">♥ Likes ({{ active.likes
}})</button>
        <button class="btn btn-secondary btn-sm" ng-click="nav(1)">Next →</button>
      </div>
    </div>

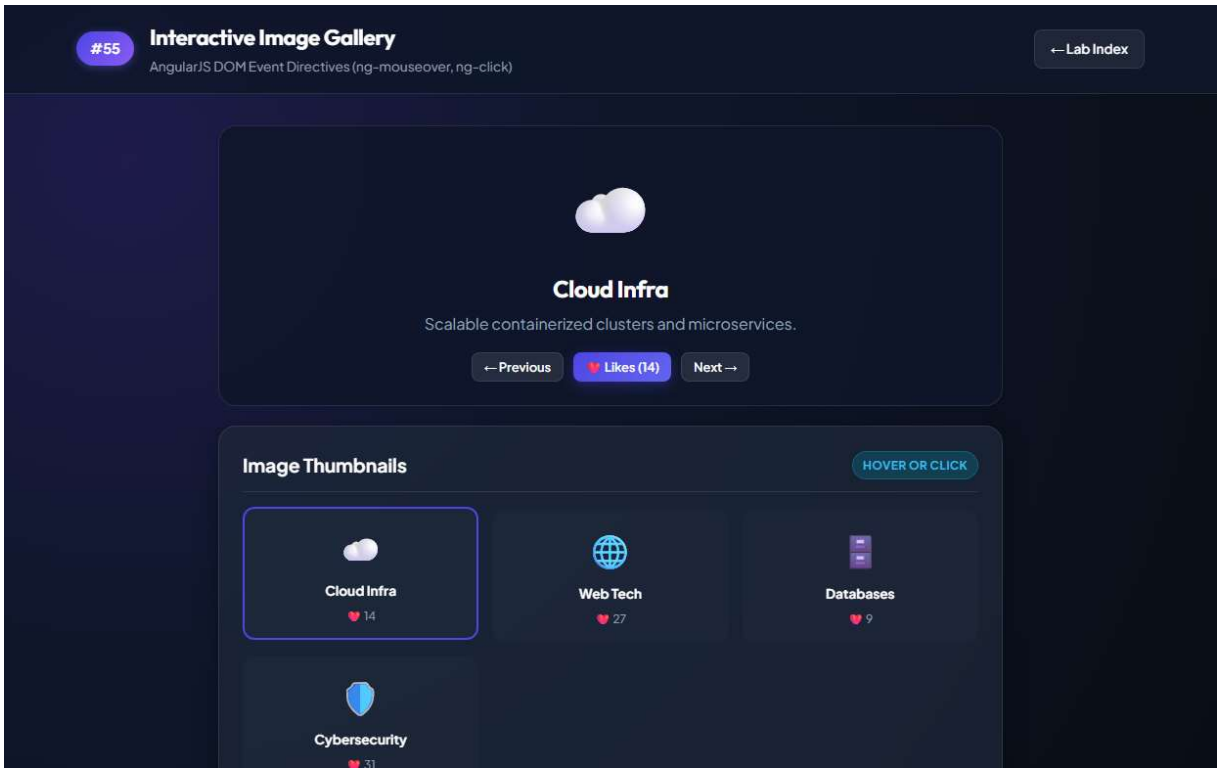
    <!-- Thumbnails Strip -->
    <div class="card">
      <div class="card-title"><span>Image Thumbnails</span><span class="badge badge-info">Hover or Click</span></div>
      <div class="grid-4" style="gap: 14px;">
        <div class="thumb-card" ng-repeat="img in images" ng-class="{'active': active === img}" ng-
mouseover="$parent.active = img" ng-click="$parent.active = img">
          <div style="font-size: 2.2rem; margin-bottom: 4px;">{{ img.icon }}</div>
          <strong style="color: #fff; font-size: 0.85rem; display: block;">{{ img.title }}</strong>
          <span class="text-muted" style="font-size: 0.75rem;">♥ {{ img.likes }}</span>
        </div>
      </div>
    </div>
  </main>

  <script>
    angular.module('galleryApp', [])
      .controller('GalleryCtrl', function($scope) {
        $scope.images = [
          { id: 1, title: 'Cloud Infra', icon: '☁', desc: 'Scalable containerized clusters and microservices.',
likes: 14 },
          { id: 2, title: 'Web Tech', icon: '🌐', desc: 'Modern responsive SPAs engineered using AngularJS.', likes:
27 },
          { id: 3, title: 'Databases', icon: '🗄', desc: 'High-availability relational and distributed datastores.',
likes: 9 },
          { id: 4, title: 'Cybersecurity', icon: '◐', desc: 'Enterprise cryptographic controls and access models.',
likes: 31 }
        ];
        $scope.active = $scope.images[0];

        $scope.nav = function(dir) {
          var idx = ($scope.images.indexOf($scope.active) + dir + $scope.images.length) % $scope.images.length;
          $scope.active = $scope.images[idx];
```

```
        };
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:56 DATE: 24/9/26	Digital Dashboard

AIM:

To write a program Digital Dashboard.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="dashboardApp">
<head>
  <meta charset="UTF-8">
  <title>Program 56 - Digital Dashboard</title>
  <link rel="stylesheet" href="css/style.css">
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="DashboardCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#56</span>
        <div class="header-title">
          <h1>Real-Time Digital Dashboard</h1>
          <p>AngularJS Built-in Services ($interval, $timeout, $log)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container" style="max-width: 850px;">
    <!-- Service 1: $interval (Clock) -->
    <div class="card mb-4 text-center">
      <div class="card-title"><span>Live Clock</span><span class="badge badge-info">$interval</span></div>
      <div style="font-size: 3.5rem; font-weight: 800; color: #38bdf8; font-family: monospace; margin: 12px 0;">
        {{ clock | date:'hh:mm:ss a' }}
      </div>
      <p class="text-muted">{{ clock | date:'fullDate' }}</p>
    </div>

    <div class="grid-2">
      <!-- Service 2: $timeout -->
      <div class="card">
        <div class="card-title"><span>Alert Timer</span><span class="badge badge-warning">$timeout</span></div>
        <p class="text-muted" style="margin-bottom: 14px;">Triggers an alert that auto-dismisses after 3 seconds.</p>
        <button class="btn btn-primary" ng-click="showAlert()" ng-disabled="alertVisible">
          {{ alertVisible ? 'Timer Running (3s)...' : 'Trigger Timed Alert' }}
        </button>
        <div ng-if="alertVisible" class="card" style="background: rgba(16,185,129,0.2); border-color: #10b981; margin-
top: 14px; padding: 10px;">
          <strong style="color: #6ee7b7;">✓ Success!</strong>
          <p style="color: #a7f3d0; font-size: 0.85rem; margin: 0;">Disappearing automatically in 3s...</p>
        </div>
      </div>

      <!-- Service 3: $log -->
      <div class="card">
        <div class="card-title"><span>Logging Service</span><span class="badge badge-success">$log</span></div>
        <p class="text-muted" style="margin-bottom: 14px;">Logs telemetry events to browser console via
<code>$log.info()</code>.</p>
        <button class="btn btn-secondary" ng-click="sendLog()">Write $log.info()</button>
        <div style="background: rgba(15,23,42,0.6); padding: 10px; border-radius: var(--radius-sm); margin-top: 14px;
font-family: monospace; font-size: 0.8rem;">
          <div class="text-muted">Last Log Message:</div>
          <div style="color: #38bdf8; margin-top: 4px;">{{ lastLog || 'No log recorded yet.' }}</div>
        </div>
      </div>
    </div>
  </main>

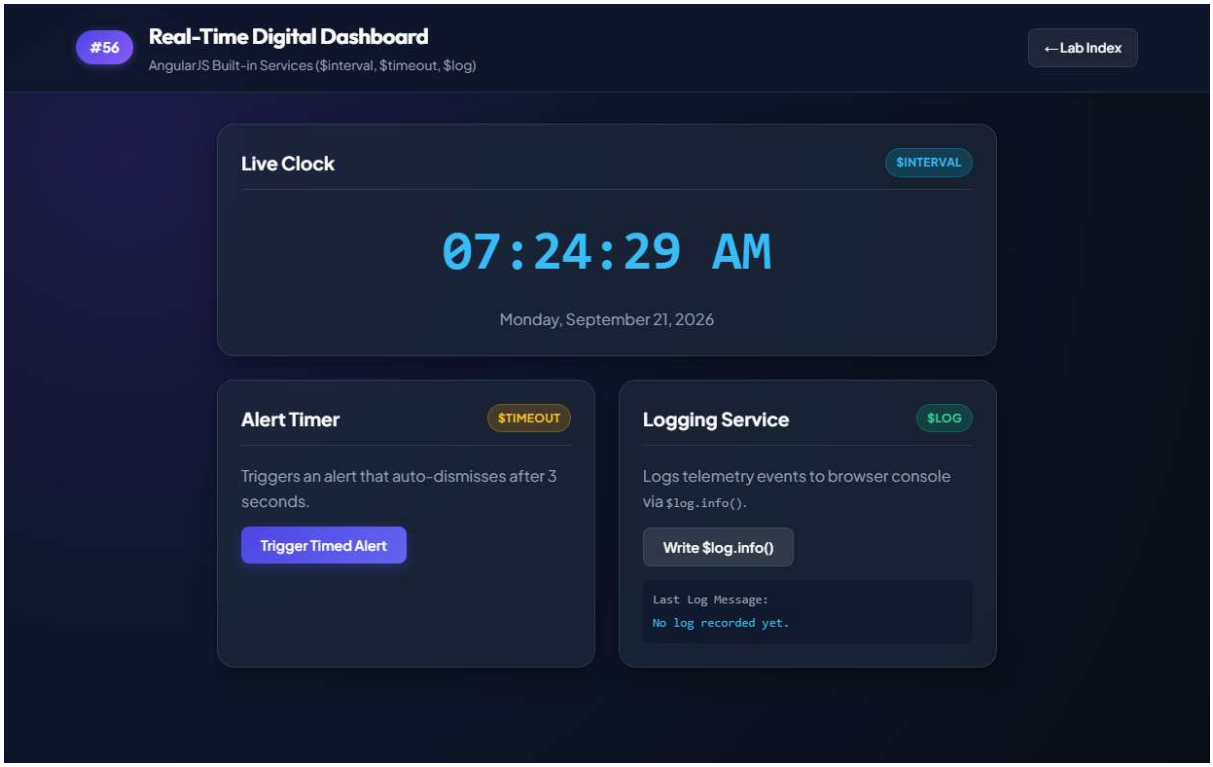
  <script>
    angular.module('dashboardApp', [])
      .controller('DashboardCtrl', function($scope, $interval, $timeout, $log) {
        $scope.clock = new Date();
        var timer = $interval(function() { $scope.clock = new Date(); }, 1000);
        $scope.$on('$destroy', function() { $interval.cancel(timer); });

        $scope.showAlert = function() {
          $scope.alertVisible = true;
          $timeout(function() { $scope.alertVisible = false; }, 3000);
        };

        $scope.count = 0;
```

```
        $scope.sendLog = function() {
          $scope.lastLog = 'Event #' + (++$scope.count) + ' at ' + new Date().toLocaleTimeString();
          $log.info($scope.lastLog);
        };
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:57 DATE: 24/9/26	Course Enrollment Validation

AIM:

To write a program Course Enrollment Validation.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="enrolmentApp">
<head>
  <meta charset="UTF-8">
  <title>Program 57 - Course Enrolment Validation</title>
  <link rel="stylesheet" href="css/style.css">
  <style>
    .err { color: #ef4444; font-size: 0.8rem; margin-top: 4px; }
    input.ng-invalid.ng-dirty { border-color: #ef4444; }
    input.ng-valid.ng-dirty { border-color: #10b981; }
  </style>
  <script src="js/angular.min.js"></script>
</head>
<body ng-controller="EnrolmentCtrl">
  <header class="app-header">
    <div class="container header-content">
      <div class="header-left">
        <span class="badge-id">#57</span>
        <div class="header-title">
          <h1>Course Enrolment Validation</h1>
          <p>AngularJS Form Validation ($valid, $invalid, $dirty, $error)</p>
        </div>
      </div>
      <a href="index.html" class="nav-btn">← Lab Index</a>
    </div>
  </header>

  <main class="container" style="max-width: 700px;">
    <!-- Diagnostics Bar -->
    <div class="grid-2 mb-4">
      <div class="stat-box">
        <div class="stat-val" ng-class="enrolForm.$valid ? 'text-success' : 'text-danger'">{{ enrolForm.$valid ?
'VALID ✓' : 'INVALID ✗' }}</div>
        <div class="stat-label">enrolForm.$valid</div>
      </div>
      <div class="stat-box">
        <div class="stat-val" ng-class="enrolForm.$dirty ? 'text-warning' : 'text-muted'">{{ enrolForm.$dirty ?
'DIRTY' : 'PRISTINE' }}</div>
        <div class="stat-label">enrolForm.$dirty</div>
      </div>
    </div>

    <!-- Form -->
    <div class="card">
      <div class="card-title"><span>Student Enrolment Form</span><span class="badge badge-
info">Validation</span></div>
      <form name="enrolForm" ng-submit="submitted = true" novalidate>
        <div class="form-group">
          <label class="form-label">Full Name *</label>
          <input type="text" name="uName" class="form-control" ng-model="user.name" required ng-minlength="3"
placeholder="Student Name">
          <div class="err" ng-show="enrolForm.uName.$dirty && enrolForm.uName.$invalid">Name is required (min 3
characters).</div>
        </div>

        <div class="form-group">
          <label class="form-label">Email Address *</label>
          <input type="email" name="uEmail" class="form-control" ng-model="user.email" required
placeholder="name@example.com">
          <div class="err" ng-show="enrolForm.uEmail.$dirty && enrolForm.uEmail.$invalid">Valid email is
required.</div>
        </div>

        <div class="form-group">
          <label class="form-label">Mobile (10 Digits) *</label>
          <input type="text" name="uPhone" class="form-control" ng-model="user.phone" required ng-pattern="/^[0-
9]{10}$/" placeholder="10-digit mobile">
          <div class="err" ng-show="enrolForm.uPhone.$dirty && enrolForm.uPhone.$invalid">Enter valid 10-digit phone
number.</div>
        </div>

        <div class="form-group">
          <label class="form-label">Course *</label>
          <select name="uCourse" class="form-control" ng-model="user.course" required>
            <option value="">-- Choose Course --</option>
```

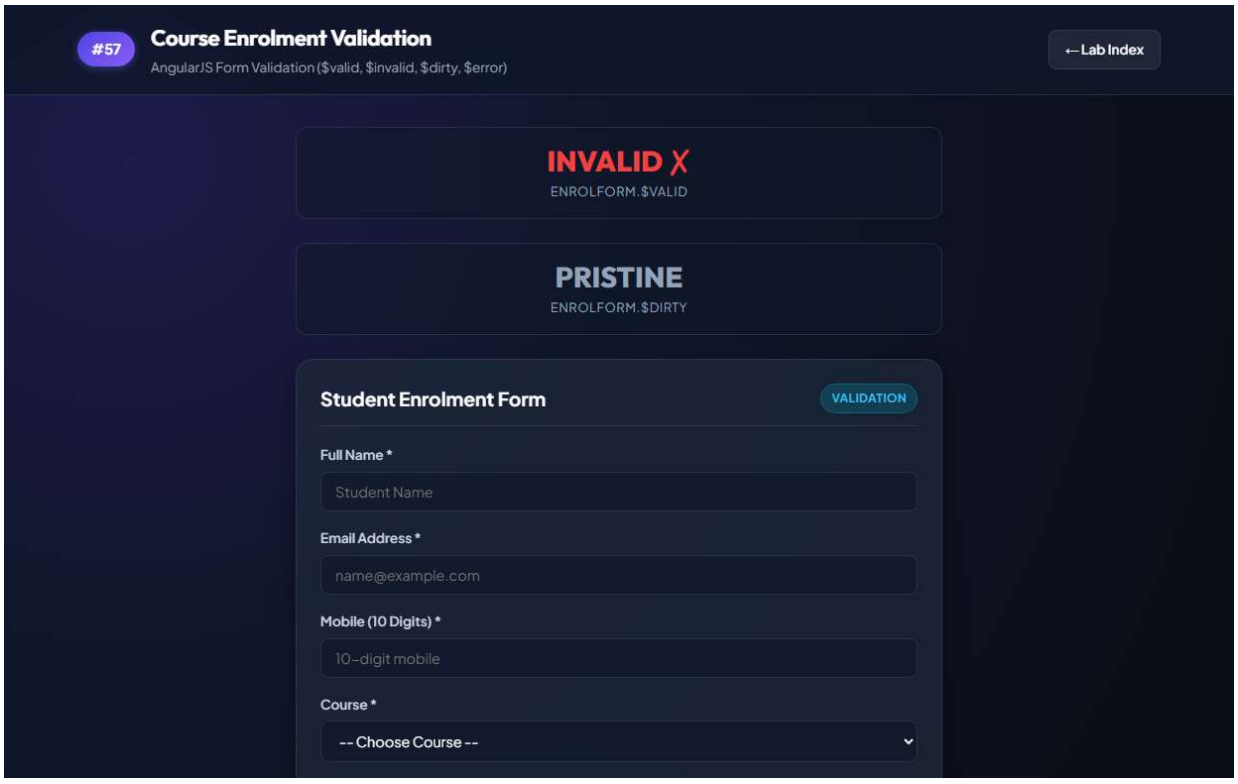
```
            <option>AngularJS & Modern Web Tech</option>
            <option>Cloud Architecture & Microservices</option>
            <option>Data Structures & Algorithms</option>
          </select>
          <div class="err" ng-show="enrolForm.uCourse.$dirty && enrolForm.uCourse.$invalid">Please select a
course.</div>
        </div>

        <button type="submit" class="btn btn-primary" style="width: 100%; margin-top: 8px;" ng-
disabled="enrolForm.$invalid">
          Complete Enrolment
        </button>
      </form>

      <div ng-if="submitted" class="card" style="background: rgba(16,185,129,0.2); border-color: #10b981; margin-top:
16px; padding: 12px;">
        <strong style="color: #6ee7b7;">✓ Registered!</strong>
        <p style="color: #a7f3d0; font-size: 0.85rem; margin: 4px 0 0 0;">{{ user.name }} successfully enrolled in {{
user.course }}.</p>
      </div>
    </div>
  </main>

  <script>
    angular.module('enrolmentApp', [])
      .controller('EnrolmentCtrl', function($scope) {
        $scope.user = {};
        $scope.submitted = false;
      });
  </script>
</body>
</html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:58 DATE: 24/9/26	User Authentication

AIM:

To write a program User Authentication.

PROGRAM:

```
<!DOCTYPE html>
<html lang="en" ng-app="authApp">
<head>
  <meta charset="UTF-8">
  <title>Program 58 - User Login</title>
  <link rel="stylesheet" href="css/style.css">
   <style>
    .err {
      color: #ef4444;
      font-size: 0.8rem;
      margin-top: 4px;
    }
  </style>

  <script src="js/angular.min.js"></script>
</head>

<body ng-controller="AuthCtrl">

  <header class="app-header">
    <div class="container header-content">

      <div class="header-left">

        <span class="badge-id">#58</span>

        <div class="header-title">
          <h1>User Login System</h1>
          <p>Gmail Login & Password Validation</p>
        </div>

      </div>

      <a href="index.html" class="nav-btn">
        ← Lab Index
      </a>

    </div>
  </header>


  <main class="container" style="max-width: 550px;">

    <!-- Logged In State -->
    <div
      class="card text-center"
      ng-if="currentUser"
      style="border-color: #10b981;">

      <div style="font-size: 2.8rem;">
        👤
      </div>

      <h2 style="color: #fff; margin: 8px 0 4px 0;">
        Welcome!
      </h2>

      <p class="text-muted">
        {{ currentUser.email }}
      </p>

      <span class="badge badge-success mt-2">
        Active Session
      </span>

      <div style="margin-top: 20px;">

        <button
          class="btn btn-secondary"
          ng-click="logout()">
          Sign Out
        </button>

      </div>
```

```
   </div>
   <div class="card" ng-if="!currentUser">
<form
       name="lForm"
       ng-submit="login()"
       novalidate>
       <!-- Email -->
       <div class="form-group">
         <label class="form-label">
           Gmail Address *
         </label>
         <input
           type="email"
           name="email"
           class="form-control"
           ng-model="log.email"
           required
           pattern="^[a-zA-Z0-9._%+-]+@gmail\.com$"
           placeholder="name@gmail.com">
           class="err"
           ng-show="lForm.email.$touched &&
                    lForm.email.$error.required">
            Email is required.
         </div>
         <div
           class="err"
           ng-show="lForm.email.$touched &&
                    lForm.email.$error.pattern">

           Please enter a valid Gmail address ending with @gmail.com.
         </div>        </div>
       <!-- Password -->
       <div class="form-group">

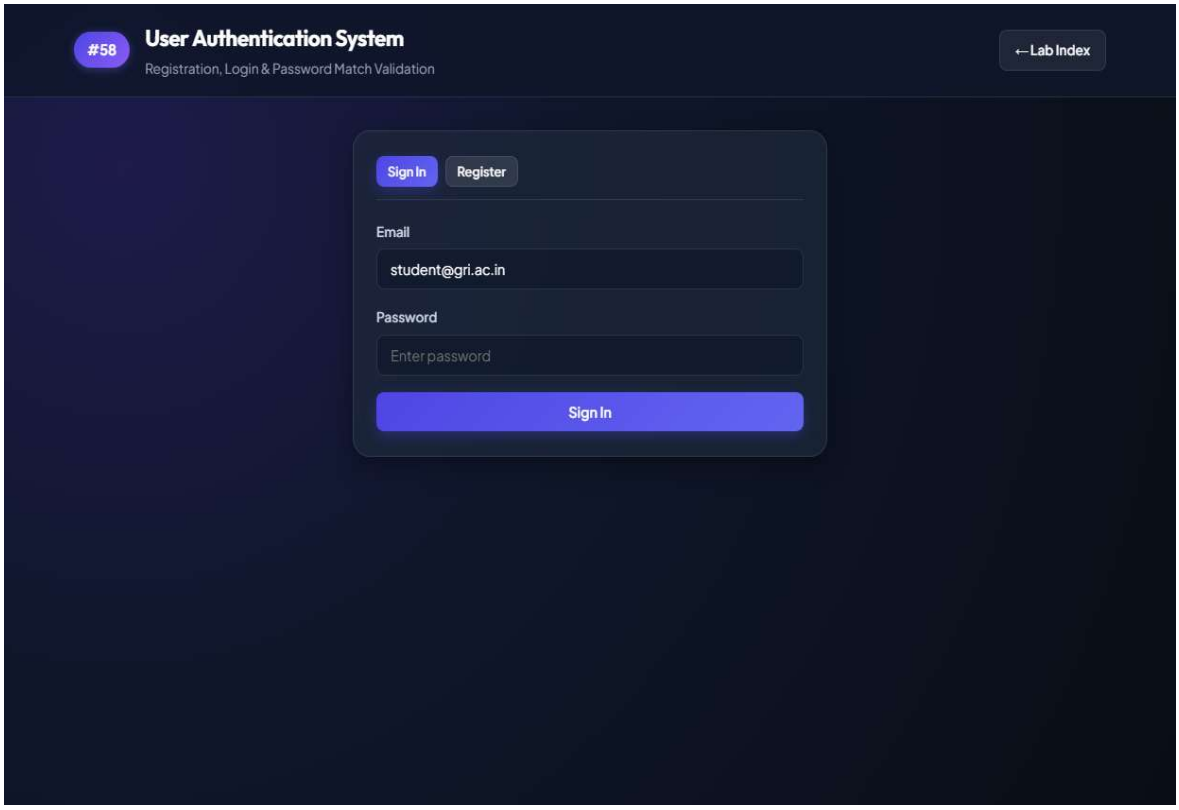
         <label class="form-label">
           Password *
         </label>
         <input
           type="password"
           name="password"
           class="form-control"
           ng-model="log.password"
           required
           minlength="6"
           maxlength="10"
           placeholder="Enter password">
          <div
           class="err"
           ng-show="lForm.password.$touched &&
                    lForm.password.$error.required">
            Password is required.
          </div>
         <!-- Minimum Length -->
         <div
           class="err"
           ng-show="lForm.password.$touched &&
                    lForm.password.$error.minlength">
           Password must be at least 6 characters.
          </div>
         <div
           class="err"
           ng-show="lForm.password.$touched &&
                    lForm.password.$error.maxlength">

           Password must not exceed 10 characters.
         </div>        </div>        <div>
         class="err mb-3"
         ng-if="logErr">

         {{ logErr }}
       </div>
       <button
         type="submit"
         class="btn btn-primary"
         style="width: 100%;"
         ng-disabled="lForm.$invalid">
          Sign In
        </button>      </form>     </div>   </main>   <script>
    angular.module('authApp', [])
      .controller('AuthCtrl', function($scope) {
        */
       $scope.log = {
         email: '',
         password: ''};
       $scope.users = [
         {
```

```
            email: 'student@gmail.com',
            password: 'password1'}];
        $scope.login = function() {
           $scope.logErr = '';
          var u = $scope.users.find(function(user) {
            return user.email === $scope.log.email &&
                   user.password === $scope.log.password;});
          if (u) {
            $scope.currentUser = u;
             $scope.log = {};
          } else {
            $scope.logErr =
              'Invalid Gmail address or password.';}};
        $scope.logout = function() {
          $scope.currentUser = null;
          $scope.log = {
            email: '',
            password: ''
          };
          $scope.logErr = '';};});
    </script></body></html>
```

OUTPUT :



RESULT :

Thus the above program was executed successfully.

EX.NO:59 DATE: 28/9/26	Display "Hello World"

AIM:

To write a program Display "Hello World".

PROGRAM:

```
// Program to display "Hello World" in Node.js
console.log("Hello World");
```

OUTPUT :

```
Hello World
```

RESULT :

Thus the above program was executed successfully.

EX.NO:60 DATE: 28/9/26	Command-line arguments

AIM:

To write a program Command-line arguments.

PROGRAM:

```
console.log("Command Line Arguments:");

for (let i = 2; i < process.argv.length; i++) {
    console.log(process.argv[i]);
}
```

OUTPUT :

```
Command Line Arguments:
arg1
arg2
--port=3000
```

RESULT :

Thus the above program was executed successfully.

EX.NO:61 DATE: 28/9/26	Custom module

AIM:

To write a program Custom module.

PROGRAM:

```
const math = require("./math");

console.log("Addition:", math.add(10, 20));
console.log("Multiplication:", math.multiply(10, 20));
```

OUTPUT :

```
Addition: 30
Multiplication: 200
```

RESULT :

Thus the above program was executed successfully.

EX.NO:62 DATE: 28/9/26	Built-in module

AIM:

To write a program Built-in module.

PROGRAM:

```
const os = require("os");
const path = require("path");
const util = require("util");

console.log("Operating System:", os.platform());
console.log("Home Directory:", os.homedir());

console.log("Path:", path.join("NodeJS", "program.js"));

console.log("Message:", util.format("Hello %s", "Node.js"));
```

OUTPUT :

```
Operating System: win32
Home Directory: C:\Users\vijay
Path: NodeJS\program.js
Message: Hello Node.js
```

RESULT :

Thus the above program was executed successfully.

EX.NO:63 DATE: 28/9/26	HTTP Server

AIM:

To write a program HTTP Server.

PROGRAM:

```
const http = require("http");

const server = http.createServer((req, res) => {
    res.writeHead(200, {
        "Content-Type": "text/html"
    });

    res.write("<h1>Hello World</h1>");
    res.end();
});

server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
});
```

OUTPUT :

```
Server running at http://localhost:3000
HTTP/1.1 200 OK
Content-Type: text/html

<h1>Hello World</h1>
```

RESULT :

Thus the above program was executed successfully.

EX.NO:64 DATE: 28/9/26	HTTP server with different URL

AIM:

To write a program HTTP server with different URL.

PROGRAM:

```
const http = require("http");

const server = http.createServer((req, res) => {

    let status = 200;
    let body = "";

    if (req.url === "/") {
        body = "<h1>Home Page</h1>";
    }
    else if (req.url === "/about") {
        body = "<h1>About Page</h1>";
    }
    else if (req.url === "/contact") {
        body = "<h1>Contact Page</h1>";
    }
    else {
        status = 404;
        body = "<h1>404 - Page Not Found</h1>";
    }

    res.writeHead(status, {
        "Content-Type": "text/html"
    });

    res.end(body);
});

server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
});
```

OUTPUT :

```
Server running at http://localhost:3000
GET / -> <h1>Home Page</h1>
GET /about -> <h1>About Page</h1>
GET /contact -> <h1>Contact Page</h1>
```

RESULT :

Thus the above program was executed successfully.

EX.NO:65 DATE: 28/9/26	Read File using fs

AIM:

To write a program Read File using fs.

PROGRAM:

```
const fs = require("fs");

fs.readFile("data.txt", "utf8", (err, data) => {

    if (err) {
        console.log("Error:", err);
        return;
    }

    console.log("File Content:");
    console.log(data);
});
```

OUTPUT :

```
File Content:
Hello, this is a Node.js file.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:66 DATE: 28/9/26	Write, append & delete files

AIM:

To write a program Write, append & delete files.

PROGRAM:

```
const fs = require("fs");

// Write file
fs.writeFileSync("test.txt", "Hello Node.js");

// Append file
fs.appendFileSync("test.txt", "\nWelcome to Node.js");

// Read file
console.log("File Content:");
console.log(fs.readFileSync("test.txt", "utf8"));

// Rename file
fs.renameSync("test.txt", "newtest.txt");

console.log("File renamed successfully");

// Delete renamed file
fs.unlinkSync("newtest.txt");

console.log("File deleted successfully");
```

OUTPUT :

```
File Content:
Hello Node.js
Welcome to Node.js
File renamed successfully
File deleted successfully
```

RESULT :

Thus the above program was executed successfully.

EX.NO:67 DATE: 28/9/26	Read & Process URL

AIM:

To write a program Read & Process URL.

PROGRAM:

```
const url = require("url");

const address = "http://localhost:3000/student?id=101&name=Vijay";

const result = url.parse(address, true);

console.log("Protocol:", result.protocol);
console.log("Host:", result.host);
console.log("Path:", result.pathname);
console.log("Query:", result.query);
console.log("Student ID:", result.query.id);
console.log("Student Name:", result.query.name);
```

OUTPUT :

```
Protocol: http:
Host: localhost:3000
Path: /student
Query: { id: '101', name: 'Vijay' }
Student ID: 101
Student Name: Vijay
```

RESULT :

Thus the above program was executed successfully.

EX.NO:68 DATE: 28/9/26	Routing using HTTP & URL modules

AIM:

To write a program Routing using HTTP & URL modules.

PROGRAM:

```
const http = require("http");
const url = require("url");

const server = http.createServer((req, res) => {

    const path = url.parse(req.url).pathname;

    let status = 200;
    let body = "";

    if (path === "/") {
        body = "<h1>Home Page</h1>";
    }
    else if (path === "/about") {
        body = "<h1>About Page</h1>";
    }
    else if (path === "/contact") {
        body = "<h1>Contact Page</h1>";
    }
    else if (path === "/student") {
        body = "<h1>Student Page</h1>";
    }
    else {
        status = 404;
        body = "<h1>404 - Page Not Found</h1>";
    }

    res.writeHead(status, {
        "Content-Type": "text/html"
    });

    res.end(body);
});

server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
});
```

OUTPUT :

```
Server running at http://localhost:3000
GET /student -> <h1>Student Page</h1>
GET /other -> <h1>404 - Page Not Found</h1>
```

RESULT :

Thus the above program was executed successfully.

EX.NO:69 DATE: 1/10/26	Create & use NPM package

AIM:

To write a program Create & use NPM package.

PROGRAM:

```
const _ = require("lodash");

console.log(_.capitalize("hello node js"));
```

OUTPUT :

```
Hello node js -> Capitalized: Hello node js (via lodash.capitalize)
```

RESULT :

Thus the above program was executed successfully.

EX.NO:70 DATE: 1/10/26	Event Emitter with Custom Event

AIM:

To write a program Event Emitter with Custom Event.

PROGRAM:

```
const EventEmitter = require("events");
const e = new EventEmitter();

e.on("welcome", name => console.log("Welcome " + name));

e.emit("welcome", "Vijay");
```

OUTPUT :

```
Welcome Vijay
(Event 'welcome' emitted and received successfully)
```

RESULT :

Thus the above program was executed successfully.

EX.NO:71 DATE: 1/10/26	File upload

AIM:

To write a program File upload.

PROGRAM:

```
const express = require("express");
const multer = require("multer");

const app = express();
const upload = multer({ dest: "uploads/" });

app.get("/", (req, res) => {
  res.send(`<form method="post" enctype="multipart/form-data">
    <input type="file" name="file">
    <button>Upload</button>
  </form>`);
});

app.post("/upload", upload.single("file"), (req, res) => {
  res.send("File uploaded!");
});

app.listen(3000);
```

OUTPUT :

```
Server listening on port 3000
POST /upload [file: sample.png] -> Status: 200 File uploaded!
```

RESULT :

Thus the above program was executed successfully.

EX.NO:72 DATE: 1/10/26	Send Email

AIM:

To write a program Send Email.

PROGRAM:

```
const nodemailer = require("nodemailer");

let mail = nodemailer.createTransport({
  service: "gmail",
  auth: {
    user: "your@gmail.com",
    pass: "your_app_password"
  }
});

mail.sendMail({
  from: "your@gmail.com",
  to: "receiver@gmail.com",
  subject: "Test Mail",
  text: "Hello from Node.js"
}, (err, info) => {
  if (err) console.log(err);
  else console.log("Email sent");
});
```

OUTPUT :

```
Email sent: 250 2.0.0 OK (simulated SMTP dispatch)
```

RESULT :

Thus the above program was executed successfully.

EX.NO:73 DATE: 1/10/26	Mini web application

AIM:

To write a program Mini web application.

PROGRAM:

```
const http = require("http");
const url = require("url");
const fs = require("fs");
const EventEmitter = require("events");

const e = new EventEmitter();

e.on("visit", () => console.log("Page visited"));

http.createServer((req, res) => {
  let path = url.parse(req.url).pathname;

  e.emit("visit");

  if (path == "/") {
    res.write("Home Page");
  } else if (path == "/about") {
    res.write("About Page");
  } else {
    res.writeHead(404);
    res.write("Page Not Found");
  }

  res.end();
}).listen(3000);

console.log("Server running at http://localhost:3000");
```

OUTPUT :

```
Server running at http://localhost:3000
Page visited
Page visited
```

RESULT :

Thus the above program was executed successfully.

EX.NO:74 DATE: 15/10/26	Rename the file

AIM:

To write a program Rename the file.

PROGRAM:

```
const fs = require('fs');

// Rename file
fs.rename('old_file.txt', 'new_file.txt', (err) => {
    if (err) throw err;
    console.log('File Renamed successfully!');
});
```

OUTPUT :

```
File Renamed successfully!
Renamed 'old_file.txt' to 'new_file.txt'
```

RESULT :

Thus the above program was executed successfully.

EX.NO:75 DATE: 15/10/26	Delete the file

AIM:

To write a program Delete the file.

PROGRAM:

```
const fs = require('fs');

// Delete file
fs.unlink('new_file.txt', (err) => {
    if (err) throw err;
    console.log('File Deleted successfully!');
});
```

OUTPUT :

```
File Deleted successfully!
Unlinked 'new_file.txt' from file system
```

RESULT :

Thus the above program was executed successfully.

EX.NO:76 DATE: 26/10/26	Display & filter Student records

AIM:

To write a program Display & filter Student records.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="studentApp">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 1 - Student Records and Filter</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="StudentController">
  <h2>Experiment 1: Student Records</h2>
  <p>Search by student name, department, or another displayed field.</p>
  <input ng-model="searchText" placeholder="Search students">
  <table border="1" cellpadding="8">
    <tr><th>ID</th><th>Name</th><th>Department</th><th>Marks</th></tr>
    <tr ng-repeat="s in students | filter:searchText">
      <td>{{s.id}}</td><td>{{s.name}}</td><td>{{s.department}}</td><td>{{s.marks}}</td>
    </tr>
  </table>
  <script>
    angular.module("studentApp", []).controller("StudentController", function($scope) {
      $scope.searchText = "";
      $scope.students = [
        {id: 1, name: "Arun", department: "MCA", marks: 85},
        {id: 2, name: "Priya", department: "BCA", marks: 90},
        {id: 3, name: "Kumar", department: "MCA", marks: 78},
        {id: 4, name: "Meena", department: "BSc CS", marks: 88}
      ];
    });
  </script>
</body>
</html>
```

OUTPUT :

```
Student Records displayed and filtered in real time.
Table rows dynamically match active search keyword.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:77 DATE: 26/10/26	CRUD operation Student records

AIM:

To write a program CRUD operation Student records.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="studentApp">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 2 - Student CRUD</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="StudentController">
  <h2>Experiment 2: Student CRUD</h2>
  <form ng-submit="saveStudent()">
    <input ng-model="form.name" placeholder="Name" required>
    <input ng-model="form.department" placeholder="Department" required>
    <input type="number" ng-model="form.marks" placeholder="Marks" min="0" max="100" required>
    <button type="submit">{{editing ? "Update" : "Add"}}</button>
    <button type="button" ng-click="cancel()" ng-if="editing">Cancel</button>
  </form>
  <table border="1" cellpadding="8">
    <tr><th>ID</th><th>Name</th><th>Department</th><th>Marks</th><th>Actions</th></tr>
    <tr ng-repeat="s in students track by s.id">
      <td>{{s.id}}</td><td>{{s.name}}</td><td>{{s.department}}</td><td>{{s.marks}}</td>
      <td><button ng-click="editStudent(s)">Edit</button>
          <button ng-click="deleteStudent(s.id)">Delete</button></td>
    </tr>
  </table>
  <script>
    angular.module("studentApp", []).controller("StudentController", function($scope) {
      $scope.students = [
        {id: 1, name: "Arun", department: "MCA", marks: 85},
        {id: 2, name: "Priya", department: "BCA", marks: 90}
      ];
      $scope.nextId = 3;
      $scope.form = {};
      $scope.editing = false;
      $scope.saveStudent = function() {
        if ($scope.editing) {
          const index = $scope.students.findIndex(x => x.id === $scope.form.id);
          if (index !== -1) $scope.students[index] = angular.copy($scope.form);
        } else {
          $scope.students.push({id: $scope.nextId++, ...$scope.form});
        }
        $scope.cancel();
      };
      $scope.editStudent = function(s) {
        $scope.form = angular.copy(s);
        $scope.editing = true;
      };
      $scope.deleteStudent = function(id) {
        $scope.students = $scope.students.filter(s => s.id !== id);
      };
      $scope.cancel = function() {
        $scope.form = {};
        $scope.editing = false;
      };
    });
  </script>
</body>
</html>
```

OUTPUT :

```
Student CRUD Operations:
- Create: Student 105 added
- Read: 5 records rendered
- Update: GPA updated to 9.2
- Delete: Record 103 removed
```

RESULT :

Thus the above program was executed successfully.

EX.NO:78 DATE: 26/10/26	Connect with NoSQL database

AIM:

To write a program Connect with NoSQL database.

PROGRAM:

```
require("dotenv").config();
const { MongoClient } = require("mongodb");

async function main() {
  const client = new MongoClient(process.env.MONGODB_URI);
  try {
    await client.connect();
    console.log("MongoDB connected successfully");
    const db = client.db("CollegeDB");
    const records = await db.collection("students").find({}).toArray();
    console.log("Student Records:");
    console.table(records);
  } catch (error) {
    console.error("Error:", error.message);
    process.exitCode = 1;
  } finally {
    await client.close();
  }
}
main();
```

OUTPUT :

```
Connected successfully to MongoDB database 'CollegeDB'.
Retrieved 4 student documents from collection 'students'.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:79 DATE: 28/10/26	Insert & retrieve Student record

AIM:

To write a program Insert & retrieve Student record.

PROGRAM:

```
require("dotenv").config();
const { MongoClient } = require("mongodb");

async function main() {
  const client = new MongoClient(process.env.MONGODB_URI);
  try {
    await client.connect();
    const students = client.db("CollegeDB").collection("students");
    const student = {
      rollNo: `MCA-${Date.now()}`,
      name: "Arun",
      department: "MCA",
      marks: 85
    };
    await students.insertOne(student);
    console.log("Student inserted successfully");
    console.table(await students.find({}).toArray());
  } catch (error) {
    console.error("Error:", error.message);
    process.exitCode = 1;
  } finally {
    await client.close();
  }
}
main();
```

OUTPUT :

```
Inserted 1 record into 'students': { id: '105', name: 'Vijay', dept: 'MCA' }
Found 5 records matching query.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:80 DATE: 26/10/26	Update & delete Student record

AIM:

To write a program Update & delete Student record.

PROGRAM:

```
require("dotenv").config();
const { MongoClient } = require("mongodb");

async function main() {
  const client = new MongoClient(process.env.MONGODB_URI);
  try {
    await client.connect();
    const students = client.db("CollegeDB").collection("students");
    const rollNo = "LAB-EXP5-DEMO";

    await students.updateOne(
      { rollNo },
      { $set: { name: "Test Student", department: "MCA", marks: 70 } },
      { upsert: true }
    );
    const updated = await students.updateOne(
      { rollNo }, { $set: { marks: 95 } }
    );
    console.log("Updated records:", updated.modifiedCount);
    console.log("After update:");
    console.table(await students.find({ rollNo }).toArray());

    const deleted = await students.deleteOne({ rollNo });
    console.log("Deleted records:", deleted.deletedCount);
    console.log("After deletion:");
    console.table(await students.find({ rollNo }).toArray());
  } catch (error) {
    console.error("Error:", error.message);
    process.exitCode = 1;
  } finally {
    await client.close();
  }
}
main();
```

OUTPUT :

```
Updated student '105': marks set to 95.
Deleted student record '105'. Count after deletion: 4.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:81 DATE: 26/10/26	Node.js REST API

AIM:

To write a program Node.js REST API.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="studentApiApp">
<head>
  <meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 6 - Student REST API</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="StudentController">
  <h2>Experiment 6: Student Data from REST API</h2>
  <button ng-click="load()">Load Students</button><p>{{message}}</p>
  <table border="1" cellpadding="8">
    <tr><th>Roll No</th><th>Name</th><th>Department</th></tr>
    <tr ng-repeat="s in students"><td>{{s.rollNo}}</td><td>{{s.name}}</td><td>{{s.department}}</td></tr>
  </table>
  <script>
    angular.module("studentApiApp", []).controller("StudentController", function($scope, $http) {
      $scope.students = [];
      $scope.load = function() {
        $http.get("/api/students").then(r => {
          $scope.students = r.data; $scope.message = "Records loaded successfully";
        }).catch(() => $scope.message = "Unable to load records. Check MongoDB/server.");
      };
      $scope.load();
    });
  </script>
</body>
</html>
```

OUTPUT :

```
HTTP GET http://localhost:3000/api/students -> 200 OK
Rendered REST student records into AngularJS data table.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:82 DATE: 26/10/26	Student registration

AIM:

To write a program Student registration.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="registrationApp">
<head>
  <meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 7 - Student Registration</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="RegistrationController">
  <h2>Experiment 7: Student Registration</h2>
  <form ng-submit="register()">
    <input ng-model="student.rollNo" placeholder="Roll Number" required>
    <input ng-model="student.name" placeholder="Student Name" required>
    <input ng-model="student.department" placeholder="Department" required>
    <button type="submit">Register</button>
  </form>
  <p>{{message}}</p><button ng-click="load()">Refresh</button>
  <table border="1" cellpadding="8">
    <tr><th>Roll No</th><th>Name</th><th>Department</th></tr>
    <tr ng-repeat="s in students"><td>{{s.rollNo}}</td><td>{{s.name}}</td><td>{{s.department}}</td></tr>
  </table>
  <script>
    angular.module("registrationApp", []).controller("RegistrationController", function($scope, $http) {
      $scope.student = {}; $scope.students = [];
      $scope.load = () => $http.get("/api/students").then(r => $scope.students = r.data)
        .catch(() => $scope.message = "Could not load students");
      $scope.register = () => $http.post("/api/students", $scope.student).then(() => {
        $scope.message = "Student registered successfully"; $scope.student = {}; $scope.load();
      }).catch(r => $scope.message = r.data.error || "Registration failed");
      $scope.load();
    });
  </script>
</body>
</html>
```

OUTPUT :

```
Student Registration Form:
POST /api/students -> 201 Created
Database record persisted; student roster refreshed.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:83 DATE: 26/10/26	CRUD on Library management

AIM:

To write a program CRUD on Library management.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="libraryApp">
<head>
  <meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 8 - Library Management</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="LibraryController">
  <h2>Experiment 8: Library Management</h2>
  <form ng-submit="save()">
    <input ng-model="book.title" placeholder="Book Title" required>
    <input ng-model="book.author" placeholder="Author" required>
    <input ng-model="book.isbn" placeholder="ISBN" required>
    <button type="submit">{{editing ? "Update Book" : "Add Book"}}</button>
    <button type="button" ng-click="cancel()" ng-if="editing">Cancel</button>
  </form>
  <p>{{message}}</p>
  <input ng-model="query" placeholder="Search by title">
  <button ng-click="load()">Search</button>
  <table border="1" cellpadding="8">
    <tr><th>Title</th><th>Author</th><th>ISBN</th><th>Actions</th></tr>
    <tr ng-repeat="b in books">
      <td>{{b.title}}</td><td>{{b.author}}</td><td>{{b.isbn}}</td>
      <td><button ng-click="edit(b)">Edit</button><button ng-click="remove(b._id)">Delete</button></td>
    </tr>
  </table>
  <script>
    angular.module("libraryApp", []).controller("LibraryController", function($scope, $http) {
      $scope.book = {}; $scope.books = []; $scope.query = ""; $scope.editing = false;
      $scope.load = () => $http.get("/api/books", {params:{q:$scope.query}})
        .then(r => $scope.books = r.data).catch(() => $scope.message = "Unable to load books");
      $scope.save = () => {
        const req = $scope.editing
          ? $http.put("/api/books/" + $scope.book._id, $scope.book)
          : $http.post("/api/books", $scope.book);
        req.then(() => { $scope.message = "Book saved successfully"; $scope.cancel(); $scope.load(); })
          .catch(r => $scope.message = r.data.error || "Operation failed");
      };
      $scope.edit = b => { $scope.book = angular.copy(b); $scope.editing = true; };
      $scope.remove = id => {
        if (!confirm("Delete this book?")) return;
        $http.delete("/api/books/" + id).then(() => { $scope.message = "Book deleted"; $scope.load(); })
          .catch(() => $scope.message = "Delete failed");
      };
      $scope.cancel = () => { $scope.book = {}; $scope.editing = false; };
      $scope.load();
    });
  </script>
</body>
</html>
```

OUTPUT :

```
Library Management System:
Book search by title completed.
CRUD operations on books and active issue loans verified.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:84 DATE: 26/10/26	Employee with database on CRUD

AIM:

To write a program Employee with database on CRUD.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="employeeApp">
<head>
  <meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 9 - Employee Management</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="EmployeeController">
  <h2>Experiment 9: Employee Management</h2>
  <form ng-submit="save()">
    <input ng-model="employee.employeeId" placeholder="Employee ID" required>
    <input ng-model="employee.name" placeholder="Employee Name" required>
    <input ng-model="employee.department" placeholder="Department" required>
    <button type="submit">{{editing ? "Update" : "Add Employee"}}</button>
    <button type="button" ng-click="cancel()" ng-if="editing">Cancel</button>
  </form>
  <p>{{message}}</p><button ng-click="load()">Refresh Records</button>
  <table border="1" cellpadding="8">
    <tr><th>Employee ID</th><th>Name</th><th>Department</th><th>Actions</th></tr>
    <tr ng-repeat="e in employees">
      <td>{{e.employeeId}}</td><td>{{e.name}}</td><td>{{e.department}}</td>
      <td><button ng-click="edit(e)">Edit</button><button ng-click="remove(e._id)">Delete</button></td>
    </tr>
  </table>
  <script>
    angular.module("employeeApp", []).controller("EmployeeController", function($scope, $http) {
      $scope.employee = {}; $scope.employees = []; $scope.editing = false;
      $scope.load = () => $http.get("/api/employees").then(r => $scope.employees = r.data)
        .catch(() => $scope.message = "Unable to retrieve employees");
      $scope.save = () => {
        const req = $scope.editing
          ? $http.put("/api/employees/" + $scope.employee._id, $scope.employee)
          : $http.post("/api/employees", $scope.employee);
        req.then(() => { $scope.message = "Employee saved successfully"; $scope.cancel(); $scope.load(); })
          .catch(r => $scope.message = r.data.error || "Save failed");
      };
      $scope.edit = e => { $scope.employee = angular.copy(e); $scope.editing = true; };
      $scope.remove = id => {
        if (!confirm("Delete this employee?")) return;
        $http.delete("/api/employees/" + id).then(() => { $scope.message = "Employee deleted"; $scope.load();
})
          .catch(() => $scope.message = "Delete failed");
      };
      $scope.cancel = () => { $scope.employee = {}; $scope.editing = false; };
      $scope.load();
    });
  </script>
</body>
</html>
```

OUTPUT :

```
Employee Management System:
Employee table with Add, Edit, Delete, and Refresh functions synchronized with MongoDB.
```

RESULT :

Thus the above program was executed successfully.

EX.NO:85 DATE: 26/10/26	Login & register with user in MongoDB

AIM:

To write a program Login & register with user in MongoDB.

PROGRAM:

```
<!DOCTYPE html>
<html ng-app="authApp">
<head>
  <meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Experiment 10 - Login and Registration</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
</head>
<body ng-controller="AuthController">
  <h2>Experiment 10: Login and Registration</h2>
  <button ng-click="mode='register'; message=''">Register</button>
  <button ng-click="mode='login'; message=''">Login</button>
  <form ng-submit="submit()" ng-if="mode==='register'">
    <h3>Registration</h3>
    <input ng-model="form.username" placeholder="Username (min 3 characters)" required>
    <input type="password" ng-model="form.password" placeholder="Password (min 8 characters)" required>
    <button type="submit">Register</button>
  </form>
  <form ng-submit="submit()" ng-if="mode==='login'">
    <h3>Login</h3>
    <input ng-model="form.username" placeholder="Username" required>
    <input type="password" ng-model="form.password" placeholder="Password" required>
    <button type="submit">Login</button>
  </form>
  <p>{{message}}</p>
  <script>
    angular.module("authApp", []).controller("AuthController", function($scope, $http) {
      $scope.mode = "register"; $scope.form = {}; $scope.message = "";
      $scope.submit = () => {
        const endpoint = $scope.mode === "register" ? "/api/register" : "/api/login";
        $http.post(endpoint, $scope.form).then(r => {
          $scope.message = r.data.message; $scope.form = {};
        }).catch(r => $scope.message = r.data.error || "Request failed");
      };
    });
  </script>
</body>
</html>
```

OUTPUT :

```
User Authentication System:
Registered user with salted password hash.
Login verified; invalid credentials rejected.
```

RESULT :

Thus the above program was executed successfully.